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Designing Micro-Mobility Services to Support Sustainable Mobility Practices in Stockholm

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Utformning av mikromobilitetstjänster för att stödja hållbara mobilitetspraktiker i Stockholm

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Sammanfattning

På grund av klimatförändringar ställs vi inför ett akut behov att minska vardagskonsumtionens miljöpåverkan. Persontransporter är en av de största konsumtionskategorierna och står för 16% av mänsklighetens ekologiska fotavtryck baserat på aktiviteter gällande personlig konsumtion (WWF 2020). För att ta itu med detta behöver vi förstå hur människor rör sig i sitt dagliga liv, undersöka deras mobilitetspraktiker och utforska nya sätt att uppnå en mer hållbar mobilitet. Detta projekt är ett 20 veckors examensarbete på avancerad nivå och genomfördes som en del av ett utbytesprogram med The University of Tokyo.

Plattformsbaserade mikromobilitetstjänster blev ett vanligt inslag i Stockholms stadsbild år 2018 när Voi introducerades på Stockholms marknad. Mikromobilitetstjänster sågs initialt som ett hållbart transportalternativ som skulle bidra till minskad biltrafik i innerstan, men ny forskning ifrågasätter deras effektivitet i att minska koldioxidutsläppen, främst då forskning tyder på att tjänster för delad mikromobilitet, särskilt e-scooters, huvudsakligen ersätter promenader och kollektivtrafik (Hollingsworth 2019).

Detta projekt syftar till att förstå hur delade mikromobilitetstjänster kan designas för att stödja invånare i Stockholm i att anta mer hållbara mobilitetspraktiker. Projektet utgår från principer för tjänstedesign och teorier om sociala praktiker och genomfördes med kvalitativa forskningsmetoder, inklusive kontextuella och djupintervjuer, en co-creative workshop och en avslutande idégenereringsprocess. Projektet resulterade i ett förslag på ett tjänstesystem som stödjer mer hållbara mikromobilitetspraktiker genom tre mål: ökad utnyttjandegrad, uppmuntran till mer ansvarsfull användning och ersättning av bilar med mer hållbara färdmedel.

Det föreslagna tjänsteystemet, Integrerad Aktiv Mobilitet, består av tio attribut som syftar till att främja hållbar mobilitet genom att erbjuda en mikromobilitetstjänst som uppmuntrar aktiv mobilitet och gör resan till ett värdefullt mål i sig. Systemet är integrerat med kollektivtrafiken genom en stationsbaserad tjänst med hybridcyklar och ett integrerat betalningssystem. Framtida arbete bör omfatta användbarhetstestning och livscykelanalyser i kombination med konsumentvalsmodeller för att utvärdera miljöpåverkan.

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Abstract

In the face of the environmental challenge, there is an urgent need to reduce the environmental impact of everyday consumption. Personal transportation stands out as a major contributor to personal consumption, accounting for 16% of humanity's ecological footprint as measured by the most impactful activities (WWF, 2020). To address this, there is a need to investigate how people move about in their daily lives, understand their everyday mobility practices, and explore possibilities for more sustainable mobility. This 20-week master's thesis project, conducted as part of an exchange programme with The University of Tokyo, aims to contribute to this understanding.

Platform-enabled shared micro-mobility services became a common sight in Stockholm in 2018 when Voi was introduced to the market. While initially seen as a sustainable mode of transport, recent research suggests that shared micro-mobility services, particularly e-scooters, lack effectiveness in reducing carbon dioxide emissions (Hollingsworth et al. 2019). These studies indicate that these services primarily replace walking as well as short vehicle lifespan and high environmental footprint associated with manufacturing these vehicles.

This project aims to understand how shared micro-mobility services can be designed to support Stockholm residents in adopting more sustainable mobility practices. Drawing on service design principles and theories of social practice, it was conducted through qualitative research methods, including contextual and deep interviews, a co-creative workshop, and a final ideation process. The project resulted in a service system that supports more sustainable mobility practices through three objectives: modal shift, utilisation rate, and more responsible use.

The proposed service system, Integrated Active Mobility, consists of ten key attributes that aim to support sustainable mobility by making everyday mobility a rewarding experience. The system offers hybrid vehicles with a point system to incentivise active mobility. Additionally, the system is integrated with public transport through a station-based layout and an integrated payment system. Future work includes usability testing and life cycle assessments combined with consumer choice models to evaluate the environmental impact.

Acknowledgements

We want to sincerely thank everyone who contributed to this thesis project. Firstly, we are deeply grateful to our supervisor, Mia Hesselgren, for her time, invaluable insights, and support throughout the entire process.

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A big thank you goes out to all participants in the interview study, we are very happy that so many people took time off their busy commutes to talk to us and share their mobility practices. And, of course, to everyone who took time from their busy schedule to participate in the in-depth and stakeholder interviews.

Finally, the workshop participants deserve acknowledgement for their valuable knowledge, expertise, and creativity. Their contributions during the workshops were instrumental in refining our thesis project and shaping its final form.

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1. Introduction

This chapter presents the background, including the problem statement, information about micro-mobility in Stockholm, and the project's purpose and objectives. Additionally, the chapter defines key terms and delimitations, including what is meant by ‘more sustainable mobility’.

1.1 Background

In an urban setting, there is an ever-present need for mobility. People move to and from work and social gatherings, leaving, picking up kids, and acquiring goods or services. While the purpose of mobility is often to be a means to something else, sometimes the trip itself can be the end goal. In Stockholm, Sweden, public transport is widely adopted and the preferred mode choice, followed by walking, cars, and biking (Trafikförvaltningen 2020). However, according to the WWF Living Planet Report (2020), personal transportation accounts for 16% of humanity's ecological footprint by activities. Its prevalence in everyday life highlights its impact in light of the environmental challenge.

The way people move matters. These patterns of movement form mobility practices, which are routinised habits and customs that people share, involving both people and the things they use. People who engage in mobility practices know what to do next without thinking about it too much (Spaargarden 2016).

In the last few years, a new mode of mobility has been introduced in Stockholm: shared micro-mobility services (SMMS). While shared micro-mobility services like bike rentals had been available for a while, platform-enabled SMMS took off in late 2018 when the electric scooter company Voi entered the market. Suddenly, bright peach-coloured e-scooters became a common sight in the inner city, and people started to get used to seeing riders whooshing by. Although e-scooters themselves are not new, a patent was filed for a kick-scooter called Autoped in 1913 (Somerville 2023), the large number of vehicles present in the cityscape was a new phenomenon. Following Voi's introduction, many SMMS providers, mainly focusing on e-scooters, entered and exited the Stockholm market.

By the time of their introduction, e-scooters felt new and exciting. SMMS was described as a promising mobility alternative for future cities, aiding a shift towards a landscape with fewer cars in the inner city, leaving space for active transport and smaller, electric-powered vehicles. Young people rode together, cruising around for fun, while others used them for quick trips within the city centre.

However, this initial enthusiasm sparked a heated debate. Since the business model allowed for free-floating parking, e-scooters were often left scattered around the city unorganised. They were accused of littering the cityscape and hindering pedestrian access on sidewalks, especially for older people, people with disabilities, or parents with strollers (Sveriges Radio 2024). Additionally, incidents occurred, leading to concerns about the safety of both riders and pedestrians (VTI 2024).

Since their introduction, several regulations have been imposed on these services, mainly regarding the number of vehicles and parking restrictions. Furthermore, recent research shows that shared e-scooters do not deliver the sustainability improvements initially envisioned. They contribute to a negative environmental impact in terms of CO_2 emissions, mainly due to high manufacturing emissions, short vehicle lifetimes, and reckless use (Hollingsworth et al. 2019). More importantly, research suggests that SMMS mostly substitutes walking instead of heavier CO_2-emitting modes of transport like cars. Ideally, SMMS would encourage a shift from cars towards more sustainable options like walking and public transport. However, this goal appears to be largely unmet at present.

On the contrary, mobility practices are inherently dynamic, influenced by new technologies, cultural trends, and shifting preferences (Scott, Bakker 2015). The COVID-19 pandemic is a prime example of how commuting patterns change when faced with more significant societal challenges (SCB 2022). Smart technologies like video conferencing, cloud computing, and project management tools paved the way for a work-from-home culture, significantly impacting mobility practices.

For these services to have a positive impact on the environment in terms of reduced carbon dioxide emissions, several factors need to be considered: lifetime, modal shift, recharging and rebalancing, and utilisation rate (Clemm et al. 2024). Furthermore, the design of these services must allow people to use more sustainable modes of transport in their day-to-day lives. The mission is to understand how SMMS can help people pursue this beyond simply observing user behaviours. By understanding user needs and motivations, this thesis explores how micro-mobility services can incentivise and enable a shift towards a future where shared micro-mobility plays a role in a more sustainable transportation landscape.

1.2 Purpose and Research Questions

The purpose of this project is to contribute to the development of more sustainable transportation options in Stockholm. To achieve this, this project aims to understand how shared micro-mobility services can be designed to support Stockholm residents in adopting more sustainable mobility practices by answering these questions:

RQ1: What role do micro-mobility services play in Stockholmer's mobility practices?

RQ2: What shared micro-mobility service attributes are required to enable more sustainable mobility practices in Stockholm?

1.3 Definitions and Delimitations

This section describes the sustainable transportation hierarchy, terminology, and abbreviations. Additionally, it establishes the project's boundaries and limitations.

1.3.1 Sustainable transportation hierarchy

This thesis defines ‘more sustainable modes of mobility’ by referencing the Sustainable Transportation Hierarchy as defined by the Institute for Sensible Transport (2018). A mode of transport is considered more sustainable relative to another less sustainable.

The hierarchy reveals two key metrics: emissions and space intensity. Emissions refer to CO_2 emissions per person per kilometre travelled, while space intensity refers to the physical space required per person. The ranking within the hierarchy primarily emphasises emissions, positioning transport modes from 'dirty' to 'clean.' As shown in Figure 1, an average Victorian car is the 'dirtiest' option, while walking is the 'cleanest.' These extremes also represent the most and least space-intensive modes, respectively.

While shared micro-mobility services are not included in the original ranking, this report positions them as more sustainable than motorcycles but less sustainable than trains and public transport. This placement is not based on emissions or space intensity, but on the notion that SMMS should complement, not compete with, public transport and walking. If an SMMS user opts for the service instead of walking, it can be seen as a less sustainable choice. However, when SMMS encourages people to use public transport instead of modes with higher CO_2 emissions, the overall outcome is reduced emissions.

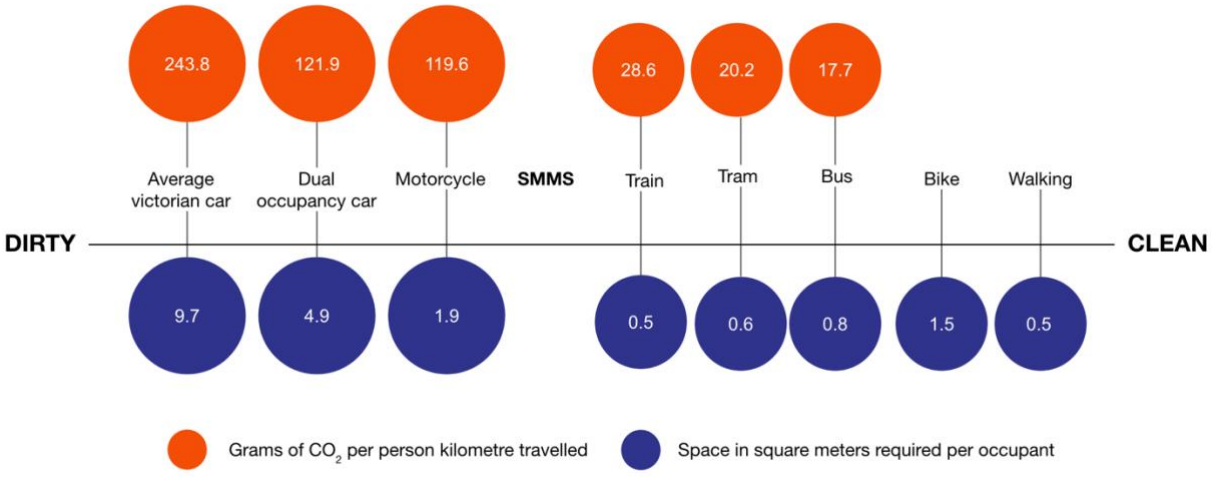


Figure 1. Sustainable Transport Hierarchy, adapted from the Institute for Sensible Transport (2018)

1.3.2 Terminology

Mobility – the ability to move freely and easily. The purpose of mobility is often to gain access to destinations, activities, services, and goods (UN-Habitat n.d.).

Transport – the movement of people and goods with a vehicle (Lee et al 2023).

Micro-mobility – small, lightweight vehicles that travel at low speeds, usually below 25 km/h, primarily used for short-distance travel.

Shared Micro-mobility Services – electric or human-powered micro-mobility vehicles offered on-demand as a service, shared among users.

First/ Last Mile Problem – refers to the difficulty of connecting journeys made by public transport with a person's origin or destination.

Modal Shift – describes a switch from less sustainable modes of transport, like cars, to more sustainable options like cycling or public transport.

Product-service Systems – tangible products and intangible services designed and combined so that they jointly are capable of fulfilling specific customer needs (Tischner et al. 2002).

Smart Product-service Systems – a combination of physical products and digital services that leverage information and communication technologies (ICT) to create a more integrated and user-centric experience.

Free-floating – something that is not docked or stationed in a fixed location, but rather available for use and left wherever the user finishes their trip.

Mobility as a Service – on-demand access to various transportation options through a single platform, allowing users to seamlessly plan, book, and pay for their trips.

1.3.3 Abbreviations

SMMS – Shared micro-mobility service

PSS – Product service system

SDL – Service-dominant logic

GHG – Greenhouse gas

CO_2 eq – Carbon dioxide equivalent

IC card – Integrated circuit card

HMW – How might we

1.3.4 Scope and delimitations

This master's thesis project, completed within a 20-week timeframe, focuses on platform-enabled SMMS. It excludes small-scale, non-platform services such as bike rentals aimed at day rentals for tourists. The project's scope is limited to everyday mobility, which refers to how people move around in their daily lives. Special occasion mobility, which might imply different requirements for SMMS systems, is not considered within the scope of this project.

While this project does not question the existence of SMMS services, it does investigate possibilities to reduce their CO_2 emissions and how SMMS can encourage a shift away from personal vehicles, ultimately leading to lower overall emissions. However, manufacturing and end-of-life emissions are outside the scope of this project.

2. Theoretical Frame of Reference

This chapter introduces the theoretical framework used in the project. It explores service design and its principles, product-service systems, and theories of social practice in the context of everyday mobility. It further defines micro-mobility and its environmental effects, as well as outlining the mobility landscape in Stockholm.

2.1 Service Design

This section introduces service design and its five core principles, including service dominant logic.

Services are a prevalent part of society, with services being a large and growing part of the economy. According to service-dominant logic, services are the 'beginning, middle, and end' of all economic activity (Bastiat 1995). The design of services, therefore, has the potential to shape much human activity.

Service Design is rooted in design thinking and customer experience and brings a creative and human-centred process to service improvement and the design of new services. A service design mindset is pragmatic and co-creative and seeks to balance technological opportunity with human needs and business relevance. The service design process is iterative at its core, working with several cycles of convergent and divergent thinking, seeking to deepen and explore knowledge with early user feedback and prototypes (Stickdorn et al. 2018).

Service design is a cross-disciplinary approach that encompasses more than just the service experience, but also the delivery, operations, business model, and the entire being of the service. There is no single definition of service design. As described in This is Service Design Doing, the authors refer to service design as 'a human-centred, collaborative, interdisciplinary, iterative approach… to create and orchestrate experiences that meet the needs of the business, the users, and other stakeholders' (Stickdorn et al. 2018, p. 27).

2.1.1 The five principles of service design

There are five principles of service design thinking: user-centred, collaborative, iterative, sequential, real, and holistic (Stickdorn et al. 2018).

Human-centred means that service design considers all human experiences by and of the service, not only the end user, including the service provider as well as other stakeholders and non-customers that might be impacted by the service.

Collaborative in the way that multiple stakeholders of various backgrounds should participate in the service design process, which can facilitate breaking down barriers to create a common language and understanding.

Iterative means that service design has an explorative and adaptive nature, with multiple cycles of iteration toward implementation.

Sequential refers to the importance of the experience, which can be visualised as a series of interrelated actions between different stages or touchpoints. Thus, this is the way the service itself should be visualised and orchestrated.

Real because the service needs to be viable, as it should in its essence be a practical approach. The object of the design must have business potential in the real world, including backstage processes and business goals for the companies. Consequently, the needs of the users must be researched in the real world, as well as with prototyping. There is also a need for evidence, making the intangible values of the service evident in the physical reality - where the folded toilet paper in a hotel bathroom acts as a classic example.

Holistic refers to the need to address the needs of all stakeholders affected by the entirety of the service and the business related to it.

2.1.2 Service-dominant logic

Service-dominant logic (SDL) challenges the traditional view of economic activity by placing services, rather than goods, at the forefront of exchange. Coined by Vargo and Lusch (2004), SDL emphasises that no products or services have inherent value. Instead, value is created in the dynamic moment of a service exchange, where the beneficiary ultimately determines the perceived value.

Five key axioms underpin SDL and provide a framework for understanding the nature of services:

1. *Service is the fundamental basis of exchange:* This axiom posits that all economic activity, regardless of the tangible goods involved, is ultimately based on the exchange of services.
2. *Value is co-created by multiple actors, always including the beneficiary:* This axiom highlights the collaborative nature of value creation, where the beneficiary plays a crucial role in co-creating value alongside the service provider and other stakeholders.
3. *All social and economic actors are resource integrators:* This axiom recognises that all actors involved in a service exchange contribute resources, tangible or intangible, that contribute to the value creation process.
4. *Value is always uniquely and phenomenologically determined by the beneficiary:* This axiom emphasises the subjective nature of value, as it is ultimately determined by the individual experiences and perceptions of the beneficiary.
5. *Value co-creation is coordinated through actor-generated institutions and institutional arrangements:* This axiom acknowledges the role of social structures and norms in shaping how value co-creation takes place.

These axioms, as outlined by Vargo and Lusch (2004), provide the foundation for SDL, which explains that the value of an object is ultimately determined by the perceived services it can provide and that these services are co-created by multiple actors, including, but not limited to, the service provider (Stickdorn et al. 2018).

2.2 Product-Service Systems

This section introduces product-service systems, followed by the subcategory of product renting and sharing. Finally, smart product-service systems are introduced.

Product-service systems (PSS) encompass a wide range of products, services, and businesses. They constitute a spectrum that extends from predominantly physical products to purely intangible services. A physical item like a book, defined solely by its material properties and not the potential services it offers users, exemplifies a pure product. Conversely, counselling exemplifies a true service, involving no physical products whatsoever. Between these extremes lie numerous examples of PSS types, as shown in Figure 2, some leaning more toward the product side and others towards the service side.

Tukker (2004) defines a product-service system as consisting of 'tangible products and intangible services designed and combined so that they jointly are capable of fulfilling specific customer needs'. He further identifies three main types of PSS with distinct value-creation mechanisms. Product-oriented PSS are mainly geared towards selling products, but the contracted services are in the realm of the responsibility of the provider. In a use-oriented PSS, the ownership and the responsibility for the usability of the tangible product lies with the provider. Lastly, in the result-oriented PSS, the provider is responsible for delivering the result, but the means to provide that result does not include any predetermined product.

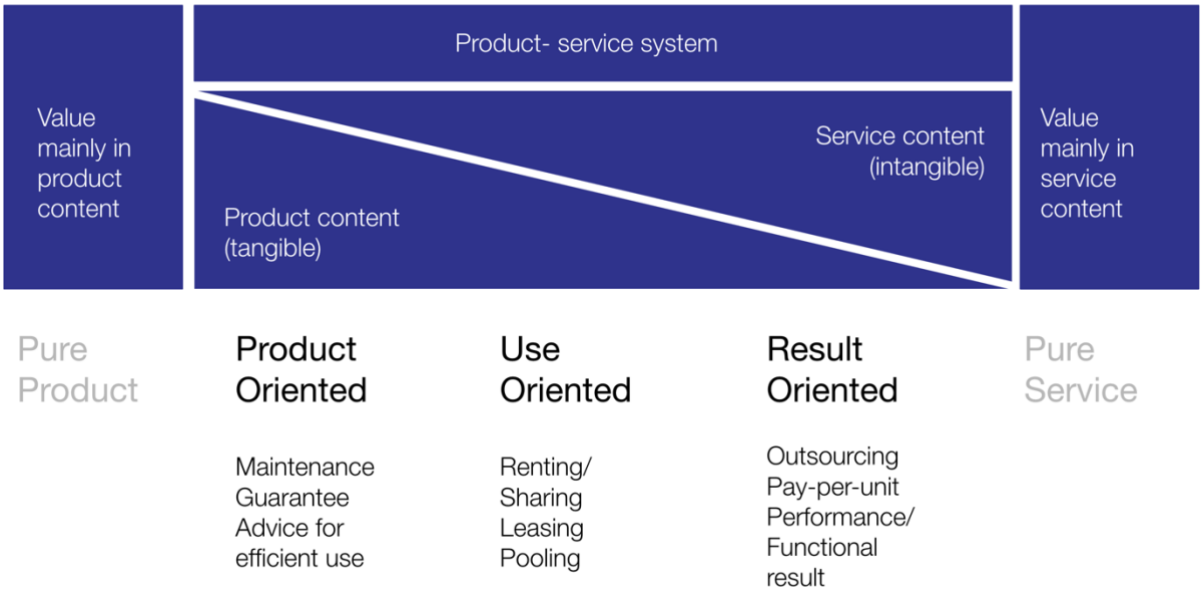


Figure 2. The PSS-spectrum as defined by Tukker (2004)

For a PSS to achieve significant environmental improvements compared to traditional product-heavy models, it must simultaneously create added value for users, justifying the business model, while also minimising material use and emissions (Tukker 2004). Tukker suggests that result-oriented PSS holds the most promise, as it allows for greater freedom in designing solutions that meet customer needs without relying on predetermined tangible products. However, as a business model becomes more abstract and service-oriented, complexity increases, making it more challenging for providers to verify the environmental impact of their offerings and for users to recognise a fair exchange of value.

Despite this increasing complexity, service designers gain more freedom to address users' true needs. This shift involves moving beyond simply asking 'What can the product do?' and delving into the user's practices, opening possibilities for more radical solutions that could

tackle wicked problems, which are complex social or environmental issues with no easy solutions, where any attempt to address them can create new challenges.

For a PSS to be attractive to the user, it has to offer both tangible and intangible value for users. The tangible value is easier to grasp, the user identifies the business offering and can make a well-informed, rational calculation on the true cost of the product before deciding if the value exchange seems fair. For intangible values, it is not necessarily as straightforward. Intangible value is geared towards higher needs such as affiliation, love, self-esteem, and self-realisation (Tukker 2004). Intangible values represent value but are not intrinsic, they cannot be touched or easily formulated (Cambridge University Press, n.d) and it is often linked to the customer experience of a product or service. According to Steiner and Harmon (2009), intangible values can be categorised into knowledge, experience, and emotions.

Knowledge is related to the customer's subjective perception of the expertise of the company, including its product and service offerings. Experience reflects customer satisfaction and can be the starting point of a strong customer-brand relationship. The experience does not have to be firsthand but can be the experience of a family member or friend. Finally, emotions in various forms arise when a customer interacts with a business, and of course, these can be both negative and positive. Furthermore, emotions are difficult to measure and control (Steiner & Harmon 2009).

In conclusion, PSS represents a dynamic approach to meeting customer needs, blending tangible products with intangible services across a spectrum of offerings. Despite the increasing complexity inherent in result-oriented models, the focus on addressing genuine user needs offers opportunities for innovative solutions and meaningful value exchanges. Furthermore, by recognising and integrating both tangible and intangible dimensions of value, businesses can better meet the diverse needs of customers while contributing to sustainable practices.

2.2.1 Product renting and sharing

SMMS falls under the PSS category of product renting and sharing. In this use-oriented business model, users do not own the tangible product but pay for its use, with the provider maintaining it. The main difference from leasing is that users do not have unlimited or exclusive access to the product.

Tukker (2004) suggests that product renting and sharing generally leads to a loss of intangible value for customers, as rental products typically do not contribute to higher self-esteem or ‘priceless’ experiences. However, he acknowledges examples like renting a car and feeling like a ‘king for a day.’

Furthermore, this business model increases the time and effort users need to invest to gain access to the product, potentially incentivising less frequent use. While a tangible sacrifice for users, the low initial cost can attract more new users. The provider, on the other hand, needs to invest more capital in human resources, primarily because ownership stays with the service provider. This can be offset by the fact that the overall capital required in the system is lower due to the shared use (Tukker 2004).

Increased product usage can significantly reduce the system's environmental impact, particularly when the manufacturing phase has a high life cycle cost. However, the business model may slightly discourage product use due to limited access. This can lead to further

environmental benefits, as users might shift to more sustainable transportation options like public transport (Tukker 2004). Nevertheless, this is only a positive outcome if the substitution pattern includes a switch to a more sustainable mode compared to the mode that is the subject of analysis.

2.2.2 Smart product-service systems

Many modern SMMSs can be defined as smart product-service systems. Valencia et al. (2013, p.16) describe a smart PSS as a system that 'integrates smart products and e-services into single solutions to jointly address needs of customers. They are defined as smart because the services make use of 'ICT such as microchips, software, and sensors to connect, collect, and process information' (Rijsdijk & Hultink 2009, p. 25). Several of the shared micro-mobility services readily available in Stockholm, including e-bike and e-scooter platforms, employ intelligent solutions, enabling users to seamlessly interact with and access information about the products before, during, and after usage through a digital platform that offers real-time updates.

When sharing items with unfamiliar individuals, both providers and users face heightened transaction costs related to searching for and coordinating rentals, as well as the potential risks involved in interacting and negotiating with strangers. To mitigate these challenges, digital platforms offer an effective solution. Through the utilisation of ratings and reputation systems, these platforms can reduce the risks associated with sharing among strangers, while also facilitating the logistical aspects of sharing or providing access to the vehicle (Frenken & Schor 2017).

To conclude, a product renting and sharing PSS has the potential to reduce climate impact compared with traditional, use-oriented business models. However, the changes might be incremental and the business model itself is insufficient to promise sustainability improvements. If it implies a mode shift towards less resource-intensive modes of transport the impact can be positive, but a less product-centred approach might be more suited to strive for more than incremental change. When designing a product-sharing and renting PSS it is important to make up for the loss in intangible value and to use smart resources to facilitate mutual communication between users, providers, and products alike.

2.3 Theories of Social Practice

This section introduces theories of social practice, followed by a framing of three types of practice intervention strategies.

When investigating mobility from a sustainability perspective there might be a need to look beyond current travel patterns and behaviours. Human behaviour is influenced by many factors such as the designed objects around us and undergo constant change. How people act, and the behaviour that can be observed, *what people do*, are just the tip of the iceberg. Starting from that might eliminate other, more sustainable alternatives from the solution space. Additionally, when faced with wicked problems such as the environmental challenge, a more systemic approach is needed.

The act of mobility is often a means to something else - people take the car to leave and pick up their kids from after-school activities. Then offering activities to the kids is the main activity, using the car is just a means to realise that activity. In a broader sense the main purpose of that

activity is to provide a meaningful childhood and equip the kids with important skills for their future. Having this viewpoint opens a bigger solution space that does not stem from current patterns of use.

Social practices are routinised habits and customs that people share, which involve people, artefacts, and technology. People who engage in social practices know what to do next without thinking about it too much (Spaargarden 2016). A practice is defined as 'a routinised way in which bodies are moved, objects are handled, subjects are treated, things are described, and the world is understood' (Reckwitz 2002). Examples of practices include cooking, swimming, doing laundry, and travelling to work.

Shove et al. (2008) introduced the stuff-image-skill model, visualised in Figure 3, which explains practices as a bundle of three elements. 'Stuff' refers to the material elements that constitute the practice, such as technology, infrastructure, spaces, and compositions. 'Image' is the social and personal meanings that can be achieved by participating in a practice, including emotions, beliefs, identity and aesthetics. Finally, 'Skill' refers to the know-how of accomplishment and involvement of a practice, such as taste and competence. These three elements are connected by dynamic linkages, forming a bundle of the three elements.

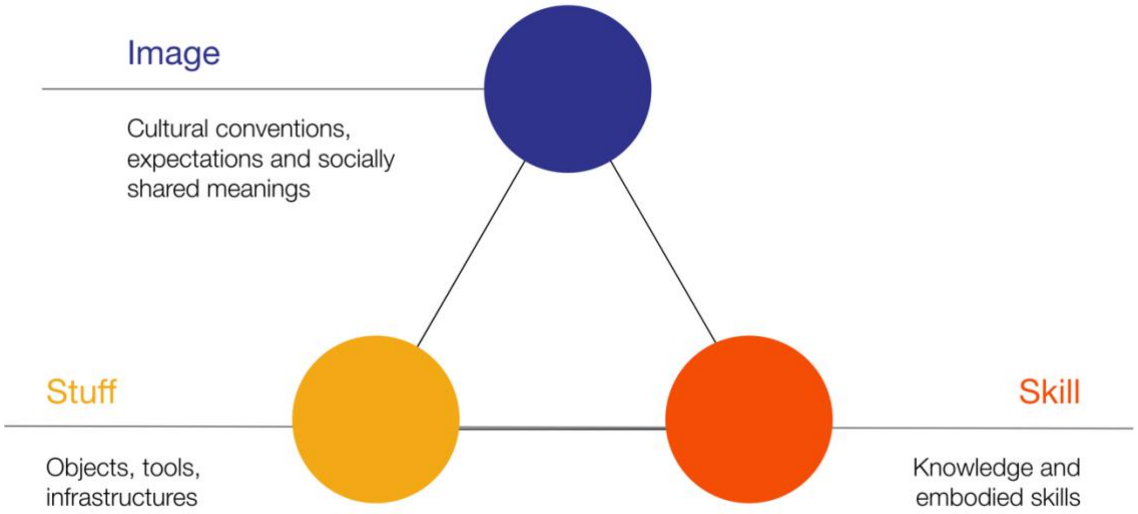


Figure 3. 'Stuff-image-skill' model, adapted from Shove et. al. (2008)

Furthermore, social practices are inherently dynamic and social, meaning that they exist only through reiterative performances over time and are learned from one another, from which norms of practice emerge (Scott, Bakker, and Quist 2012). When several practices are bundled together, they form lifestyles. Considering the dynamic nature of social practices, the linkages between practice elements can be strengthened or weakened through performance. This suggests that practices can be unstable, and potentially fade from use over time. Conversely, innovations in practice can happen when existing links are broken, and new ones are formed. New technology, ideas, or procedures, or even a novel combination of existing elements, can create these new connections (Scott, Bakker, & Quist 2012).

Furthermore, technology itself can create a lock-in effect on practices. As technology becomes more sophisticated ('smarter' as Scott, Bakker, & Quist, 2012 put it), the tools and artefacts used become so integrated into daily life that the practice itself becomes dependent on the

specific product. This dependence can make it difficult to switch to alternative practices, even if they might be more enjoyable or desirable.

Scott, Bakker, & Quist (2012) further argues that design should move beyond a focus on the product itself (e.g., cars) and instead examine the underlying practices (e.g., driving). This shift in perspective allows designers to challenge conventional consumption patterns and the established norms associated with those practices. By highlighting the dynamic nature of social practices, designers can question the 'normal' and taken-for-granted behaviours, rather than assuming user needs and behaviours are fixed.

2.3.1 Mobility practice interventions

Intervention in everyday mobility practices can be done in several ways, with different time limits and aspirations for change. As established in the section about theories of social practice, there is a need to look beyond technology fixes and behaviour change. Instead, interventions need to target the mobility practises themselves. This includes practices like driving and commuting, along with the practices of policymakers and governing bodies who influence these behaviours (Strengers & Maller 2016).

Spurling (2015) suggests three different types of 'intervention framings' that highlight three different types and scales of ambitions, where the last one is seen as the most promising. The first type of intervention framing is one of 'recrafting resource-intensive practices'. This focuses on reducing the resource intensity of existing practices by changing the elements that they are composed of to make them less resource intensive. By intervening in the constitutive elements of practices, the aim is to reduce the overall resource intensity without challenging the scale or extent of the practice itself.

The second intervention framing is 'substituting practices'. This involves discouraging current unsustainable practices and replacing them with existing or new alternatives. Substituting aims to change how particular 'needs' or 'wants' are met by identifying one practice that could be substituted with another, one more sustainable than the other. The idea is to target both practices simultaneously and shift the balance in favour of the more sustainable one. By recrafting multiple practices simultaneously, policymakers can promote the adoption of more sustainable alternatives. This approach changes the practices such that more or fewer performances occur, rather than focusing on individual behaviour.

The third and final intervention framing is 'Changing how practices interlock'. This connects to the notion of practice bundles being reproduced through reiterative performances over time (Shove et al. 2012) This approach acknowledges that our 'need' for travel is not fixed. By looking at the interlocking practices, interventions may target a wider system of practices that produces the need for mobility, thereby challenging the need itself. This opens doors for interventions that go beyond traditional transport policies. Policymakers could target other household needs, such as how groceries are acquired and how work and leisure are connected.

By applying social practice theories, policymakers and designers can move beyond simply introducing shared micro-mobility and instead focus on how it integrates into the existing web of everyday practices. Embracing these strategies offers the potential to create a more sustainable and efficient urban mobility system, reducing resource consumption and enhancing the quality of life in urban environments.

2.4 The Environmental Impact of Micro-mobility

This section acts as a frame of reference for understanding micro-mobility and the environmental impacts of shared micro-mobility services.

Micro-mobility refers to small, lightweight vehicles that travel at low speeds, usually below 20 km/h. They are primarily used for short-distance travel and can function as an alternative to owning a car and complement public transportation in urban settings. Micro-mobility vehicles can be either human-powered or electric, and they may be privately owned or offered as a service, shared among users (Institute for Transportation and Development Policy 2020).

Several factors determine whether a shared micro-mobility service contributes positively, negatively, or conditionally to the environment. A literature review by Clemm et al. (2024) identified four key factors influencing the environmental impact of these services: modal shift, vehicle lifetime, utilisation rate, and efficient recharging and rebalancing. All four factors have a positive correlation with environmental sustainability.

Modal shift refers to substituting another mode of transport. The environmental impact of modal shift depends on the emissions profile of the replaced mode of transport. Shifting from a high-emission mode, like a personal car, can significantly reduce environmental impact. However, replacing an already low-emission mode, such as walking or cycling, would not yield environmental benefits.

Several studies provide insights into substitution patterns related to SMMS. Studies conducted in Paris (Christoforou et al. 2021) and Zurich (Reck et al. 2022) indicate that e-scooters most frequently replace walking trips (35% of trips) and public transport use (37% and 23%). A Study in Oslo showed that as much as 60% of shared e-scooter rides replaced walking, followed by 23% for public transport (Fernley et al. 2020). Additionally, a study made in Vienna also showed that walking was the most substituted, followed by public transport (Laa B., Leth U. 2020). All these studies show that SMMS does not replace personal car usage in any significant capacity.

E-bikes and e-scooters are common modes for SMMS, these vehicles are also available for personal ownership. One study that compared the CO_2-eq emissions per passenger mile for personal vehicles and shared services showed that personal ownership of the vehicle resulted in a decrease in emissions while shared services increased (Reck et al. 2022). This was true for both e-bikes and e-scooters and goes against the notion that the sharing economy always yields environmental benefits. However, Rick argues that SMMS can help spark sustainable mobility transition long term, even though they are not environmentally beneficial in a short-term perspective.

Studies suggest that increased utilisation rates of shared micro-mobility vehicles can reduce their environmental impact, as measured by CO_2 emissions per passenger kilometre (Sun & Ertz 2021). Sun and Ertz (2021) specifically highlight the negative influence of low utilisation rates on shared bicycles, arguing that this factor contributes to higher greenhouse gas (GHG) emissions than personally owned bicycles. A higher utilisation rate allows a single vehicle to provide more transportation value throughout its life cycle, reducing the environmental impact per trip.

Some researchers highlight the potential of planning SMMS so that they synergise better with public transport and encourage multi-modal trips (Berg Wincent et al. 2023; Severengiz S et al. 2020). Using SMMS in combination with public transport might substitute personal car usage to a greater degree and help solve the first/last mile problem with public transport.

A life cycle assessment of shared electric e-scooters showed that 50% of the CO_2-eq emission comes from material and manufacturing (Hollingsworth et al. 2019). The result also showed that the life cycle emissions of e-scooters are highly sensitive to their operational lifetime. This stresses the importance of a long e-scooter lifetime through robust design, proper maintenance practices, and encouraging considerate use. Reducing the need for frequent production cycles will considerably lower the associated greenhouse gas emissions. Most studies lack reliable information on the actual lifetime of vehicles operating in a shared micro-mobility service. However, assumptions have been made as low as three to 15 months (Clemm et al. 2024).

To counter the lack of lifetime information, some life cycle studies assume vehicle lifetime based on battery lifetime, as it is assumed to be the decisive factor for the e-scooters lifetime (Hollingswoth et al. 2019). In shared Product-Service Systems (PSS), such as SMMS, reckless user behaviour can shorten product lifespans. This risk stems from the absence of a sense of ownership, as users do not bear the direct consequences of damage (Tukker, 2004).

Finally, efficient recharging and rebalancing refers to operational strategies that produce carbon emissions. This mainly involves picking up and redistributing vehicles according to demand and picking them up for charging or replacement of batteries. The extent of these procedures is affected by how well the system of vehicles balances itself naturally and battery life. Using electric vehicles when managing the service can reduce the environmental impact of these operations (Clemm et al. 2024).

2.5 The Mobility Landscape in Stockholm

This section frames Stockholm's mobility landscape, including several modes of mobility that extend the use of SMMS. It then explores the history of SMMS in Stockholm, including key policy interventions, and concludes with an overview of past e-scooter user behaviour patterns.

In the Stockholm region, all public transport is provided by the municipality and procured by the company Stockholms Lokaltrafik, SL. Public transport in Stockholm includes traffic with metro, buses, commuter trains, local railways, and commuter boats, all operated by SL (Region Stockholm 2024).

A travel habits survey conducted in 2019 (Trafikförvaltningen, 2020) revealed public transport as the dominant mode of transportation in central Stockholm, accounting for 35% of trips. Walking followed closely at 28%, with car use at 25% and cycling at 10%, as seen in Figure 4. However, the survey also highlighted a significant disparity in travel patterns between central Stockholm and its adjacent municipalities. Cars were the preferred mode of transport in these areas, constituting 52% of all trips. Public transport use was less popular at 24% while cycling and walking were even lower at 6% and 15%, respectively (Trafikförvaltningen, 2020).

The survey further revealed a correlation between age and car usage, with car popularity increasing as age increases. Additionally, the data suggests a gender difference in travel

preferences, with men exhibiting a higher propensity for car use compared to women. The three dominating reasons for making a trip in Stockholm are work and school (36%), leisure (15%), and grocery shopping (10%). Lastly, the survey showed that the start times for weekday trips are concentrated in two peaks with high traffic in the morning and afternoon. (Trafikförvaltningen, 2020).

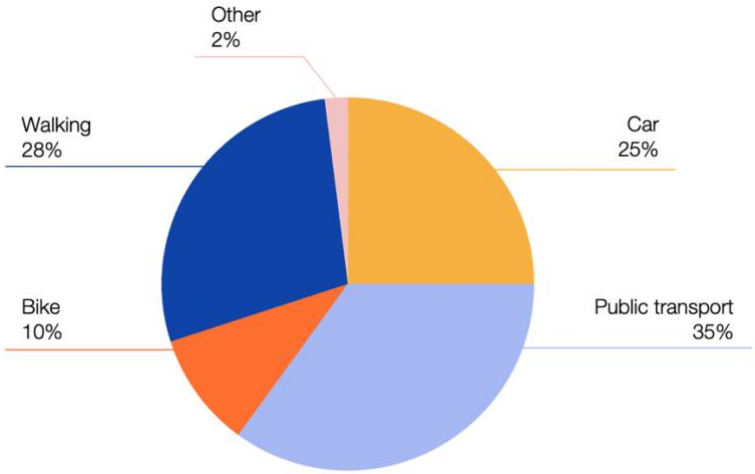


Figure 4. Modes of transport in Stockholm (Region Stockholm, 2020)

In Stockholm, the micro-mobility market has been subject to many different companies operating in the same area. The Swedish company Voi was the first to launch an SMMS offering e-scooters in 2018, followed by the American company Lime the same year. As of 2024, four companies are operating in the Stockholm area: Voi, Lime, Bolt, and RideMovi. All offer shared e-scooters, with RideMovi additionally providing free-floating e-bikes, and Bolt expanding its services to include ride-hailing. Berg Wincent (2024) compiled a comprehensive timeline of the companies that have entered and left the market since Voi first launched in Stockholm, as seen in Figure 5.

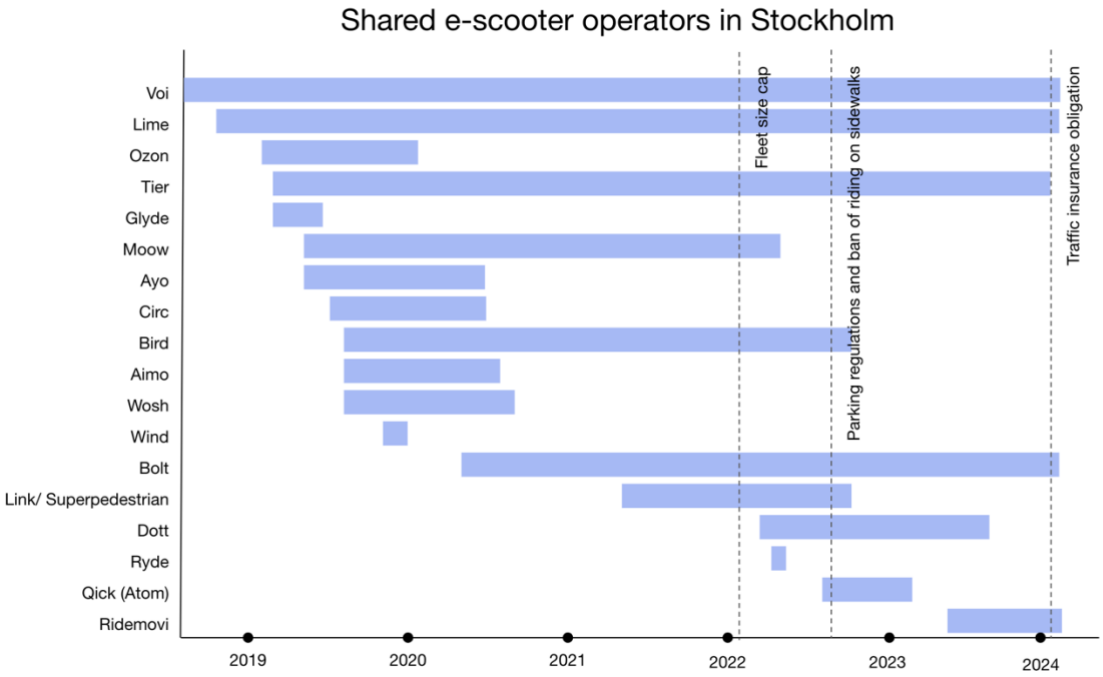


Figure 5. E-scooter operators in Stockholm, entry and exit dates, adapted from Berg Wincent (2024)

In Figure 5, Berg Wincent also highlights major policy implementations that have shaped the micro-mobility landscape. A significant development was the introduction of a fleet size cap in early 2022, which regulates the number of vehicles available for rent on the streets. Currently, Stockholm allows a maximum of 12 000 e-scooters (Stockholms stad 2024). Additionally, 2022 saw the implementation of parking regulations, followed by more recent requirements for traffic insurance. Stockholm's parking regulations mandate that shared e-scooters be parked in designated spots. Stockholm offers a mix of parking options for e-scooters, with 700 designated stands and 100 shared stands across operators. Additionally, parking in regular bike stands is permitted.

The usage of e-scooters and the number of vehicles in Stockholm for all operating companies have been documented since late 2020, as seen in Figure 6 (Miljöbarometern 2024). The vehicle data before this date was manually collected from the e-scooter companies that operated during that time. The data shows a steady decline in trips taken since 2021. There is also a clear seasonal correlation affecting the number of trips, where wintertime has the least number of trips. This is expected due to Stockholm's weather conditions.

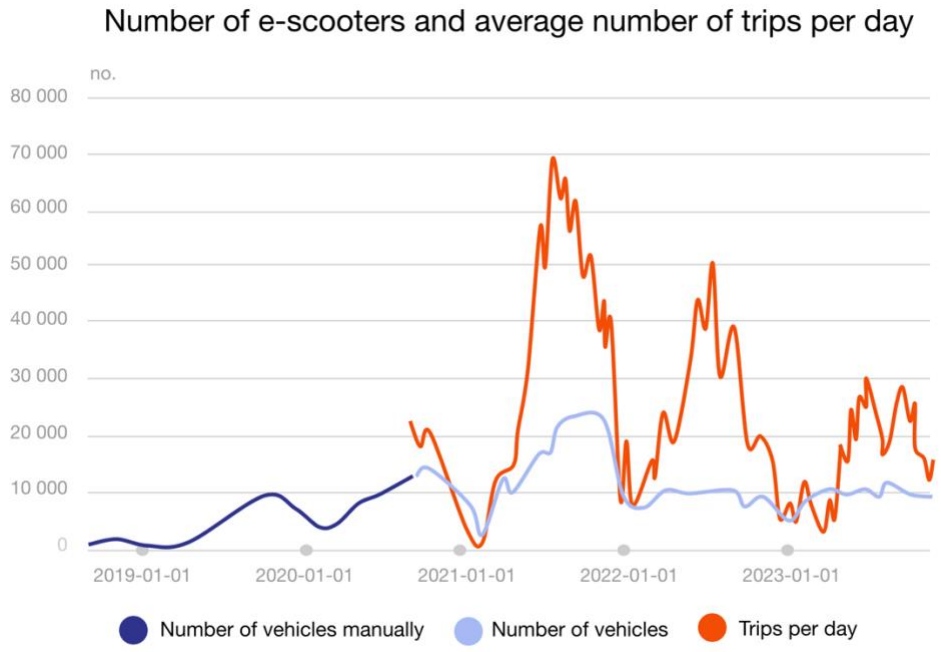


Figure 6. Number of shared e-scooters and travels made with shared e-scooters per day, adapted from Miljöbarometern (2024)

In addition to e-scooter services in Stockholm, there have been a few micro-mobility services offering bikes and e-bikes. In 2006, Stockholm's first public bike-sharing program, Stockholm City Bikes, launched. The bikes used a station-based system and were funded by the city with advertising support from Clear Channel. In 2018, there was a legal issue, and the city was left without any official bike-sharing program (Berg Wincent 2024). However, in May 2022, Stockholm launched a new e-bike sharing system called Stockholm e-bikes, facilitated by the company Marfina. This was a station-based system of electric bikes where users could pre-reserve bikes, pick them up, and leave them in one of the 100 geofenced locations around Stockholm. These locations were marked by a specific sign and a parking square. However, the system faced technological issues, malfunctioning bikes, and difficulties with scaling up. These

issues culminated in safety concerns that ultimately led to the city terminating the contract in May 2023 after a year of operation (Berg Wincent 2024).

In Sweden, national law dictates that e-scooters with a maximum speed of 20 km/h and a maximum motor power of 250 watts follow the same rules as bicycles. This implies riders can operate these e-scooters on bike paths and adhere to bicycle traffic regulations. These regulations include mandatory helmet use for riders under 15 years old. Additionally, the e-scooters must be equipped with functional brakes, a bell, and front and rear lights. Conversely, e-scooters exceeding these specifications are classified as mopeds, requiring riders to follow moped regulations (Polisen 2023).

Similar laws also dictate bikes with electric assistance. For these e-bikes, the electric motor should engage when the pressure on the pedals increases. The motor may only amplify the power from the pedals, it cannot provide any additional power at speeds over 25 km/h, and the rated power must not exceed 250 watts (Transportstyrelsen 2024). Micro-mobility service providers in Stockholm are responsible for ensuring that their vehicles meet the legal requirements. In addition, they typically enforce an 18-year-old age limit for using their vehicles and services (Voi 2024; Lime 2023).

2.5.1 E-scooter user behaviour in Stockholm

Between 2019 and November 2023, Sweden witnessed a concerning rise in e-scooter accidents, with 9 271 incidents involving at least one e-scooter, both shared e-scooters and personal. Tragically, ten of these accidents resulted in fatalities, according to VTI (2024). While men under 35 years old represent the most frequently affected group, analysis based on usage frequency reveals an even higher risk for young people. Additionally, these accidents often include drivers under the influence of alcohol (VTI 2024). These statistics show a safety risk regarding e-scooter use and shed light on demographic trends.

Shared e-scooters see a higher incidence of reckless rider behaviour, leading to damage and misuse within the city. In an article by Sveriges Radio (2024), Fredrik Johansson, diver and chair of the association Rena Mälaren, details his efforts retrieving discarded e-scooters from the waters around Lake Mälaren, which has its' outlet in the Baltic Sea, spanning the islands of Stockholm. According to Johansson, the organisation has retrieved around 1400 e-scooters from the bottom of Lake Mälaren since 2018. This is an environmental concern because the electronics in the e-scooter are a hazard to the environment.

Additionally, Stockholm municipality moved 3000 wrongly parked e-scooters during the month of April 2024 (Sveriges Radio 2024). A 2020 resident survey conducted by Stockholm Stad revealed that a significant majority of residents held negative views on e-scooters. Only 20% of respondents expressed a very positive or somewhat positive attitude, while 59% indicated a very negative or somewhat negative attitude (Miljöbarometern 2024).

The presence of e-scooters extends beyond physical transportation infrastructure. Notably, the Swedish company Voi was referenced in a 2024 song by the artist Fröken Snusk, with lyrics depicting e-scooter use under the influence of alcohol. This example, publicly condemned by Voi's CEO (Sveriges Television 2024), highlights the culture of normalisation of irresponsible e-scooter behaviour, further reinforcing concerns about a culture of recklessness surrounding these micro-mobility vehicles.

2.6 Mobility as a Service

The transportation sector is undergoing a significant transformation, driven by technological advancements, environmental concerns, and evolving user preferences. In this context, Mobility as a Service (MaaS) has emerged as a promising concept that aims to revolutionise how people travel. MaaS goes beyond simply providing access to transportation; it offers a holistic approach that integrates various modes of transport into a seamless and user-centric service (Heikkilä 2014).

Heikkilä (2014) defines MaaS as an added level to the organisation of service provision that could transform the passenger transport sector by focusing on convenient provision of a versatility of attractive mobility services. This definition underscores the importance of user experience and the need to offer a diverse range of mobility options.

MaaS platforms typically function through mobile applications that allow users to plan, book, and pay for their journeys across various transportation modes, including public transportation, shared bicycles, e-scooters, car rentals, and ride-hailing services. Heikkilä (2014) proposes a monthly payment where the user can add a bundle of these services and compared it to telecommunication and media service sectors. This integration eliminates the need for multiple ticketing systems and applications, streamlining the user experience and promoting multimodal travel. However, implementing a successful MaaS system requires overcoming certain challenges. Integrating different transportation providers into a single platform requires collaboration and data sharing, which can be a complex achievement. Additionally, ensuring equitable access across different socioeconomic demographics is crucial for MaaS to reach its full potential. Several failed attempts at successfully integrating MaaS have been recorded, one being UbiGo, which tried to implement MaasS in Sweden (Sochor et al. 2015).

Examining the Japanese transportation landscape reveals a successful case of mobility integration with unique characteristics compared to other models. Japan's extensive public train network comprises several distinct rail systems facilitated by different operators. Within central Tokyo, most above-ground trains are operated by Japan Railways (JR), and the underground subway system is serviced by two primary operators: Tokyo Metro and Toei Subway. Additionally, a network of private train lines extends outwards from central Tokyo, serving the Greater Tokyo Area (Go Tokyo 2024). Despite Tokyo's intricate system of public transport, a single form of payment streamlines travel across the entire area: Integrated Circuit (IC) cards. These IC cards are available both as physical cards and as digital versions through services such as Apple Pay and Google Pay. Since 2013, ten major IC card providers have achieved successful interoperability, allowing seamless travel across different train lines and operators (Kurosaki and Higashino 2019).

Some bike-sharing operators also use the IC cards in Tokyo, not for payment but for membership registration. Given the widespread adoption of IC cards in Japanese society, this approach streamlines the user experience of SMMS. While alternative payment methods are typically registered through mobile apps, certain bike-sharing providers enable users to tap their IC cards at a station to initiate a ride (Hello Cycling 2024; Docomo 2024). Additionally, the IC card can also be used in several restaurants and convenience stores.

3. Method

This chapter outlines the project's approach, detailing the processes and methods utilised throughout, including data collection, synthesis and analysis, ideation, and concept development.

3.1 Approach

This project undertook a service design approach, utilising human-centred methods and tools, focusing on the mobility user experience in Stockholm. This was combined with theories of social practice, examining how shared practices, like commuting or errands, influence Stockholmers' daily lives.

Users and non-users of micro-mobility services were targeted to gather qualitative data from Stockholm residents. Qualitative research methods were used throughout the project to gain deeper insights into the lived experiences and social contexts surrounding mobility- and micro-mobility practices in Stockholm. To improve the richness and rigour of the research, a triangulation of methods has been used, combining literature studies, participant approaches, and co-creative workshops. This ensures that the insights produced rely on an extensive and diverse dataset, providing a solid groundwork for design choices (Stickdorn et al., 2018). Furthermore, triangulation ensures thoroughness through validation of interpretations of data findings, strengthening confidence in the findings (Thurmond 2001).

To develop service attribute suggestions, qualitative data was combined with insights from the frame of reference, which primarily consisted of quantitative data on the environmental effects of SMMS and other mobility modes. This approach enables a comprehensive understanding of mobility practices in Stockholm within the project's timeframe while integrating established data on sustainable mobility to inform design decisions.

3.2 Process

This project followed the double-diamond process, an iterative design process with two cycles of converging and diverging. The diverging phases seek to create opportunities, and the converging phase is aimed at decision-making (Designcouncil n.d.). The process is divided into four overarching phases of divergent or convergent thinking: *discover, define, develop, and deliver*, as displayed in Figure 7 below.

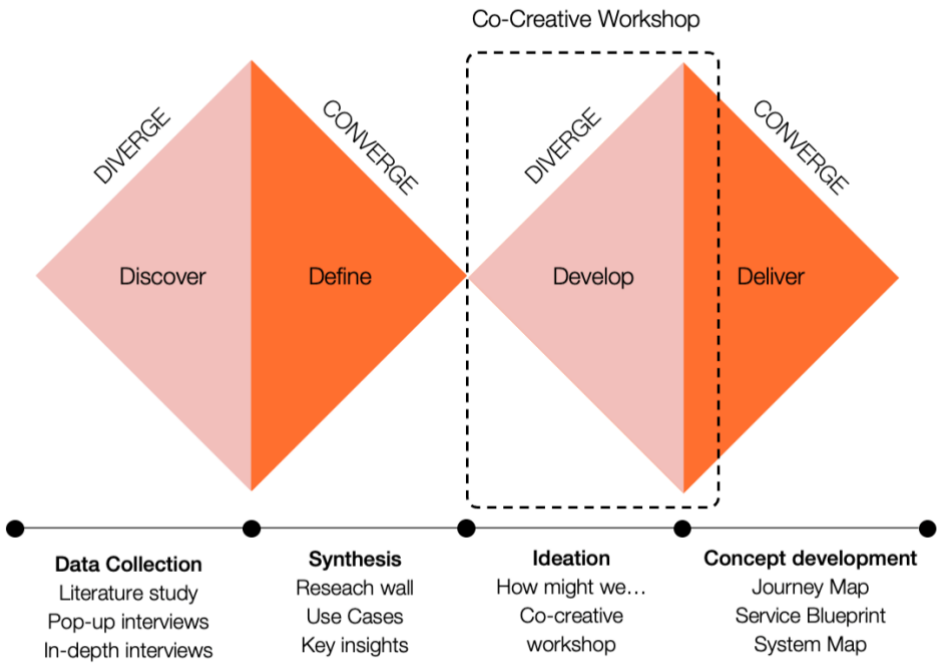


Figure 7. Double diamond process overview

The first diverging phase, the *discover* phase, was aimed at generating substantial knowledge about the problem through several different types of research methods. Before aiming to develop any solutions, the diverging nature of this phase allowed for challenging initial assumptions and understanding and empathising with the people indirectly and directly affected by these services (Stickdorn et al. 2018). The methods used in the first phase included a literature study to gain an understanding of the existing knowledge on micro-mobility, social practice theories, and sustainable mobility, as well as in-depth and contextual interviews with users and non-users of shared micro-mobility services and stakeholder interviews with experts in the field.

In the first converging phase, the *define* phase, the knowledge gained in the prior phase is organised and focused on extracting key insights (Stickdorn et al. 2018). This phase aims to define the right problem before ideation, which will improve the depth of the proposed solution and avoid only scratching the surface of a complex project. The data gathered in the prior phase was synthesised on a research wall, and key insights were identified. The key areas to focus on for ideation were determined based on the literature study on the environmental effects of SMMS.

The second diverging phase, the *develop* phase, started with drafting 'How Might We…' (HMW) statements from the key insights to transform problems into opportunities. It further defined sustainability objectives to guide the ideation based on the knowledge gained in the literature study. A co-creative workshop was held with experts in the field of SMMS, where they ideated on interventions corresponding to each HMW statement.

The second converging phase, the *deliver* phase, aimed to organise and prioritise the ideas from the ideation into a final service system concept. This was done by compiling and further developing the result from the co-creative workshop, pinpointing service attributes and mapping out the service.

3.3 Data Collection

The following section describes the first phase in the double diamond process, the discover phase. The methods used were pop-up contextual interviews, in-depth interviews and stakeholder interviews. Furthermore, sample size and demography are presented, along with descriptions of the methods used.

3.3.1 Pop-up contextual interviews

Contextual pop-up interviews were conducted to get an overview of the mobility landscape of Stockholm, as seen in Figure 8. These were brief, informal interviews with individuals encountered in locations where micro-mobility services were readily available.

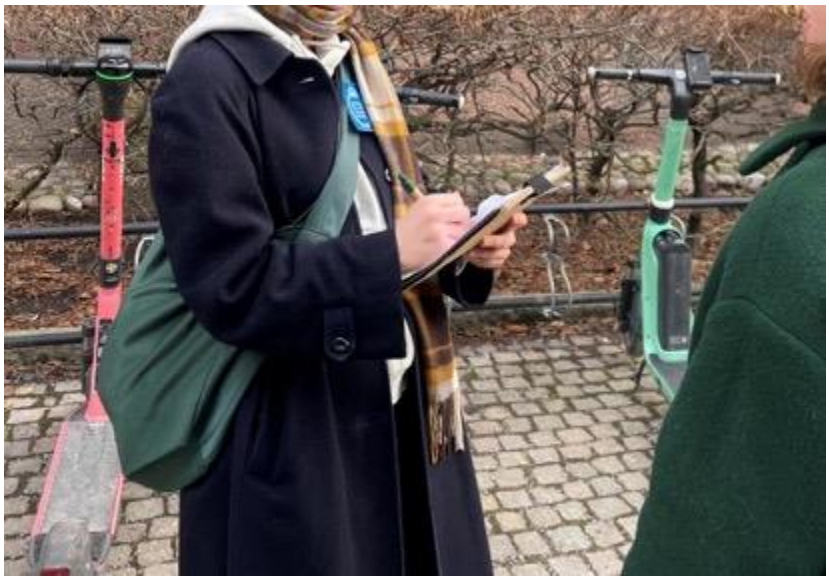


Figure 8. Pop-up interviews

The pop-up interviews were conducted to capture a broader demographic within the given timeframe and identify interesting areas to focus on for the following in-depth interviews. By approaching people contextually, i.e. while they are travelling within Stockholm, they are more likely to think of specific details linked to their experience and to give more tangible answers related to their surroundings. There is also the benefit of having the possibility to point and show specific elements (Stickdorn et al. 2018).

A total of 50 pop-up interviews were conducted on the streets of Stockholm, engaging with Stockholmers as they went about their everyday mobility practices. The interviews lasted five to ten minutes each and adhered to a predefined set of questions regarding mobility practices and opinions on shared micro-mobility in Stockholm. The complete questionnaire can be found in Appendix A. Interviewees were approached at various mobility sites such as train stations and parking lots. The sampling aimed to encompass a diverse range of respondents across various demographic factors. The final sampling is presented in Table 1, and the detailed version is given in Appendix B, Table 1.

Table 1. Sampling population, pop-up interviews

		Number	Share
Gender	Female	23	46%
	Male	27	54%
Age	20 or younger	9	18%
	21-30	13	26%
	31-40	7	14%
	41-50	10	20%
	51-64	9	18%
	65 or older	2	4%
Transport Mode	Commuter train or tram	9	18%
	Car	2	4%
	Bus	8	16%
	Metro	16	32%
	Bike	8	16%
	Walk	5	10%
	Personal e-scooter	1	2%
	Shared e-scooter	1	2%
N		50	

3.3.2 Stakeholder interviews

Four stakeholder interviews were conducted with individuals expected to provide deeper insights on the topic of mobility and shared micro-mobility services. As seen in Table 2, four interviews were conducted with various stakeholders, three of whom were researchers active in the field. One stakeholder interview was conducted with a SMMS entrepreneur who was part of the founding team of one of Stockholm's larger e-scooter service providers. The interviews were loosely structured around a set of customised questions for each stakeholder, aiming to maximise their unique perspectives. This approach allowed the interviews to adapt and extract the most relevant knowledge from each interviewee without predetermining discussion topics.

Table 2. Sampling population, stakeholder interviews

N	Stakeholder
S1	Transportation analyst, researcher in autonomous vehicles
S2	Researcher in micro-mobility and e-scooters
S3	Micro-mobility entrepreneur
S4	Region Stockholm
S5	Researcher in mobility services and sustainable futures

3.3.3 In-depth interviews

After assessing the pop-up interviews and the stakeholder interviews, the knowledge gained was used to identify areas that could be deeper explored in in-depth interviews. The in-depth interviews were semi-structured conversations with participants, both users and non-users of micro-mobility, as seen in Figure 9.

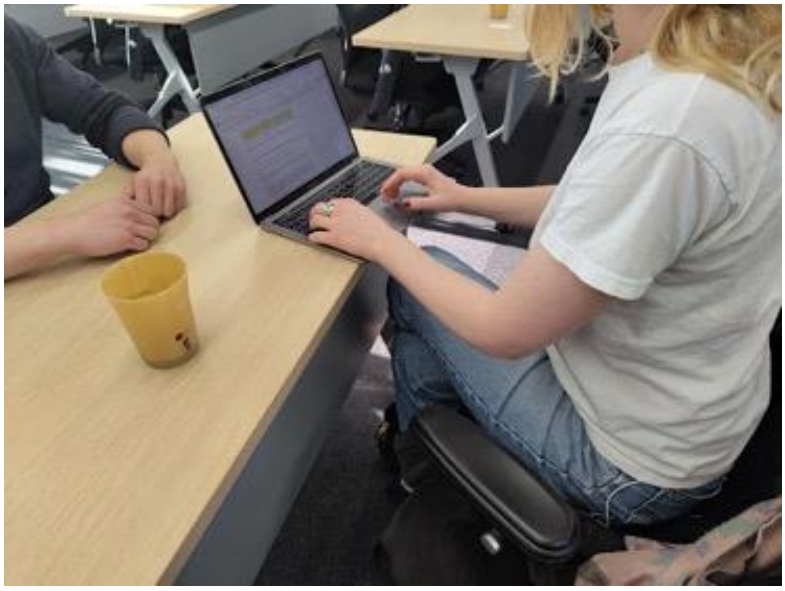


Figure 9. In-depth interviews

To ensure focused discussions and obtain the desired insights, an interview guide was prepared. However, having a semi-structured approach allowed the conversational flow to unfold naturally. Consequently, allowing the interviewee to express their perspective freely in their preferred manner. Most of the in-depth interviews were conducted in person, at a location chosen by the interviewee, and some were conducted remotely. To understand the interviewee's mobility practises, the interview guide was divided into five sections: Demography, General Mobility, Shared Micro-mobility services, Sustainability, and Dreams. See Appendix C for the complete interview guide.

The in-depth interviews aimed to understand Stockholmer’s mobility practices in terms of the social and material elements that go into the practice. To reach potential interviewees, snowball sampling was used. This method entails contacting a few initial participants with relevant backgrounds and requesting them to suggest additional contacts who may also be willing to

participate in the study (Stickdorn et al. 2018). This method was applied iteratively; when one snowball did not yield new contacts, another initial participant was contacted to continue the sampling process. A total of 16 people were interviewed, and a demography description of the final sample is presented in Table 3. A detailed description of the sample size is given in Appendix B, Table 2. Inner and outer city boundaries are defined by the congestion tax system employed by the Stockholm municipality (Transportstyrelsen n.d.). All interviews were recorded, and Microsoft Word aided the transcriptions. All names have been replaced.

Table 3. Description of demography of in-depth interview sample

		Number	Share
Gender	Female	7	44%
	Male	9	56%
Age	20 or younger	1	6%
	21-30	12	75%
	31-40	2	13%
	41-50	1	6%
	51-64	0	0%
	65 and older	0	0%
Residential area	Inner city	5	69%
	Outer city	11	31%
N		16	

3.4 Synthesis and Analysis

This section describes the methods of synthesis and analysis of the interview data, including research wall, use cases and key insights.

Following the initial phase of divergence, during which a large volume of data was collected, the material was analysed and focused using data visualisation methods. These methods aimed to empathise with users by structuring the data to uncover patterns and pinpoint areas where further research is necessary (Stickdorn et al. 2018). The interviews were analysed mainly by

employing research walls to find emerging themes. Based on the interview material, need groups were defined, and key insights were extracted.

3.4.1 Research wall

The synthesis of the raw material was primarily performed through research walls, as seen in Figure 10, enabling visualisation of the data and identification of emerging patterns. Since the goal was to understand user needs and experience gaps, an inductive clustering method was employed. This allowed categories and themes to emerge from the data rather than being driven by preconceived assumptions or hypotheses. The data from the three different types of interviews were visualised and analysed separately. However, since the in-depth interview questions were based on the outcomes of the pop-up and expert interviews, the result of the analysis of the in-depth interviews was used to extract key insights.

For the in-depth interviews, interesting participant responses were documented on post-it notes. Each post-it note was given an indexation of the interviewee number and line number from the transcription. After reviewing a set of two interviews, the post-it notes were organised on the research wall. Patterns and basic themes emerged inductively, and any post-it notes that did not fit existing themes were placed in separate categories to form new themes. As all in-depth interviews were coded, 32 basic themes were identified and sorted into five key insights.



Figure 10. Data visualisation through research wall

3.4.2 Use cases

To answer the first research question, "*What role do micro-mobility services play in people's everyday practices*?', use cases were developed based on interview data. These use cases illustrate how Stockholmers currently integrates SMMS into their mobility practices and highlight situations where the need for these services arises.

The use cases were developed to describe situations or needs that can present themselves among different types of users and are not based on a fixed personality type, lifestyle, or demography.

A need may change in different situations, meaning that one person can experience a need on one day but another need on a different day. On the contrary, lifestyle factors are more rigid, go deeper, and might change very slowly.

3.4.3 Key insights

The data and patterns identified in the interviews were rephrased as key insights to concisely summarise the findings and facilitate communication of the main problems discovered during the analysis. Key insights serve to precisely articulate the most valuable learnings from the research and act as a springboard for the subsequent ideation phase (IDEO 2015). Once defined, key insights represent the final convergence point, 'pinpointing' the problem before the following divergent phase begins.

Effective key insights are grounded in raw data, such as quotes, photos, or recordings. Their phrasing is crucial as it guides further research processes (Stickdorn et al. 2018). There are various ways to structure them, but a common approach focuses on three levels: customer segmentation or situation, the need or desired outcome, and the associated problem or friction. This three-level structure allows for ideation in diverse areas.

The five final key insights emerged from the 32 basic themes, and the insights were then prioritised based on their relevance to the research question and their potential to reveal areas for improvement. Not all basic themes could be effectively integrated within the key insights, but those deemed most valuable and reflective of the design challenge and sustainability objectives were prioritised. Additionally, each key insight was paired with user needs and feelings related to the problem statement.

3.5 Ideation

This section describes the ideation methods used, including phrasing 'How-might-we...' statements and hosting a co-creative workshop with experts in fields relevant to SMMS.
The third project phase focused on identifying SMMS attributes that could promote more sustainable everyday mobility practices. This phase was initiated with a pre-ideation method called "How Might We..." to transform problems into opportunities for innovation. These HMW statements then guided a co-creative workshop to ideate mobility interventions that could influence or change existing mobility practices.

3.5.1 How might we…

The key research insights were rephrased into three actionable 'How might we' (HMW) statements to focus research on key areas for improvement. HMW is a pre-ideation method that transforms challenges and issues into design opportunities by formulating trigger questions based on insights and user stories (Stickdorn et al. 2018). Ideally, HMW statements are broad enough to stimulate innovative thinking while avoiding limitations to specific solutions (IDEO 2015).

Each HMW statement targeted a specific area of potential improvement identified in the literature study, which will now be referred to as a 'sustainability objective.' To clarify the underlying issue behind each HMW statement, it was further elaborated with a discussion of its implications and potential strategies for achieving the defined sustainability objectives.

3.5.2 Co-creative workshop

A co-creative workshop is a collaborative session that brings together individuals with different expertise to ideate and facilitate collaboration and creativity involving multiple perspectives. If executed properly, this will foster innovative solutions that might not emerge through individual efforts. The people invited to the workshop and their expertise will determine its outcome. Thus, it is essential to select participants carefully (Stickdorn et al. 2018).

The purpose of the co-creative workshop was to collect relevant stakeholders to discuss how SMMS can help Stockholmers adopt more sustainable everyday mobility practices. The workshop aimed to ideate possible mobility interventions to influence or change existing mobility practices. The suggested mobility interventions will bridge the gap between the environmental concerns of SMMS identified in the literature study and the identified sustainability objectives, as illustrated in Figure 11.

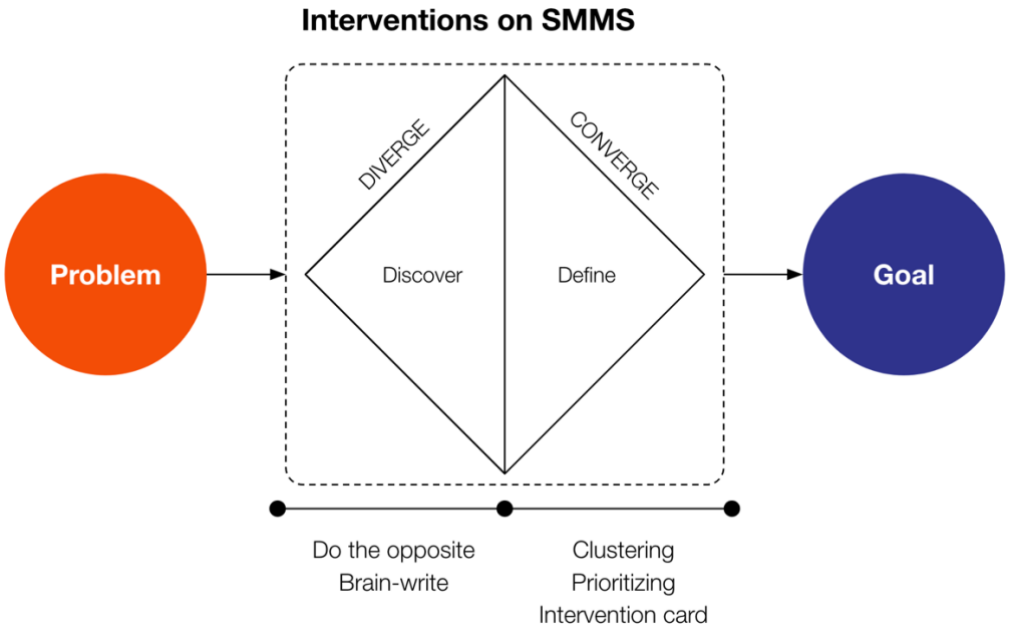


Figure 11. Workshop process

Eight researchers with expertise relevant to SMMS, such as sustainability assessments and sustainable transport and whose research areas are detailed in Table 4, were invited to participate in the workshop. The workshop lasted 2.5 hours and took place at the University of Tokyo. The full workshop outline is presented in Appendix D. During the workshop, templates for each exercise were handed out, and all templates are attached in Appendix E.

The workshop began with a brief lecture on SMMS and the findings from the data collection. This introductory session aimed to provide context, establish a shared understanding of the problem, and cultivate empathy toward the users. After the lecture, the three HMW statements were presented, and the participants were divided into two groups of three and one group of two. These groups were pre-assigned to ensure a diverse mix of knowledge and demographics within each team. Each group was randomly assigned one of the three HMW statements for focused ideation.

A warm-up exercise called ‘do the opposite’ or reverse brainstorming was conducted to kickstart the ideation process. Participants individually brainstormed the worst possible solutions to their assigned HMW statement. This approach capitalises on the notion that identifying problems can sometimes be easier than finding solutions. Additionally, it provided a broader base for identifying potential improvements and transforming ‘bad ideas’ into innovative solutions.

The warm-up exercise was followed by a brainwriting session, where each participant individually and in silence wrote down many ideas corresponding to their HMW statement. Brainwriting was chosen instead of brainstorming since it allows for more reflection, facilitates when ideas are more complex, and empowers less extroverted participants (Stickdorn et al. 2018).

The following exercise was clustering and combining ideas, where the participants created suggestions based on the ideas from the brainwriting session. To converge the number of ideas into one suggestion, the participants were asked to prioritise the suggestions based on two parameters, usability and feasibility, allowing the experts to use their knowledge to assess the suggestions.

Finally, as the last exercise, the participants compiled the intervention that was most feasible and usable on an intervention-card template, describing the idea, the user and their needs, and relevant stakeholders. Figure 12 shows a picture taken from the workshop.

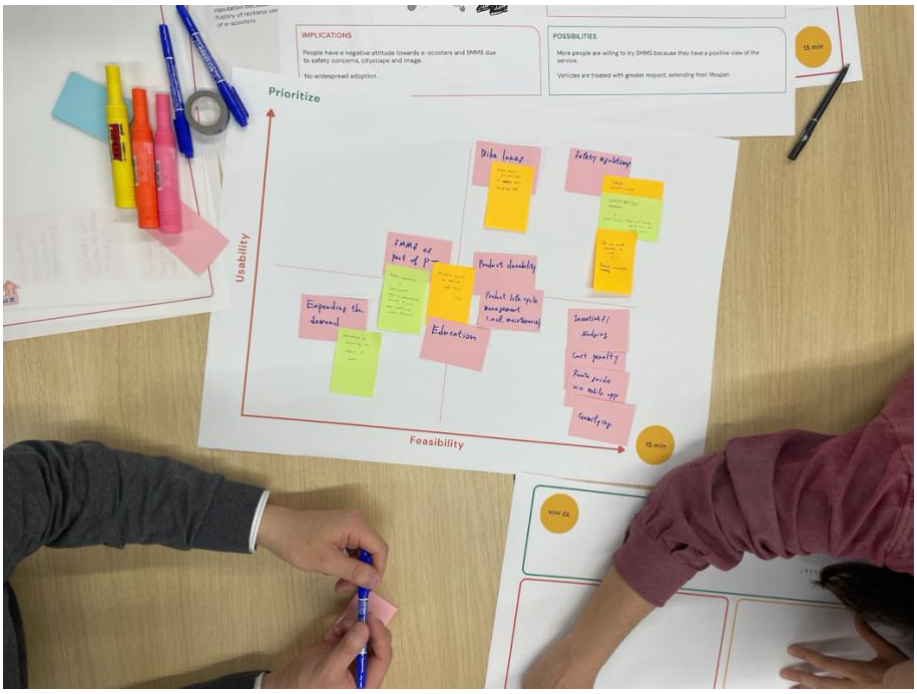


Figure 12. Co-creative workshop

Table 4. Workshop participants, title and research field

N	Research field

E1	Researcher, Sustainable production and consumption
E2	Researcher Life Cycle Engineering and Design Engineering
E3	Researcher, Artificial Intelligence
E4	PhD Student, Circular Economy
E5	Researcher, Sustainability Assessment
E6	PhD Student, Sustainable Transport
E7	PhD Student, Sustainable Transport
E8	Researcher, Sustainability Assessment

3.6 Concept Development

This section presents the methods used to refine the workshop results and develop the service system concept. Building upon the co-creation workshop's output, key service attributes that would facilitate more sustainable everyday mobility practices were identified. These attributes were then integrated to create a cohesive service system concept.

3.6.1 Customer journey map

A customer journey map was developed to visualise users' primary steps when engaging with the proposed service system concept. This map illustrates the intended experiences of users of the SMMS service over time. By making intangible experiences tangible, the customer journey map fosters a shared understanding among team members, acting as a boundary object. Although it cannot capture the full complexity of a service, it simplifies the experience, making customer interactions with and around the service easier to understand (Stickdorn et al. 2018).

3.6.2 Service blueprint

A service blueprint was created to map the user experience alongside the various components that orchestrate it, including physical evidence, customer actions, service provider actions (both frontstage and backstage), and support processes, serving as an extension to the journey map

Physical evidence includes tangible and intangible components the user encounters when using the service, such as physical artefacts and digital elements. Customer actions refer to the steps the customer takes while engaging with the service. Frontstage and backstage actions are carried out by the service provider. Frontstage actions involve people and processes with which the user directly interacts, while backstage actions involve processes hidden from the user's view. Finally, support processes are activities performed by external partners (Stickdorn et al. 2018).

3.6.3 System map

A system map was developed to comprehensively illustrate the components of the service system concept. This map encompassed three key levels: service, organisational, and institutional.

A system map is a visual representation that depicts the main parts of a system within which an organisation operates. It can encompass various components across different levels, including people, stakeholders, processes, and institutional arrangements. As highlighted by Stickdorn et al. (2018), system maps are particularly effective in visualising complex systems and making them more understandable.

4. Interview Insights

This chapter presents the insight from all interviews, including pop-up interviews, stakeholder interviews and in-depth interviews, along with use cases identified through the interviews. Finally, the chapter ends with a description of the key insights derived from the interview data.

The qualitative insights presented in this chapter provides an understanding of the role of micro-mobility services within the existing mobility practices of Stockholm residents, thereby providing a response to research question one, ‘what role do micro-mobility services play in Stockholmer's mobility practices?’.

4.1 Pop-ups Insights

This section presents the insights gathered from the pop-up interviews. The responses to each question were analysed thematically, and the findings are presented below, categorised by the specific questions asked to participants.

4.1.1 Current journey

The first pop-up interview question aimed to capture participants' top-of-mind thoughts and feelings by asking about the strengths and weaknesses of their current travel mode. The insights are presented in Table 5.

Table 5. Strengths and weaknesses of the participants’ current journey

Mode	Strengths	Weaknesses
Public Transport	Frequent, fast, convenient, reliable (when on time), extensive network, affordable (with monthly pass).	Delays, crowdedness (especially during peak hours), dependence on schedules.
Cycling	Fast (especially compared to buses in some cases), healthy, allows freedom and control over travel time, convenient for short distances, good for summer weather	Unpleasant or impractical in winter (weather dependence), lack of secure parking (theft concerns), pedestrians not respecting bike lanes.
Car	Point-to-point travel without stops.	Traffic congestion, expensive (parking and gas), less environmentally friendly.
E-scooters	Fast, fun, and convenient (especially when readily available).	Limited usability in winter, safety concerns, not suitable for carrying large items.

4.1.2 Trip planning

The second question aimed to understand how much planning participants did to perform their trips and how they went about planning their journeys. For public transport users, three themes emerged:

- App Users: Some public transport users rely on apps to plan their journeys, especially when they need to arrive at a specific time.
- Frequency and Routine: Some public transport users feel planning is unnecessary due to frequent departures or established routines.
- Contingency Planning: Some public transport users plan occasionally to check for disruptions or unfamiliar routes.

These insights indicate a link between familiarity and planning where familiar routes rarely requires planning. However, the need for planning can still arise when there is a need to arrive with precise timing. This planning is used for reassurance of arriving on time.

For other modes, the respondents rarely expressed the need to plan their journey. Respondents who travelled by e-scooter and cyclists generally travelled spontaneously, with some planning for unfamiliar routes or checking for bike lanes.

4.1.3 Competencies and skills

To assess the competencies required for their chosen travel mode, respondents were asked to identify the knowledge or assets necessary to undertake such a journey. For public transport users, these competencies were highlighted:

- Obtaining a valid ticket, either through an app or through a physical ticket vendor.
- Navigational knowledge and understanding of the public transport network, including stations, and routes.

While not explicitly mentioned as a requirement, understanding the local language might be advantageous for navigating stations. Furthermore, the participants generally considered public transport to be accessible to everyone.

Respondents travelling by bike or e-scooter mentioned similar competencies. These include:

- The ability to operate the chosen mode (bike, e-bike, e-scooter).
- Knowledge about and ability to follow traffic rules.
- Knowledge of safe routes and bike lanes as well as awareness of surrounding traffic.
- Knowledge in using a navigation app for unfamiliar routes.
- User interface and mobile app knowledge for platform-enabled shared e-scooters and e-bikes.

Several cyclists emphasised the value of having experience when riding in Stockholm's traffic. This experience helps navigate unexpected situations and identify good bike lanes.

4.1.4 Norms and travel behaviour

To understand cultural conventions and socially shared meanings associated with travel in Stockholm, participants were asked about social norms and proper etiquette for their chosen

mode of transport. Public transport users emphasised public decency and respect for others as core values. These values, as elaborated by participants, translate into specific behaviours, such as:

- Showing courtesy and offering seats to those who might need it.
- Avoid making loud noises and refrain from talking on the phone or speaker.
- Avoid taking up unnecessary extra space and respect others' personal space.
- Refraining from eating food that smells.
- Not resting one's feet on the seat.
- Being mindful of personal hygiene.
- Paying attention to one's surroundings.

One participant equated public decency with avoiding excessive comfort, emphasising that public transport isn't an extension of one's living room. In other words, behaviours acceptable at home might not be appropriate when sharing a limited space with others on public transport.

For cyclists and e-scooter riders, adhering to traffic rules emerged as the most frequently mentioned aspect of proper conduct. Respect for others, particularly pedestrians and maintaining a safe distance from fellow cyclists, were also highlighted as key considerations.

However, one outlier who travelled by bike testified that rules came secondary to getting to their destination most efficiently and that being result-oriented meant for example not always stopping at a red light. This participant referred to this as a Stockholm mentality, suggesting that there might be different subcultures with different priorities. Furthermore, one e-scooter rider expressed confusion on what rules apply for e-scooters when navigating in traffic, suggesting that the rules are not addressed or that some people disregard educating themselves on what rules apply.

4.1.5 SMMS perception

To gain insight into the participants' perceptions of shared e-scooters and e-bikes, respondents were asked about their general thoughts on these services. This question aimed to understand expectations and socially shared meanings in conjunction to using or avoiding using micro-mobility services. While negative views on e-scooters predominated among respondents, some positive aspects were also identified. These positive aspects included:

- Convenient for short trips.
- Flexibility by on-demand travel.
- Convenient for summer nights when public transport is less frequent.
- Having fun and feelings of freedom.

Conversely, negative views and concerns frequently stemmed from a perception of typical riders engaging in reckless behaviour. Safety emerged as the primary concern, with participants expressing a range of negative views, including:

- Concerns about inexperienced riders.
- Low helmet use.
- Drunk driving associated with e-scooter use.
- Inconsiderate riding and parking near pedestrians, cyclists, and car traffic.

Two participants even associated e-scooters with criminality, further enhancing the negative view of the rider base. Furthermore, many respondents dislike how e-scooters are left scattered and parked poorly, which makes for an ugly addition to the cityscape. Some respondents expressed frustration with shared e-scooters occupying bicycle parking spaces, particularly those designated with secure bike lock facilities. Although municipal regulations permit parking shared e-scooters in communal bicycle spaces, some personal bike riders perceive this as problematic due to competition for limited parking. However, many respondents viewed the introduction of parking regulations positively as they think that it has resulted in reduced clutter of e-scooters. Lastly, the price of SMMS made some respondents reluctant to use SMMS as they thought that it was too expensive compared to alternatives.

These negative perceptions were mostly directed to shared e-scooter services, when extending the question to include e-bikes, many respondents view e-bikes and bikes as a safer, more comfortable, and more approachable alternative. Some respondents did not feel comfortable with the e-scooter as a vehicle and expressed that they would not personally use the e-scooter services due to this. They perceive e-scooters as dangerous vehicles and refer to reported accidents and the feeling of a flimsy construction.

For respondents under the age of 20, an additional question explored perceptions of e-scooters as cool or aesthetically pleasing. While some expressed concerns about looking awkward while riding, suggesting e-scooters might be less attractive or confidence-boosting compared to other options, these younger participants also noted a shift in perception over time. Initially seen as cool, shared e-scooters are now often viewed as more practical than trendy. This suggests the initial novelty of e-scooters may diminish as they become a regular transportation option.

Despite acknowledging the convenience and usefulness of e-scooters, especially for nighttime or short trips, younger respondents also highlighted safety concerns and cost as drawbacks. Finally, gender stereotypes emerged, with shared e-scooter use perceived as more common for boys.

4.2 Stakeholder Interview Insights

This section introduces the insights gathered through the stakeholder interviews, revealing different perspectives on the role of SMMS in Stockholm.

The need for access points near public transport stations to facilitate first-and-last-mile connectivity emerged as a common theme. As the transport analyst pointed out, most SMMS trips begin or end at public transport stations, suggesting SMMS are used for first-and-last-mile connections but not necessarily for truly multimodal journeys.

Another perspective highlighted by the e-scooter researcher was how SMMS can complement public transport, not compete with it. If users know they have the option to use SMMS when needed, such as for time-sensitive trips, SMMS can increase user confidence in public transport and address its shortcomings. Furthermore, shared mobility services were also highlighted as a strategy to delay car ownership. Access to shared mobility can address occasional car-dependent situations, potentially deterring individuals from car purchases, not addressing the modal shift from car to SMMS but using SMMS to circumvent future car purchases. Station-based services were seen as more suitable for commuting, while free-floating systems were viewed as better for complementing public transport on unexpected trips.

An employee at Region Stockholm highlighted challenges with providing SMMS regionally. Since most SMMS require some physical ability, they may be inaccessible to some people, making public funding a concern. Interviews also commonly identified a greater need for SMMS outside the city centre but highlighted it as a more challenging business model due to higher operational costs.

Finally, the Region Stockholm employee emphasised the need to mitigate car usage through other complementary services, such as conveniently located post boxes. This could encourage residents outside the city centre to use SMMS for everyday errands like picking up mail instead of driving. Another key complementary service would be providing phone charging opportunities for platform-enabled services.

The micro-mobility entrepreneur cited several factors driving the decision to offer e-scooters instead of bikes as a transportation option, including a lower capital investment and lower manufacturing costs.

The entrepreneur promoted e-scooters as a more sustainable option, highlighting their ability to reduce car traffic. This connects to the issue of space since e-scooters require significantly less space than cars. Consequently, they can contribute to creating cities with more space for people, enhancing quality of life, and reducing car dominance.

According to the micro-mobility entrepreneur, one of the key elements of a functioning SMMS is high density of vehicles, and for this to work there also should be frequent parking spaces, to make it easier to park correctly. They also stated that public relations are an important aspect of a successful SMMS, meaning having strong relations with individuals in charge at the municipal level.

When addressing the reckless behaviour and safety risks connected to e-scooters, the micro-mobility entrepreneur highlighted the potential of integrated smart processes in the e-scooter. These could include cameras detecting sidewalk riding, sensors identifying multiple riders, and systems assessing rider behaviour through handling data. They also highlighted that one measure that had been taken to tackle reckless behaviour is simply making the vehicles heavier, since a heavier or bulkier vehicle is physically harder to throw away.

4.3 In-depth Interview Insights

This section presents insights from the in-depth interviews, divided into five themes: SMMS offers increased freedom at the expense of responsibility, e-scooter services are used spontaneously, the value of time and personal space, convenience and car dependency, and finally skill gaps and accessibility challenges.

Table 6 presents the five themes and its subcategories, sorted into the ‘stuff-image-skill’ model along with supporting quotes. Additionally, each theme is described in detail in the following subsections.

Table 6. In-depth interview insights on mobility practices along with sub-categories sorted into the 'stuff-image-skill' model

SMMS offers increased freedom at the expense of responsibility	
Stuff	SMMS offers reduced need for deposit and safekeeping of vehicles Not having to pedal disconnects from the feeling of being control of speed SMMS vehicles cluttering the cityscape
Skill	SMMS requires less skill for repairs SMMS offers less immediate vehicle access
Image	Not being bound to a transport network entails a feeling of freedom Less responsibility entails more reckless behaviour E-scooters feel toy-like SMMS vehicles does not provide enough intangible value
Quote	*'We would race each other and it was really cool and fun as hell. But then I grew up and moved on from that. There was no thought of safety. Sometimes we would ride three people on one e-scooter. We would try to beat 20 km/h, the maximum speed'*
E-scooter services are used spontaneously	
Stuff	SMMS offers vehicles faster than public transport Pre-paid public transport tickets deter from use of SMMS
Skill	Knowing how to navigate quicky through the city
Image	Public transport feels unsafe at night It is not frowned upon to use SMMS under the influence of alcohol Spontaneous use entails reckless behaviour SMMS helmet use is seen as transgressive
Quote	*'I realised that there was no subway running because it was late and I had to or I just want to go home now. Okay, what are my options? I can take a scooter [...] I really only take a scooter in emergencies.'*
Value of time and personal space	
Stuff	Distractions in public transport makes it hard to be productive Public transport feels unsanitary and crowded Active mobility commuting requires access to showers and/or informal dress code
Skill	Ability to relax or focus in hectic environments
Image	Commuting feels like a chore and a waste of time Using active mobility equals being productive Productive ideal in big cities, laziness is frowned upon
Quote	*'I strive for efficiency in my everyday life, and transportation is rarely efficient in my opinion. I'm not particularly good at making it efficient either. I could read a book or study, but I dislike spending time just transporting myself'*

Convenience and car dependency	
Stuff	Owning a car for special occasion trips makes people use it often Extra steps and stops deter from the feeling of a seamless journey
Skill	Minimising transfer and transport time
Image	Stockholm is a public transport city Cars is a part of the image of a happy family Need to be in control over vehicle and feel forward movement Seamless experience important for multi-modal journeys
Quote	*'My partner owns a car and never takes public transportation. So you get used to the advantages that come with that. And then you might walk less or distances seem much bigger in your head. You become very lazy when you have the option'*

Skill gaps and accessibility challenges	
Stuff	Users do not want to pay to learn using SMMS vehicles Acquiring a privately owned SMMS vehicle for commuting requires a big capital investment
Skill	Many people lack confidence in traffic Need to know the vehicle before trying SMMS, and less people know e-scooters from childhood Trying SMMS for the first time requires prior knowledge or confidence Need to learn managing the SMM services, and the multitude of providers makes it more difficult Acquiring a privately owned SMMS for commuting requires knowledge
Image	SMMS is seen as elitist since it is perceived as an unnecessary expense
Quote	*'In Stockholm, you have to take up a bit more space in traffic, I think. You also have to be aware of so many things, there's always something happening everywhere. Many roads are one-way and there's a lot going on, so you need to take it easy but you also have to be assertive. If you're not assertive in Stockholm, you won't get anywhere'*

4.3.1 SMMS offers increased freedom at the expense of responsibility

Several interviewees highlighted the benefits of using SMMS, with freedom being a key advantage. However, the interviewees also revealed drawbacks associated with this freedom, including reckless user behaviour and a perceived loss of intangible value.

When using an SMMS vehicle, the user does not need to care for the vehicle when it is not in use. Thus, the user can take the SMMS vehicle from point A to B, but not necessarily from point B to C meaning that there is no responsibility for the deposit of the vehicle since it does not need to be returned to a specific location. Consequently, the user can take the SMMS vehicle to their workplace, without the responsibility of needing to take it back home again. This implies freedom to do something spontaneous after work, maybe in companionship with others using other transport modes, without having to bring the vehicle or having to worry about its safekeeping. One of the interviewees explains it as the following:

> ‘*But it was still the freedom, not having to think: 'Where do I put my bike now?' 'I have to take it home'. Because that's what's preventing me a bit now from taking the bike, or I think it will prevent me when I go to the office, it's twice as fast to cycle for me here, but I still haven't cycled yet, because I don't know if I'm going to go somewhere else later. But I think that maybe eventually I'll park my bike here and then take the bus the next day if I'm going to do something.*’ *(Elsa)*

SMMS offer users freedom from both the vehicle itself and their subsequent transportation choices. For example, a user could take an SMMS for a trip from point A to B, then switch to public transport for B to C. This flexibility eliminates the need to adjust plans based on their current mode of travel. There's no "making up" for leaving the vehicle somewhere, as they don't need to worry about returning it the next day.

Furthermore, using SMMS entails extra freedom for the users in the sense that the increased flexibility widens their range of options for temporal and geographical distribution. This improves their sense of being in control, both regarding when to move about, but also in a more literal sense as in controlling the movement of the vehicle:

> *‘You trust yourself in a different way when you drive your own vehicle. You know that you will be able to get around or change roads, but it feels really hopeless to sit on a commuter train and be stuck... I guess flexibility is the biggest advantage of having your own vehicle. And the biggest source of irritation is being bound to the traffic within a vehicle network.’ (Viktor)*

The decoupling of responsibility from the use of micro-mobility vehicles such as bikes had many other advantages to the users. When not owning the vehicle themselves, they did not need to worry about the risk of theft. Some interviewees highlighted the material needs of keeping a vehicle safe, there is a need for a lot of space, preferably a private space or a space shared among others, and investments in a good locking system. These are materials that people don't feel equipped with, or do not feel like they want to invest in. Besides the material needs the user is spared the mental effort of worrying. Furthermore, the interviewees highlighted other positive aspects such as not needing to learn the skills to care for the vehicles such as knowledge about bike repairs.

> *‘However, they do maintain the bikes, which I think is necessary if you're going to be using them in rough conditions like gravel. I'm not sure if winter tires are included, but I think they are. It's nice not having to maintain the bike. I don't know much about bikes, so I would need to learn if I owned one. I don't have the energy for that right now.’* (*Daniella)*

A drawback of SMMS is the initial effort required to access them, involving finding an available vehicle, reaching it, and unlocking it through the app. This mitigates the increased sense of freedom and flexibility and proposes a tension between the increased flexibility of renting a vehicle and having immediate access and thus extra flexibility by owning your vehicle.

However, there is a backside to the improved sense of freedom. Because of the lack of responsibility when the rental time is over, people tend to use the SMMS more recklessly compared to their possessions. The shared e-scooters are conceived as especially appealing to young users, often men, who seek freedom and enjoy pushing boundaries. This is likely due to the e-scooter feeling like a toy-like vehicle, as well as the service unlocking new opportunities to socialise with others, described as:

'A few years ago, when I was younger, everyone thought e-scooters were really cool. It was like having a driver's licence, and it gave you a sense of freedom to be able to ride around on one. I think it was because they were new, and I was younger. It was like, 'Let's just take e-scooters and ride around. Or let's see how fast we can go.' We would race each other and it was really cool and fun as hell. But then I grew up and moved on from that. There was no thought of safety. Sometimes we would ride three people on one e-scooter. We would try to beat 20 km/h, the maximum speed.' (Olle)

Other service characteristics, such as the lack of physical effort required to operate them, were linked to the perceived recklessness of e-scooters. Not having to pedal the vehicle decouples the user from their understanding of pace and might exhort reckless use, unlike bicycles where pedalling creates a physical connection to the speed of the vehicle.

Furthermore, the e-scooters and previously available e-bikes did not contribute with added intangible values to the users. The vehicles were described as dirty and worn and having a lack of aesthetic appeal. When riding SMMS vehicles some people did not feel sporty, or cool, and said that the branded aesthetic of many vehicles makes you feel like you are promoting the company by using it in public. This is opposed to owning a vehicle, where the user can identify with the vehicle's aesthetic and form a type of relationship with the vehicle itself. One interviewee expressed a desire to feel like the vehicle was their own, even just for a little while:

'The Stockholm e-bikes also had more advertising on them. The colour doesn't really matter, but it's the same thing as having a car with an ad on it. You want to be a little incognito in a way, and maybe feel like it's your vehicle right there and then.' (Viktor)

Additionally, the history of SMMS vehicles lying around in excess, cluttering the city and hindering access on sidewalks, contributes to negative perceptions of the service among some interviewees. These factors have fostered a general sense of distrust towards SMMS and discouraged their use. People are concerned that these services detract from the city's aesthetics and feel apprehensive about safety issues surrounding e-scooters. Two examples of SMMS vehicles cluttering the cityscape are displayed in Figure 13.



Figure 13. E-scooters cluttering the cityscape, to the left: designated parking space, to the right: bike parking

4.3.2 E-scooter services are used spontaneously

The in-depth interviews strongly suggest that the current design of e-scooter services encourages spontaneous use. Riders primarily utilise them during time-pressured situations, late nights, and to avoid crowds. Furthermore, late-night riding coincided with reports of reckless user behaviour.

Several interviewees indicated that they primarily use shared e-scooter services as a backup option when their usual mode of transportation is unavailable or not fast enough, such as when running late. When the user realises that they will not make it to their destination on time, a quick and spontaneous decision can be made if an e-scooter is readily available at that time. This indicates that e-scooter services can provide a fast and on-demand solution. One participant who frequently uses e-scooter services stated:

> *'When I go to school, I either walk or take public transportation. If I'm in a hurry and I'm late, I take a [shared e-scooter]. Because it takes 8 minutes by [e-scooter] to school and it takes over 20 minutes by public transportation or walking. So when I'm late, I use [e-scooter services] a lot. A lot.' (Vera)*

Another instance where the interviewees are using e-scooters is unplanned trips during nighttime. Long wait times due to less frequent public transportation and perceived unsafe environments in the nighttime can make riders feel anxious and frustrated. This can incentivise users to use a micro-mobility service to ensure a quick, convenient, and safe journey home. The decision to make these trips is often momentary, as one interviewee described:

> *'I realised that there was no subway running because it was late and I had to or I just want to go home now. Okay, what are my options? I can take a scooter [...] I really only take a scooter in emergencies.' (Elsa)*

Some interviewees testify to taking these nighttime trips under the influence of alcohol, as this often goes together with being out in the city at late hours when public transport is less frequent or unavailable. Amongst some Stockholmers there is a perception that drunk driving on e-scooters is not viewed as something that serious, one interview stated:

> *'It happens [that I ride e-scooters drunk]. But not when I'm so drunk that I know it won't work, and you can't justify it, because it's not something you should do. ... But if you've had a few glasses and you're in a good mood, then definitely, a summer night in the middle of the night when you know there's no one out.*
>
> *There are no buses and it is cheaper to take an e-scooter than an Uber.*
>
> *I don't think people take it too seriously, to be honest. I know a lot of people who do it and most of the people know that they probably shouldn't do it, but it's kind of like riding a bike drunk like it's not that serious.' (Vera)*

E-scooter riders under the influence of alcohol pose a safety risk to themselves and others. Alcohol impairs judgement and increases impulsivity, making riders more likely to take unnecessary risks and disregard safety regulations. Some e-scooter providers have integrated responsiveness tests in the app to prevent drunk driving, however, one interviewee testified that it is possible to circumvent these measures. Several interviewees mentioned witnessing carelessly parked e-scooters, particularly on weekend mornings. This reckless behavior,

potentially linked to alcohol use, aligns with concerns about unsafe riding practices. Figure 14 illustrates an example observed during the study.



Figure 14. E-scooter lying beside a bike lane

Interviewees identified cost as a factor contributing to the spontaneous nature of e-scooter trips. Many users with existing monthly or annual passes for public transport (e.g., SL-pass) or previous Stockholm e-bike subscriptions perceive those pre-paid services as 'free'. Once these individuals have made a financial commitment, they are less likely to consider using an additional micro-mobility service unless there's a significant perceived benefit. The interviews suggest that for these users, the extra cost needs to be outweighed by a clear advantage, such as avoiding being late or getting home safely at night. This is illustrated by one interviewee's internal monologue:

> *'I would have used [e-scooters] much more often if they were cheaper. Like, Should I pay 30 SEK or be 5 minutes late?'(Lily)*

Conversely, ownership of a vehicle is typically driven by a premeditated need, resulting in more deliberate and routinised usage. This then ties into the financial commitment, in this case put into the personal vehicle, that can discourage people from paying extra for SMMS.

Additionally, the spontaneous nature of many e-scooter trips contributes to the low helmet use rate. E-scooter riders often do not wear helmets because it contradicts the unplanned nature of their trips. Since these trips are unplanned, carrying a helmet "just in case' becomes impractical. Additionally, the low prevalence of helmet use among other riders creates a social norm where wearing a helmet feels strange. This is highlighted by one interviewee:

'I think the reason I don't wear a helmet is because [e-scooter] rides are so spontaneous. Like, I might not have planned to take a Voi at 12 at night. So, I don't bring a helmet with me just in case I might ride a [e-scooters]. And then, once I'm riding, I don't really think about the fact that I should probably be wearing a helmet. It also feels a bit silly, somehow. Because I don't think anyone else wears a helmet, it's kind of the norm.' (Ivar)

Due to the lack of trip planning associated with e-scooter use, these insights suggest difficulty in integrating them into established mobility routines. Consequently, e-scooters become a separate category for specific situations rather than a seamless part of everyday mobility.

4.3.3 The value of time and personal space

The in-depth interviews revealed that daily commutes are often seen as time-consuming, highlighting the difficulty of utilising commute time for activities perceived as valuable. Ultimately, the interviews suggested that a long and unfulfilling commute needs to be balanced by a destination or goal perceived as worthwhile.

Discussions regarding everyday mobility frequently highlighted the significance of time. Interviewees consistently emphasised the need to save time, indicating that one of the most critical factors in choosing a mode of transport was its ability to minimise travel time. Daily commutes perceived as time-consuming were often described as a waste of valuable time, hindering participation in more fulfilling activities.

The interviewees described daily life as a series of trade-offs between essential tasks like education, work, and housekeeping, and more fulfilling activities such as spending time with friends, exercise, and rest. Limited time necessitates effective management to prioritise activities that contribute to self-fulfilment rather than simply completing daily chores. Since many perceive daily commutes as a chore, one strategy involves time optimisation. For example, catching up on work emails can help reclaim free time for activities individuals genuinely enjoy. As one interviewee illustrates, her daily commute takes away valuable time from activities she considers more meaningful in her day-to-day life:

'I strive for efficiency in my everyday life, and transportation is rarely efficient in my opinion. I'm not particularly good at making it efficient either. I could read a book or study, but I dislike spending time just transporting myself. It bothers me because I think I could use those three hours to exercise or be with my dog.' (Stina)

Several interviewees identified potential for productivity within their perceived wasted commute time. However, they highlighted the challenges of utilising this time effectively due to the numerous distractions encountered during daily commutes, particularly on public transport. These distractions contribute to a general feeling of unease, hindering opportunities for both focused work and relaxation. On public transport, various factors disrupt focus, frequent stops create a constant flow of people and noise, while the ever-changing environment is perceived as distractive:

'Commuting by train and doing something else at the same time has never worked for me. Not even reading. There is too much noise and too many stops. It doesn't get quiet and peaceful in that environment. You get interrupted all the

time. Audiobooks don't work for me either, there are too many distractions. So the only thing you can really do on the train is use your phone.' (Viktor)

Another factor that was highlighted as a contributor to feelings of unease was the physical closeness to other people due to crowding. People feel like they are in the way, or like other people are obstructing them. Furthermore, some people refrain from using the metro and public transportation because they perceive it as unsanitary. They believe that the enclosed space of the metro is filled with bacteria and viruses, and they are uncomfortable being in close proximity to other people who may be sick.

Despite the acknowledged challenges, the in-depth interviews revealed instances where participants felt they could utilise commute time productively. One strategy highlighted was active mobility, such as cycling or walking to work. This approach offers an opportunity to maximise productivity by checking off two tasks – commuting and exercise – simultaneously. While laziness is described as an unappealing image, exercise is generally viewed as desirable and the antithesis of laziness:

'I think [bike commuters] are aware, they are taking care of themselves, while also taking care of the planet. So I see it very positively, really, and I wish I was like that.' (Stina)

Consequentially, active mobility can turn a necessity into something beneficial. For those with longer commutes, using a non-electric vehicle can allow for a thorough workout during travel. However, this practice presents a need for both material and meanings not available to all users. Access to workplace showers and the availability of sufficient quality vehicles are not universal. Additionally, some workplaces may have dress codes or hygiene standards that make arriving sweaty impractical. Participants specifically mentioned workplace dress codes as a significant downside to active mobility, hindering their ability to maintain a desired professional appearance upon arrival.

'However, you still need to look good as a consultant, you know, not show up with a sweaty face and your makeup all melted. And I can imagine that there are many situations where you might not be able to cycle for such reasons.' (Daniella)

Some interviewees expressed that they managed to designate their commuting time to relaxing activities such as reading or listening to audiobooks, however, the accumulated amount of time designated for these activities could also be perceived as too long:

'Listening to audiobooks and being able to close my eyes and actually use the time to do something I want to do, instead of ending up on social media or whatever else I might be doing, has been great. I've read a lot of books, but it's still a lot of time. Listening to audiobooks for two hours a day is quite a lot of time that you could have spent doing something else, I think' (Daniella)

This ties into the perceived trade-off between pursuing self-fulfilling activities and everyday mobility. It also refers to the theme of freedom, where some people feel like going to and from work and spending time on public transport is seen as something they would rather not do at all. Several interviewees highlighted the concept of a 'worthy' destination. They felt that a longer commute could only be justified if the destination or work experience offered a sufficient payoff to outweigh the inconvenience of the travel itself. Furthermore, some interviewees

mentioned that they prioritise remote work options because they perceive the physical workplace as an undesirable destination, failing to justify the investment of commute time. However, some interviewees appreciated going to the workplace to gain social interactions and achieve a more productive work environment.

The above-mentioned paragraphs highlight the image of the urban citizen as fast paced and efficient, maximising productivity in both personal and professional lives. Stockholmers, embracing this ideal, avoid wasting time and see this effectiveness as key to success. It is desirable to work hard and to exercise - wasting time on the phone while commuting, and being lazy is not desirable.

In contrast to the challenges of optimising everyday mobility, micro-mobility presents a valuable alternative. It allows people to circumvent current traffic networks and avoid overcrowded public transportation. A common theme emerged among the interviewees, many expressed their enjoyment of being outdoors, getting fresh air, and experiencing the city firsthand. This is particularly true during the summer months when people treasure the limited stretches of pleasant weather. Micro-mobility transforms the commute from an unpleasant or unwanted experience into an opportunity to gain something more from the trip, beyond mere transportation or wasted time:

> *'[When riding an e-scooter] you didn't feel like you were missing out on anything because of the transportation. It felt like another good part of the day. It wasn't a compromise. Yes, it was something more than just getting around.' (Josef)*

The challenge remains to propose how micromobility can aid in making everyday mobility feel like a valuable experience to people, as opposed to just aiding in situations of urgent need.

4.3.4 Convenience and car dependency

This section presents public transport as the preferred mode of mobility in Stockholm. However, car ownership is seen as justifiable for certain trips, leading to increased car use. Additionally, the interviewees emphasised the importance of a seamless journey with minimal disruptions, highlighting a challenge for first- and last-mile adoption.

Several interviewees confirmed that public transport is the most popular mode of transportation in Stockholm. One participant even stated, *'Taking the bus or subway happens automatically' (Olle)*. For those living in central Stockholm, car ownership often seems unnecessary and excessive. They highlight factors such as the cost of car ownership, parking difficulties, and congestion, which makes using a car unreliable and time-consuming. The interviews suggest that public transport is the norm and the default option for many in the inner city. One interviewee, who moved from Stockholm to Malmö, described that people around you influence your mode choice:

> *'If I still lived in Stockholm, I wouldn't have cycled as much. It also puts you in a kind of, what do you call it? The social context of transportation. It's easier when everyone else is cycling. But for example, if I lived in Stockholm, the majority of people would always take the metro. And then it's kind of boring to cycle around. You might have parked your bike somewhere and then you go with others somewhere else.' (Martin)*

This aligns with the previously described limitation of being bound to a mode choice. The risk exists of encountering unforeseen social situations where using a bike would be inconvenient.

Stockholm is a city that prioritises public transportation. Several interviewees expressed that there is no need for cars and that they are simply a burden. However, this does not necessarily mean that these individuals will never acquire a car. There are certain stages in life where car ownership becomes justifiable. When discussing car ownership with interviewees who own a car, it becomes evident that many justify car ownership with specific needs. These needs are often created from infrequent trips, such as going to a vacation home outside of the city centre, but also perceived "dealbreakers' where the car is justified, such as having children or long commutes. However, car ownership often leads to increased car usage for other purposes, even when alternative modes of transportation are viable. Owning a car can contribute to laziness and a lack of motivation to seek alternative modes of transportation, as one of the interviewees explained:

> *'My partner owns a car and never takes public transportation. So you get used to the advantages that come with that. And then you might walk less or distances seem much bigger in your head. You become very lazy when you have the option [...] if it's bad weather and you have limited time, you'll always choose the easy way out. When you're paying for it anyway.' (Daniella)*

One of the justifications of owning a car amongst the interview sample is owning a vacation home. The interviews suggest desired ideals that some aspire to, where vacation homes and children fit in. This ideal picture often includes owning a car for its perceived convenience in such situations. Some car owners recognised that using the car is not a necessity, however, they utilise it for convenience in other aspects of their lives than the initial justification:

> *'I do see that I might actually have a need for a car. I would say especially when you go out to the vacation home, for example, it is very difficult to get there by public transportation. Then there is the issue of shopping... Well, yes, partly it is difficult to get there by public transportation, but absolutely, it would have been possible, that's not it. It's more of a convenience thing, I would say. So I would say maybe in 80% of cases it would have been possible to get there by public transportation anyway. But it would have been more inconvenient.' (Ivar)*

> *'Exactly, and that's what's sad, that it wouldn't make a big difference [if we did not use the car]. I mean, then we'll have to take the bus instead of the car. It's not a life-changing event for us, so in that way, it's very easy, and it's also very sad that we don't do it.' (Helena)*

Interviewees who live outside of the city centre own a car and use it for commuting perceive public transport as insufficient. This can be particularly frustrating for trips between suburbs, which often require unnecessary detours through the city centre. Interviewees expressed dissatisfaction with travel times and the inconvenience of transfers. They value the seamlessness, point A to B solution, that car ownership provides. The convenience of 'point A to B' is a recurring theme for all interviewees, car owners and non-car owners alike, a key aspect that goes beyond simply reaching the final destination. Interviewees, regardless of car ownership, emphasised the importance of minimising disruptions during the journey. For those who rely on public transport, this translates to a preference for minimising transfers. One participant who is a frequent bike rider describes this:

'It's annoying that [public transport] takes so long, that the distances are so long, and that there's so much transfer time, I would say... It's just the feeling of moving forward, you know. It's the psychological aspect, that you feel like you're moving forward during that time. Not stopping.' (Martin)

This finding suggests a preference for seamless and uninterrupted journeys. Interestingly, the frustration appears less pronounced when the interruption occurs at the end of the trip, such as walking the last mile. While SMMS can address gaps in public transport networks, the need to switch modes at the beginning or end of a trip, even for a short distance, can be perceived as an unwelcome disruption to the desired seamless travel experience. This presents a significant challenge for the adoption of first/last-mile multi-modality solutions offered by SMMS.

4.3.5 Skill gaps and accessibility challenges

This section outlines the accessibility challenges for SMMS to feel approachable by more people. The interviewees expressed concerns about navigating busy traffic and a lack of prior experience using e-scooters. Furthermore, acquiring and maintaining a suitable commuting vehicle was also seen as a skill-based challenge.

For shared micro-mobility services to become widely accessible, both the vehicles and the user experience need to feel approachable and manageable for a broader population. Several interviewees who were not regular users of MMS, particularly e-scooter services, expressed a lack of confidence due to concerns about their ability to navigate busy Stockholm traffic on a vehicle they didn't fully trust themselves to control. This apprehension likely stems from multiple factors, one of which is a lack of prior experience manoeuvring similar vehicles during childhood.

Even though e-scooters were previously described by the stakeholder interviews as user-friendly and easy to learn, according to the interview data, this skill may not be as widely developed as cycling proficiency. In contrast, nearly all interviewees reported knowing how to ride a bike before trying any shared micro-mobility service offering bicycles.

Consequently, a service that provides a vehicle that most people can master from the start will have a larger target group. Otherwise, users face the hurdle of needing to invest both time and money to learn how to manoeuvre the vehicle. This can be a significant barrier, as many people already feel they lack these resources in abundance.

'But [e-scooters] seem difficult and scary, and they didn't exist when I was a kid. They're a new modern thing that I don't understand, and I'm not very technical. It seems like you need to connect to an app to use them, and I can't be bothered to figure it out... I don't know when I would have time to go out and practise around the block either.' (Helena)

The fact that these services are mostly used spontaneously means that people feel they have even less time to carefully test or get to know the service, which discourages use:

'Yeah, the few times I've thought about trying an electric scooter, it's always been when I've been stressed and then I have to start this damn app and it takes longer to get the app going with my credit card, and everything than it takes for me to walk there. Then it ends up that I've walked there anyway. So that's why I haven't tried one of those services.' (Josefine)

According to the interviews, users need either confidence or experience to try an SMMS. Drivers who do not engage in traffic often may find it more challenging to navigate Stockholm's roads. The city has many different types of traffic and a hectic atmosphere, which can be intimidating for inexperienced drivers. To get around effectively, drivers need to be confident and assertive, whether they are driving an e-scooter, bicycle, or car. Users of SMMS must be daring enough to try something new in an uncertain situation if they do not already feel that they have mastered the vehicle:

> *'In Stockholm, you have to take up a bit more space in traffic, I think. You also have to be aware of so many things, there's always something happening everywhere. Many roads are one-way and there's a lot going on, so you need to take it easy but you also have to be assertive. If you're not assertive in Stockholm, you won't get anywhere.' (Josefine)*

Increased usage of MMS leads to greater user confidence, and perceptions of vehicle safety can vary based on recent experiences. Beyond manoeuvring the vehicle itself, users develop various service-related skills as they engage with MMS. This includes understanding operational areas, slow zones, and vehicle access procedures. While these skills are acquired over time, they can act as a barrier to entry for individuals lacking time, experience, or confidence. Additionally, downloading and using the app may be a particularly significant barrier for older or those who are not technologically savvy:

> *'But I think about the fact that for me to download an app, start paying, stand there, and struggle with how to start it, it's too much for me to do. Then I think it's much more convenient to just have my [public transport] app. Like, I Know how it works' (Stina)*

The multitude of different SMM providers on the market makes the barriers to entry even higher. The sheer number of options can contribute to feelings of uncertainty and required skill, as different providers may have varying pricing structures and operational nuances. This complexity can particularly discourage users who are already pressed for time. As Josefine mentioned, the time investment required to download and register for new apps can negate the perceived time-efficiency benefit of e-scooters compared to walking. This is especially true for short trips, where the app setup process might outweigh the time saved by using an e-scooter.

Additional insights emerged from non-users of SMMS who expressed interest in active mobility commuting. They emphasised the importance of a high-quality vehicle that can withstand the demands of daily use. Therefore, a micro-mobility vehicle intended for frequent commuting should be durable and comfortable. Acquiring such a vehicle requires knowledge of vehicle types, financial resources for the initial investment, and resources for ongoing maintenance and storage. If users lack this knowledge, they may need support from their social network or invest additional time and effort to acquire it. The significant financial investment associated with purchasing a high-quality vehicle further underscores this point. One interviewee described the difficulties of acquiring a quality micro-mobility vehicle as the following:

> *'Yeah, I don't really know anything about bikes. That's the problem. That's what's stopping me [from buying a bike] right now, from just looking on [Reseller websites] because it seems like it's much more complicated than you think. Like, if you're going to buy a really good bike, you have to check it out, make sure it's fine, that certain things are right, and I don't know anything about that.' (Daniella)*

Another factor affecting accessibility beyond user skill is the perceived cost of some SMMS, particularly e-scooter services. People are often hesitant to pay extra for SMMS when they already have monthly public transport passes or other everyday mobility expenses. If the service isn't seen as a viable alternative for everyday commuting, users perceive it as an unnecessary expense on top of existing transportation costs. Consequently, not everyone can justify the additional cost of saving a few minutes compared to walking, which is one of the more mentioned types of use cases. This perception of high cost can contribute to the image of e-scooter users as somewhat elitist or snobbish:

> *'It's funny, sometimes when you see a guy in a suit riding an e-scooter, you just think... come on... you could have just walked'(Lily)*

Finally, the user's residential location is one big factor determining the accessibility of these services. The use of micro-mobility services depends on their availability in the area where you live or spend a lot of time. People who find public transportation the most difficult to use often lack access to micro-mobility services, meaning they live outside the city centre.

To conclude, SMMS users need to feel that the services are safe and easy to adopt. User confidence, accessibility, cost, and inclusivity need to be addressed. By focusing on these areas, micro-mobility services can unlock potential as a sustainable and accessible transportation option for Stockholm's residents.

4.4 Use Cases

Informed by the pop-up and in-depth interviews, this section explores how Stockholm residents currently utilise SMMS in their daily lives, synthesising the interview insights and providing a tangible answer to research question one: 'What role do micro-mobility services play in Stockholmer's mobility practices?'.

Four different use cases emerged from the analysis of the interviews. These aim to mediate different scenarios where micro-mobility plays a role. A commonality across all use cases is that the need for micro-mobility often arises spontaneously, as presented in Figure 15.

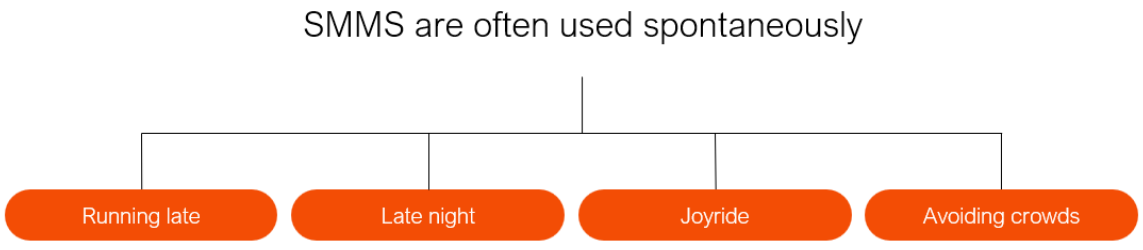


Figure 15. Use cases of micro-mobility in Stockholm

The first use case centres around *People running late*. The need for micro-mobility arises from time pressure when users perceive a risk of arriving late at their destination. Using SMMS in these circumstances is often a spontaneous choice made by the user when they realise that they are running late. Micro-mobility can facilitate quick decision-making for users pressed for time, if there are vehicles available in their vicinity at these moments. SMMS's appeal lies in its

ability to often provide quicker journeys than public transport, particularly for shorter distances. This highlights the perceived value of SMMS as a fast and on-demand transportation option.

The second use case is for *Late night travel*. From the analysis of the interviews, it became apparent that micro-mobility plays a role at night when public transport does not run as frequently. This need often coincides with nighttime activities, such as visiting bars or restaurants. Stockholm's public transportation schedule offers fewer departures late at night, leading to frustration for users who miss connections or face long wait times when returning home. When the user finds themselves in these situations at night, they can decide to use a micro-mobility service to get home faster. Beyond reducing travel time, some users perceive travelling by micro-mobility as a safer alternative to walking or waiting at stations, particularly due to concerns about encountering unpleasant or potentially threatening situations with strangers. This sense of fear or unease is more apparent for female users travelling alone.

The third use case identified is recreational riding or *Joyriding*, primarily associated with e-scooters. Participants reported using these services not just for practical transportation, but also for leisure activities with friends. This social aspect is particularly prevalent among younger users. Some associate riding shared e-scooters with a sense of freedom, as they can go around the city in a fast way together with their friends. Analysis suggests this use case is more common during pleasant summer weather. Additionally, micro-mobility services can be used in social contexts within groups where some members have bicycles and others do not have an equivalent personal vehicle.

The last use case presented is *Avoiding crowds.* While less frequently mentioned in interviews, micro-mobility services emerged as a solution for avoiding crowded public transport, particularly during the summer months. Riding SMMS rather than going underground in the subway in the summer months is something that some users value. The frustration that is associated with travelling on public transport during rush hours, when users must share limited space with a lot of passengers is prominent in the interviews. Micro-mobility services thus offer an alternative for users seeking a less crowded and potentially more enjoyable travel experience, especially during summer weather, which often is highly valued by Stockholm residents.

4.5 Key Insights

From the five themes extracted through the in-depth interviews, three key insights emerged, displayed in Figure 16. In the following subsections, each key insight will be described along with their corresponding user needs.

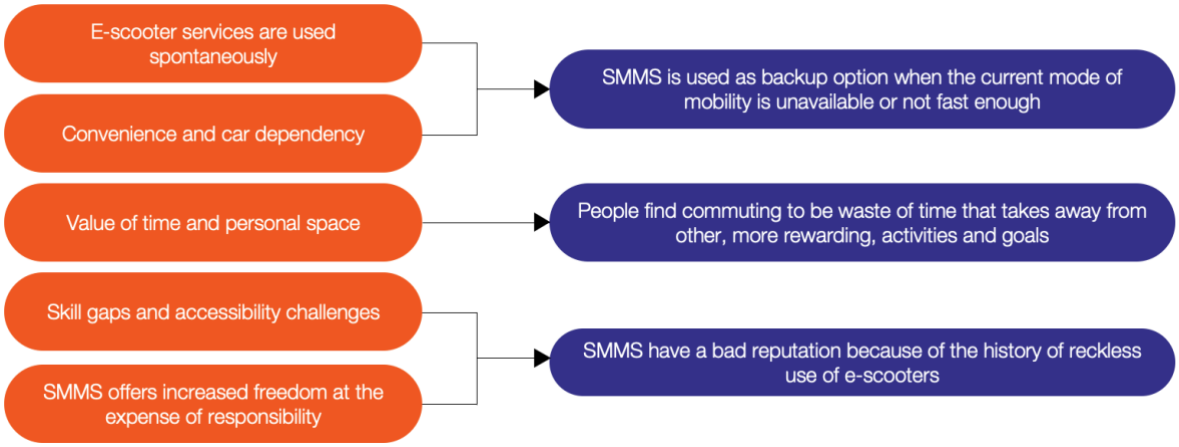


Figure 16. Key insights and corresponding in-depth interview themes

4.5.1 SMMS is used as a backup option when the primary mode of transportation is unavailable or not fast enough

As seen in the use cases, a common denominator between the different insights was the spontaneous use of the SMMS available in Stockholm.

Using SMMS with planning, and intent, and in a recurring pattern seemed to be an uncommon feat. On the contrary, the use and need for SMMS seem to arise when the current or preferred mode of transport is unavailable or not fast enough. More specifically, SMMS is used when public transport was deemed undesirable, mainly because it would be too slow and the users risked running late, or if it was deemed unsuitable to go underground or to be in a crowd, e.g. risking exposure to viruses.

Several factors were identified in the spontaneous use of SMMS, illustrated in Figure 17. The interviewees mentioned feeling impatient when a commute had many steps and mode transfers. The interviewees also expressed that adding an extra step for completing the first- and last-mile felt tedious and that the problem itself already was the number of transfers and modes, which poses a challenge for first- and last-mile adoption. Furthermore, having already paid for public transport makes users reluctant to pay extra for SMMS - which makes use only legitimate in extreme cases where spending extra money feels like a fair exchange. Another factor that contributed to the feeling of spontaneity was the frustration of having many different operators around. This poses difficulties in streamlining the adoption of these services since it was perceived as tedious to register for new services, and furthermore interviewees mentioned this as a possible dealbreaker for using the services at all. Furthermore, this contributes to a feeling of not knowing availability or access contrary to other modes of everyday mobility, like public transport, cars, or privately owned micro-mobility vehicles.

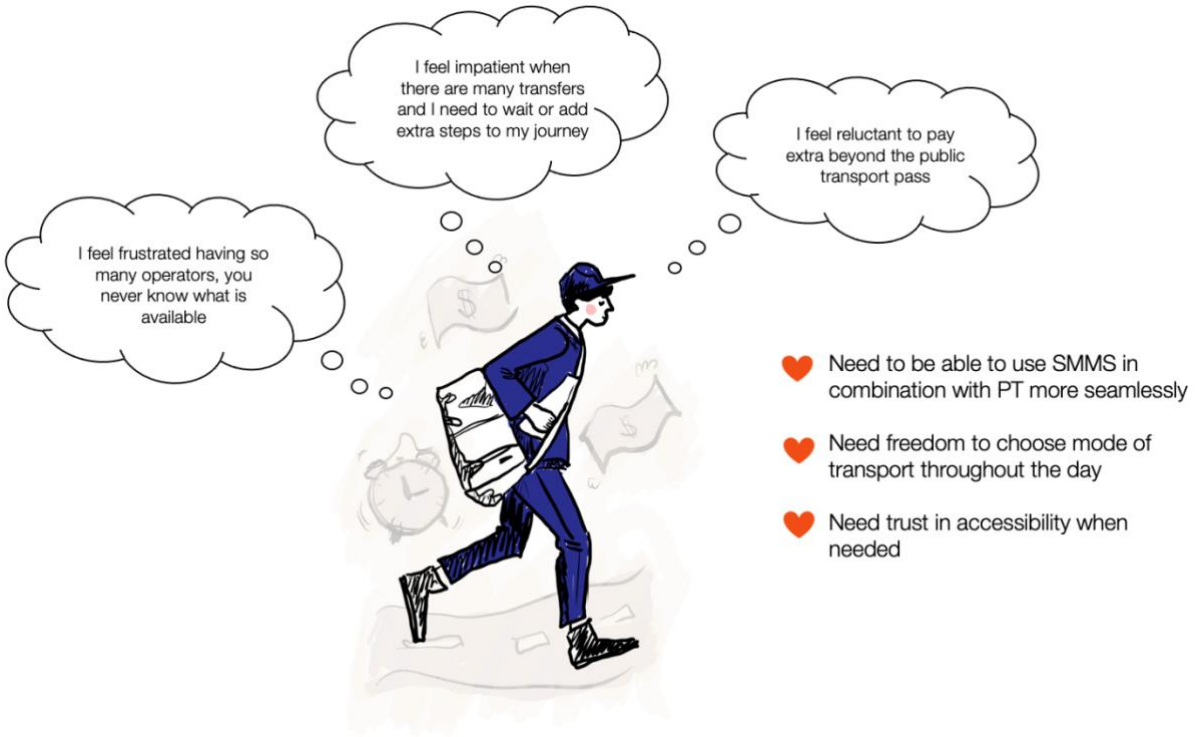


Figure 17. Factors contributing to the spontaneous use of SMMS, and corresponding user needs

These concerns highlight the following user needs that need to be addressed:

Need to be able to use SMMS in combination with public transport more seamlessly.
As mentioned, the feeling of a seamless journey is important to facilitate first-last-mile adoption. If the users are to add an extra step to this journey it is important to make the switch between modes as hassle-free as possible and to provide users with trust of availability and effectiveness.

Need the freedom to choose mode of transport throughout the day.
Many interviewees expressed the desire to use SMMS to keep freedom of mode choice throughout the day and relieve them of the responsibilities associated with operating a privately owned vehicle such as a car, e-scooter, or bike.

Need trust in accessibility when needed.
To ensure a more strategic integration of SMMS into day-to-day life it was deemed as important for users to know that SMMS services are available when the need arises. If SMMS are to be used, as an example, for commuting to work, the user needs to know where and how to access the vehicle. It is not likely that the user will spend time every morning looking for accessible vehicles in different places or using SMMS from different providers depending on what is most accessible.

4.5.2 SMMS have a bad reputation because of the history of reckless use of e-scooters

During the interviews, several interviewees expressed a distrust towards SMMS, both providers and vehicles, as illustrated in Figure 18.

This was primarily due to concerns about SMMS safety for riders and pedestrians, the potential for e-scooters to clutter cityscapes, and the perceived negative image (e.g., 'uncool' or 'unsporty') associated with SMMS use. While this insight is most relevant to e-scooters, the dominant SMMS option at the time, it also applies to e-bikes, particularly regarding safety concerns and user apprehension. Furthermore, many interviewees, particularly non-users, expressed feelings of lacking necessary experience or knowledge for using SMMS.



Figure 18. Factors contributing to distrust and fear towards SMMS and corresponding user needs

These concerns highlight the following user needs that need to be addressed:

Need to feel trust in shared micro-mobility services
This statement corresponds to the main theme of the insight, being the bad reputation of SMMS. The vehicles need to feel safe, be in good condition, and usable amongst other elements in traffic and to blend in with existing infrastructure. Additionally, a key aspect is fostering a sense of responsibility among SMMS users to minimise risks for themselves and pedestrians.

Need SMMS to be more approachable
This need aligns with the insight that people lack confidence in mastering the e-scooters, and that there is a lack of space, perceived time, or willingness to pay for practice. Consequently, inexperienced users are hesitant to even try the service due to a lack of confidence. On the other hand, if the user lacks experience, they need to make up for it through more risk-taking behaviour or lack of consequence thinking. Therefore, for SMMS to feel approachable, the vehicles should be readily mastered by a wider range of users.

Need a vehicle to identify with

Finally, SMMS needs to provide some extra intangible value for the users to find the services attractive. This value should make users feel like they can seamlessly integrate into the SMMS culture, avoiding feelings of awkwardness or projecting an unwanted image. Additionally, using an SMMS shouldn't feel like an involuntary advertisement.

4.5.3. People find commuting to be a waste of time that takes away from other, more rewarding activities and goals

The interviews revealed a recurring theme of feeling like daily commutes is a waste of time and a trade-off from other, more rewarding activities, presenting itself as a clear area of improvement.

Furthermore, many interviewees expressed feelings of discomfort in public transport, coming from factors such as hygiene, sound disturbances, and feelings of intrusion on personal space, illustrated in Figure 19. Uncomfortable and frustrating experiences on public transport, especially during rush hours, were identified as a push factor towards the perceived comfort of driving a personal car. The interviewees described disliking the feeling of standing still, and not being able to control their movements and route. They felt that the car was an easy way out, especially if they perceived commuting as inconvenient.



Figure 19. Factors contributing to commuting feeling inconvenient and uncomfortable and corresponding user needs

The corresponding needs to tackle these problems are therefore the following.

Need to feel like every day travel is used for something valuable, like work, exercise, relaxation, or quality time.
A frequently suggested solution for the feeling of wasted time was having the ability to use the commuting to either something self-fulfilling, like reading, listening to music, exercising, or to

do something productive like working or taking the time to rest. In turn, this relieves time around work hours to do other, more rewarding activities.

Need to feel happy while in transport.
Since people spend a lot of time to and from work or school, even if they do something productive or self-fulfilling, they need a positive experience. As of now, many see commuting as a chore, and there is a possibility here to change a perceived negative experience to a positive.

Need to feel in charge.
Finally, the interviewees expressed a need to feel in control, and that using time effectively during the day was a way to reinforce that sense of control. When using time wisely, users can allocate the desired time and resources to more fulfilling activities outside of work and day-to-day chores.

5. Ideation

This chapter presents the ideation activities, starting with the formulation of sustainability objectives to guide the ideation, based on the frame of reference. It further presents 'How might we' (HMW) statements phrased to transform the problems identified among the key insights into opportunities. Finally, it presents the resulting mobility interventions created in the co-creative workshop.

5.1 Objectives for Sustainability Improvements

To address the environmental aspect of SMMS through a service design lens, three sub-objectives to the overarching objective of more sustainable everyday mobility practices were defined. To bridge the gap between the problem and the goal, the means of pursuing the goal were the subject of ideation.

As identified in the frame of reference, current SMMS usage results in a net increase in CO_2 emissions. This is primarily due to the high life cycle impact associated with vehicle manufacturing, the short lifespan of vehicles, and a modal shift pattern that can lead users from low-emission to higher-emission modes of mobility (Hollingsworth, 2019).

Through a literature review, Clemm et al. (2024) identified five key factors that determine the environmental impact of SMMS: lifetime, modal shift, recharging, rebalancing, and utilisation rate. These serve as possible opportunities to improve SMMS in terms of associated carbon emissions.

With these five key factors in mind, the following three objectives were set to address the environmental impact of SMMS and provide a more sustainable and functional service system:

- *Objective 1: Increased utilisation rate*. This aims to increase the number of trips per vehicle, reducing the environmental impact per produced vehicle (Sun & Etz, 2021).
- *Objective 2: Encouragement of more responsible use.* This promotes practices that extend vehicle lifespan and minimise damage, reducing the need for frequent production cycles (Hollingsworth, 2019).
- *Objective 3: Substitution from cars to more sustainable modes of mobility*. This encourages users to switch from car trips to lower-emission options like SMMS or public transport, leading to an overall reduction in CO_2 emissions (Institute of Sensible Transport, 2018).

These three objectives were chosen because they are directly linked to the use of SMMS and offer clear potential for CO_2 emission reduction. While recharging and rebalancing are crucial aspects of the service system, they are not explicitly addressed as goals because they are less directly related to the lifestyle choices and everyday mobility behaviour of end users.

5.2 'How Might We…' Statements

This section describes the 'How might we' statements and their connection to the sustainability objectives.

The key insights were used to transform problems into spaces for innovation and were re-phrased as actionable 'How Might We' statements that the workshop participants could use as a base for ideation. Furthermore, the three HMW statements were closely linked to the three sustainability objectives to ensure that the interventions suggested in the workshop would align with the strategies to reduce emissions related to SMMS.

The HMW statements were compiled into cards that were handed out in the co-creative workshop, and are presented in Appendix F.

5.2.1 How might we encourage more intentional use of SMMS in people's everyday lives?

The first statement is related to the first key insight, namely, *SMMS is used as a backup option when the primary mode of transportation is unavailable or not fast enough*.

If the interventions suggested would facilitate a shift from the current state of use, which has been identified as mainly spontaneous, this allows the users to use the services with more intent, as a part of their day-to-day mobility. There is no need to eliminate the positive aspects of spontaneous use, such as the ability to use SMMS as a complement to public transport at times when public transport is unavailable but extending the use to facilitate routine integration into everyday life.

If SMMS is more integrated into daily life, utilisation rates will increase as they serve a wider purpose, connecting to the first sustainability objective. Furthermore, if SMMS can be integrated into commutes while providing the feeling of a seamless journey it will facilitate first- and last-mile adoption which aids towards objective two, promoting substitution from cars to more sustainable modes of transport.

Finally, it can be argued that if users see SMMS as a resource for everyday life, it can impose objectives to treat the services with greater respect as opposed to using the vehicles in situations of stress and late nights, possibly under the influence of alcohol. This can decrease SMMS vehicle damage and improper parking which can obstruct other vehicles parked in public spaces.

5.2.2 How might we make people use micro-mobility services with more respect?

The second statement addresses the second key insight, *SMMS has a bad reputation because of the history of reckless use of e-scooters*, and translates into the following 'How might we' question: How might we make people use micro-mobility services with more respect?

If the suggested interventions would improve how the services are used, this would mitigate negative perceptions caused by safety concerns, cluttering of the cityscape, and the general image of SMMS. This corresponds to the first sustainability objective, increased utilisation rate, since if more people, both users and non-users, have a more positive image of SMMS, more people will be willing to try the service.

Furthermore, promoting more responsible use and incorporating more intangible value into the service offerings can lead to users treating the vehicles with greater care, ultimately extending their lifespan. This strictly corresponds to sustainability objective number two: encouraging responsible use.

The third objective, the promotion of substitution from cars to more sustainable modes, is not explicitly addressed through this statement but might have a positive correlation to the other two goals. A more positive general perception towards SMMS can help users opt for SMMS as an alternative to cars or in combination with public transport.

5.2.3 How might we use SMMS to make everyday mobility feel like a valuable experience?

Building on the final key insight, *people find commuting to be a waste of time that takes away from other, more rewarding activities and goals*, the third 'How Might We' statement emerged.

The workshop interventions aimed at addressing this opportunity have the potential to tackle the negative aspects of the daily commute, an activity most Stockholmers participate in but rarely enjoy. This presented itself as an interesting area for innovation, and the interventions could aid towards presenting a positive, more environmentally friendly alternative to cars for those who are dissatisfied with the current commuting experience. This strictly relates to sustainability objective number three, the promotion of substitution from cars to more sustainable modes.

The frame of reference suggests that a direct switch from cars to SMMS is a difficult feat. However, car users who are not interested in public transport might be more inclined to try SMMS. Furthermore, this could potentially relieve congestion in public transport during peak hours and would also improve the user experience for those using public transport for their commutes.

Furthermore, this statement aligns with the objective of increasing utilisation rates. By enticing car users who wouldn't consider public transport, SMMS can indirectly contribute to an overall rise in ridership.

5.3 Workshop Results

This section presents the mobility interventions crafted in the co-creative workshop.

The co-creative workshop resulted in three rough drafts for interventions for more sustainable everyday mobility practices, one intervention corresponding to each HMW statement. The result from the workshop was compiled into intervention cards, which will be described below.

5.3.1 Integrated Smart SMMS

The first intervention proposed by the first workshop group was the one addressing the HMW statement of *How might we encourage more intentional use of SMMS in people's everyday lives*? The idea consists of a 'super-app' integrating SMMS providers, maps and navigation apps, and public transport providers. The full suggestion as well as prior ideas is available in Appendix G.

The app will allow for an integrated mobility experience, connecting public transport with SMMS offering a mobility-as-a-service. The user will be able to plan trips through one single app, facilitating planning and multi-modal journeys. The app will furthermore offer integrated

payment through the app or some other prepaid contactless electronic money system like the IC cards available in Japan.

Furthermore, the app will use big-data-based AI demand predictions to facilitate the relocation of vehicles and to guarantee vehicle availability. The app will offer a reservation system for SMMS vehicles to further guarantee availability and facilitation of usage for day-to-day commutes.

The stakeholders involved in providing the solution are government, national, regional, or local who are responsible for public transport provision. These will have to collaborate with SMMS providers to ensure integration and easy transfers between public transport and SMMS. Finally, this app will have to collaborate with navigation apps and location providers to ensure easy planning.

This ‘super-app’ will target current users of SMMS as well as potential users who can make use of multi-modal journeys.

5.3.2 Safety regulations

The second intervention stems from the second HMW-statement, *How might we make people use micro-mobility services with more respect*? The proposed intervention from this group was one of reinforced safety regulations. The group proposed several regulations that would improve safety around the use of SMMS, which in turn would improve the general image of SMMS and attract users who are more risk-avert. The full suggestion as well as prior ideas is available in Appendix H.

The specific regulations that were proposed by the group are speed regulations, such as a vehicle speed cap and general speed limit regulations on the streets. They also proposed compulsory insurance and compulsory use of helmets. There were other suggestions on interventions on a regulatory level, related to infrastructure such as separate lanes for SMMS. Another suggestion to increase safety was special traffic lights for SMMS as well as special traffic stops for SMMS.

The stakeholders involved in this intervention are end users who are the subject of these regulations, government, and transport networks, which SMMS is a part of.

5.3.3 Incentivising Active Mobility

The final intervention suggestion answers the third and final HMW statement: *How might we use SMMS to make everyday mobility feel like a valuable experience*? The suggested intervention is called incentivising active mobility and is an idea for a new type of SMMS that aims to use micro-mobility to promote more active transport. The full suggestion as well as the prior ideas is available in Appendix I.

The proposed service is a smart SMMS vehicle that combines battery-powered and active mobility options. This service offers real-time feedback during trips, showcasing the user's physical contribution alongside the battery's power usage. Additionally, the digital platform would provide information on health benefits like the number of calories burned and environmental benefits like CO2 reduction. Furthermore, the platform offers rewards based on the CO_2 reduction that can be used at local coffee shops.

This intervention targets healthy users, or users who aim to be healthier, who need a rewarding exercise and commuting experience. Other stakeholders are SMMS providers and local businesses, such as coffee shop owners. This intervention suggestion connects the user and the local community and boosts small businesses while offering a healthy option for everyday mobility. The HMW question is addressed by offering a rewarding activity, exercise, and fresh air, as well as the possibility for further rewards such as discounts at local coffee shops.

6. Service System Concept

This chapter presents the final service system concept design, the Integrated Active Mobility (IAM) system, built on the results of the co-creation workshop and the insights from the interview- and literature study. The mobility interventions created in the co-creative workshop were integrated into a cohesive system and further ideated and elaborated on to create a detailed and comprehensive service system design, fulfilling the user needs that were presented with the key insights.

This service system concept serves as a response to research question two, 'What shared micro-mobility service attributes are required to enable more sustainable mobility practices in Stockholm?', proposing a range of attributes and outlining a conceptual service design that could enable more sustainable mobility practices in Stockholm.

This chapter includes an overview of the service system, the intended use cases, and the ten key attributes. The result is further visualised through a customer journey map, a description of the payment models, two service blueprints, and a system map. Lastly, this chapter concludes with a summary of the service system concept and its potential environmental benefits.

6.1 Integrated Active Mobility System: Overview

The Integrated Active Mobility System presents a platform-enabled station-based shared e-bike service, illustrated in Figure 20. The service is made to integrate seamlessly with public transportation, offering enhanced safety and a more rewarding user experience. The service offers hybrid vehicles to encourage active mobility while providing a rewarding commuting experience. Using active mobility yields points that can be used in local coffee shops for discounts, supporting local business and community feeling.

The system integrates with public transportation by allowing for integrated payment through IC cards, with three payment options catering to three different use cases. Additionally, multi-modal trip planning is facilitated by integration with widely adopted location providers like Google Maps. These features improve planning and accessibility, offering a service that supports more sustainable micro-mobility practices through three objectives: modal shift, utilisation rate and more responsible use.



Figure 20. Visualisation of users in the service system

6.2 Use Cases

This section outlines the target use cases for the Integrated Active Mobility service system.

To support the sustainability objectives of increased utilisation rates, positive modal shift patterns, and more responsible use, three visionary use cases were developed as a foundation for the service system. These use cases acted as a stepping stone for envisioning how the service system concept would be used and helped form the remaining attributes in the result. The aim was to cater the system design to these three different use cases described in this section and presented in Figure 21.

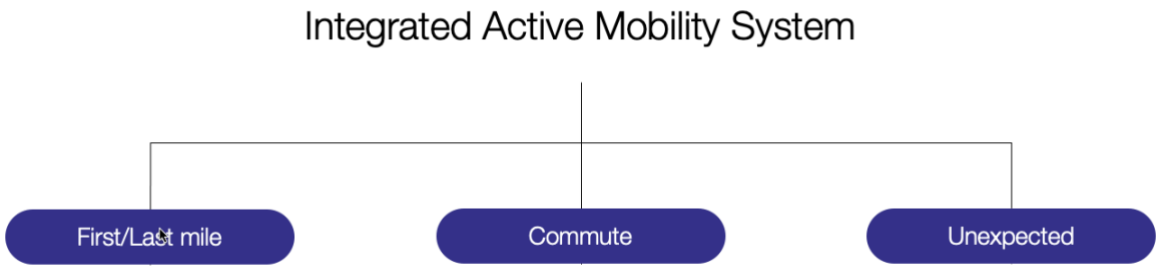


Figure 21. Visionary use cases for the IAM system

First/last-mile journeys involve bridging the gap between public transport stations and the user's destination or original departure location. The aim was to integrate the service provider with public transport to better facilitate these needs.

Commuting journeys, where the user searches for an alternative commuting mode of transportation. The aim was to offer a service that is viable as a primary commuting option.

Unexpected journeys, where the user encounters a situation where they unexpectedly need micro-mobility. The aim was to preserve the spontaneous use of micro-mobility for occasions when SMMS does not compete with public transport but acts as a complementing service, providing value that public transport fails to do at certain times, such as when running late, missing public transport departures or travelling during nighttime.

6.3 Key Attributes

This section details the key attributes of the Integrated Active Mobility service system, ranging from service level to organisational and institutional level. These attributes describe the main characteristics of the service system and are a result of the final concept development.

Figure 22 presents the IAM system's ten key attributes. The physical and service attributes are on the service level, and the support attributes are on the organisational level. High-level collaborative attributes such as governance and infrastructure are presented on the institutional level.

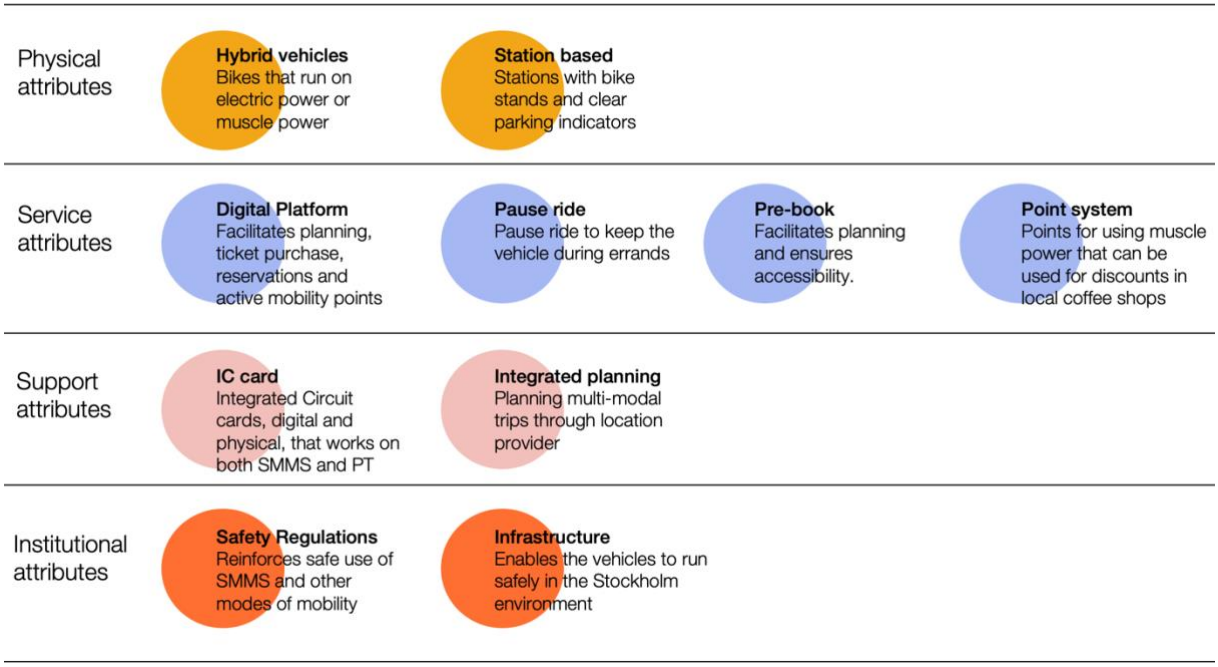


Figure 22. Key attributes of the IAM system

Digital platform for interacting with the service. The app holds functions such as pre-booking, ticket purchase, and information throughout the ride, as well as active mobility points, which will be further explained below.

IC card, or integrated circuit cards, facilitate payment and vehicle renting. These cards are available both as digital and physical cards and can be charged either in the SMMS app or in convenience stores. The card is personally linked to the user's account in the app, and it is tapped on the bike to start the ride. This IC card is proposed as an integrated payment card compatible with public transportation and the e-bike service. This facilitates seamless travel for users seeking flexibility or needing first/last-mile solutions when travelling by public transport. By eliminating the need for separate payment methods, the card encourages the integration of e-bikes into users' daily routines.

Stations that offer clear parking indications to decrease the cluttering of bikes throughout the city. E-bikes are equipped with GPS trackers, ensuring rides can only be completed at designated stations. Compared to free-floating models that require searching for available vehicles, a station-based system offers a fixed access point, promoting routine use for commuting purposes. This predictability streamlines the user experience and fosters reliability, making the service a more attractive commuting option.

Integrated planning is done through a location provider such as Google Maps, which provides multimodal and trip options and ensures interoperability.

Pause ride option that makes it possible to keep renting the vehicle while doing errands. This is to cater to the user's need for transportation to places that do not offer nearby stations. The user can ride the e-bike closer to their destination and pause the ride instead of looking for a station and booking a new vehicle when continuing the journey. This is made possible by removing the pay-per-minute model commonly found in the e-scooter services available in Stockholm today.

Pre-booking option to ensure vehicle availability for planned trips, making it a viable primary option for commuting. Booking in advance makes it possible for the user to pick up their vehicle directly when arriving at the station. A pre-booking system becomes particularly valuable if one station is experiencing high demand. By booking a vehicle in advance, users can avoid the inconvenience of arriving at a station to find no bikes available. This facilitates a more reliable and predictable departure and gives the user time to make other plans if there are no available vehicles.

Hybrid vehicles, specifically e-bikes that can be powered both by electrical power and muscle power depending on user preference. This caters to the users that want to combine commuting with daily exercise, without the need to possess, maintain and store a personal bike. The option to turn on and off electrical assistance on the e-bike can be controlled either in the app or with a switch on the bike so the user can change preference while riding.

Point system that calculates points for using muscle power. The points can be redeemed for discounts in affiliated local coffee shops. This incentivises active mobility and allows users to interact with small businesses in their local community. Furthermore, life cycle analysis for electric vehicles estimates vehicle lifetime based on battery lifespan (Hollingsworth et al. 2019). Encouraging active riding through incentives can extend this lifespan by reducing strain on the battery.

Safety regulations are an essential part of providing more safe and respectful use of SMMS. The safety regulations should not only regulate the use of SMMS but also other modes of mobility to ensure a safe user experience. The safety regulations proposed are enforced by institutional agents, such as governing bodies, and also the service provider. Safety regulations include general speed limits and compulsory insurance, as well as vehicle speed caps. These features are already part of existing SMMS but are listed as key attributes since they are essential to providing a safe and meaningful user experience.

Infrastructure is also an important supporting attribute in the SMMS system. To be a functioning and safe service that incentivises active transport, functioning road networks such as separate bike lines and other road safety measures such as traffic lights are a key feature.

6.4 Customer Journey Map

This section presents the customer journey map for the Integrated Active Mobility system, including the steps the user undertakes while interacting with the service.

Figure 23 below illustrates the customer journey map, starting when a user realises the need for SMMS. The customer journey map offers an overview of the suggested steps the user goes through when using the service. The journey involves the three different use cases and some preparatory steps before starting the trip.

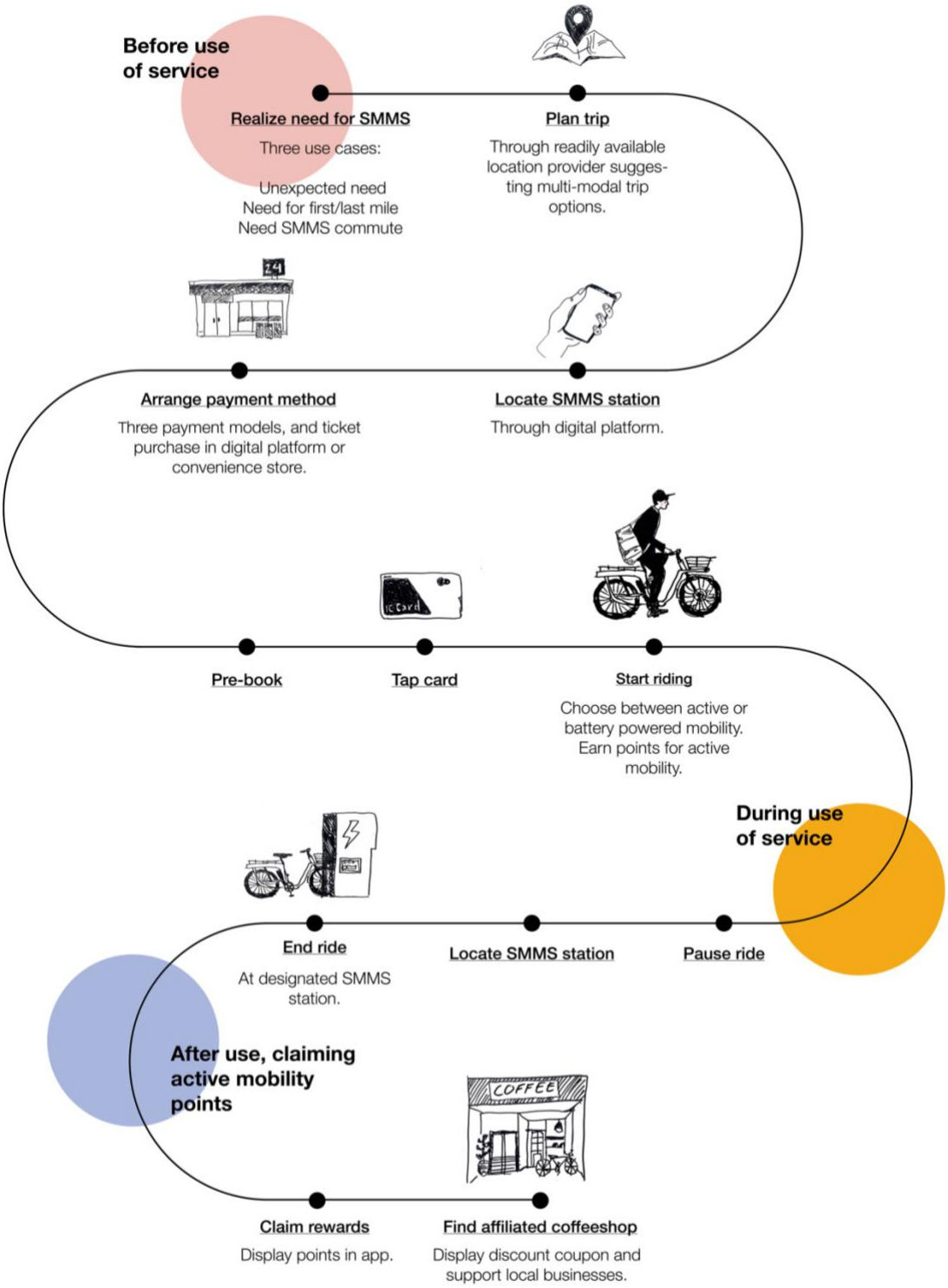


Figure 23. Customer Journey map

The first step is to plan the trip using a location app like Google Maps, which will show multi-modal trip options and available vehicles and station locations.

Second, the user locates the SMMS vehicle. Before the ride begins, they need to ensure their IC card is charged with a ticket. The payment system will be explained in more detail in the following section.

The user can pre-book the vehicle through the SMMS app and then tap their card on the bike to start the ride. While riding, they can choose between electric or muscle power. Using muscle power generates points that are displayed in the app. The user can pause the ride if needed to run errands. To finish the ride, they locate a station and tap their card again.

After the ride, the points earned are displayed in the app. These points can be redeemed for discounted coffee at local coffee shops listed within the app. To redeem points, the user shows the discount coupon displayed in the app at an affiliated coffee shop. Finally, they can use the coupon to purchase their coffee.

6.5 Payment Models

This section explores the payment methods available to users when accessing the service.

The payment models for using the Integrated Active Mobility system play a big role in its usage and in catering to the different use cases. The service system concept offers three distinct payment models to address the three use cases.

Monthly SMMS pass is for commuters who want to use the service as their primary mode of transportation. This payment method offers monthly payments with unlimited rides. According to the interviews, usage of a service is perceived as free if the payment has already been made. Users who subscribe to this method are not faced with an economical choice every time they need transportation. Having already paid for the month can also incentivise a higher utilisation rate.

Public transport add-on for users seeking first/last-mile solutions. Facilitated by the IC card, the option to purchase a monthly pass that works on both public transport and the SMMS service is implemented. This payment method is implemented to better integrate the service into users' daily lives. When public transport falls short, the service can bridge the gap, potentially making public transport in combination with micro-mobility a more attractive choice than using a personal car. Furthermore, integrated payments make SMMS commuting a viable everyday option.

Fixed price for unexpected journeys. This payment method offers a one-time payment to unlock the bike. This is to incentivise use for users who only want to use the service occasionally, and it is also a way for users to try out the service. The fixed price comes with a limited-time rent, where an additional cost will be added if the time limit is exceeded. The fixed price will also ensure that the service does not compete with public transport cost-wise, discouraging short, spontaneous trips that would substitute both walking and public transport. This option also caters to tourists.

The IC card facilitates all payment methods, and it can be charged with an SMMS ticket for all methods. Payment can be made in affiliated convenience stores or in the digital platform. Figure 24 presents a flowchart describing the different payment methods for their respective use cases.

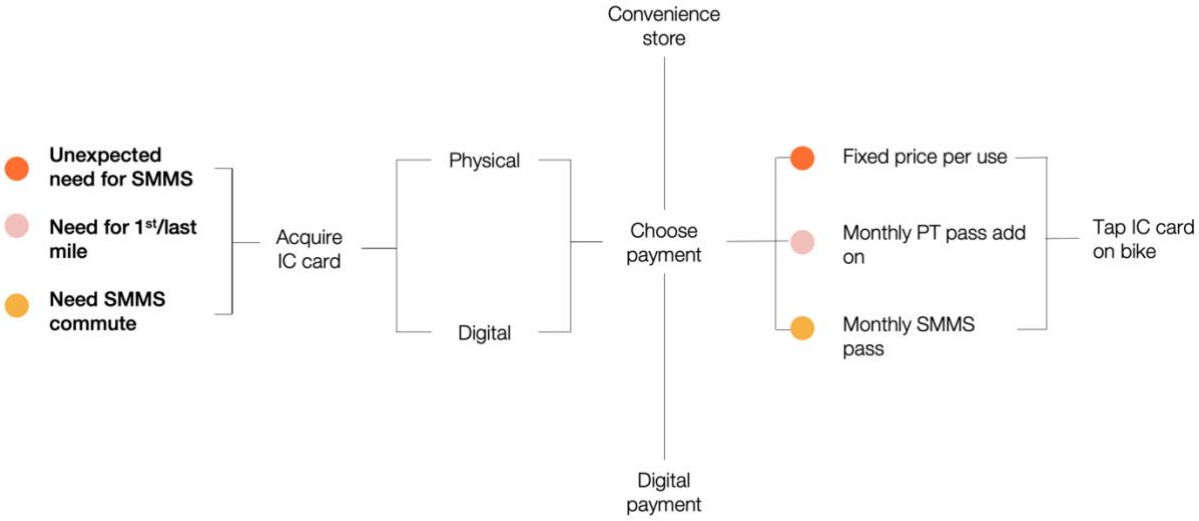


Figure 24. Payment method flowchart

6.6 System Map

This section presents an overview of the stakeholders included in the Integrated Active Mobility system and their relationships. Using a system map, the components that help realise or influence the service are illustrated on three different levels. As seen in Figure 25, these levels include the service level, the organisational level, and the institutional level.

The digital platform and the physical artefacts that make up the service, including hybrid bikes and parking stations, lie within the service boundary.

On an organisational level, collaborators are presented. To enable ticket purchases in physical stores, convenience stores act as collaborators to provide this option. The digital platform facilitates user interaction with participating local coffee shops, offering active mobility rewards. This establishes coffee shops as collaborators at the organisational level.

To further incentivise multi-modal trips, location apps for navigation are proposed as collaborators. The location apps should propose multi-modal trips, including both public transport and SMMS. Location providers should include the option to redirect users to the SMMS digital platform for ticket purchases when desired. Similarly, the IC card works as a component to streamline multi-modal trips. The IC card is set as a separate component supporting the service, as interoperability must be implemented to facilitate integrated payment on public transport and SMMS.

On an institutional level, regulations apply to both vehicles and SMMS use in the city. At this level, the municipality provides infrastructure such as bike lakes and space for SMMS stations.

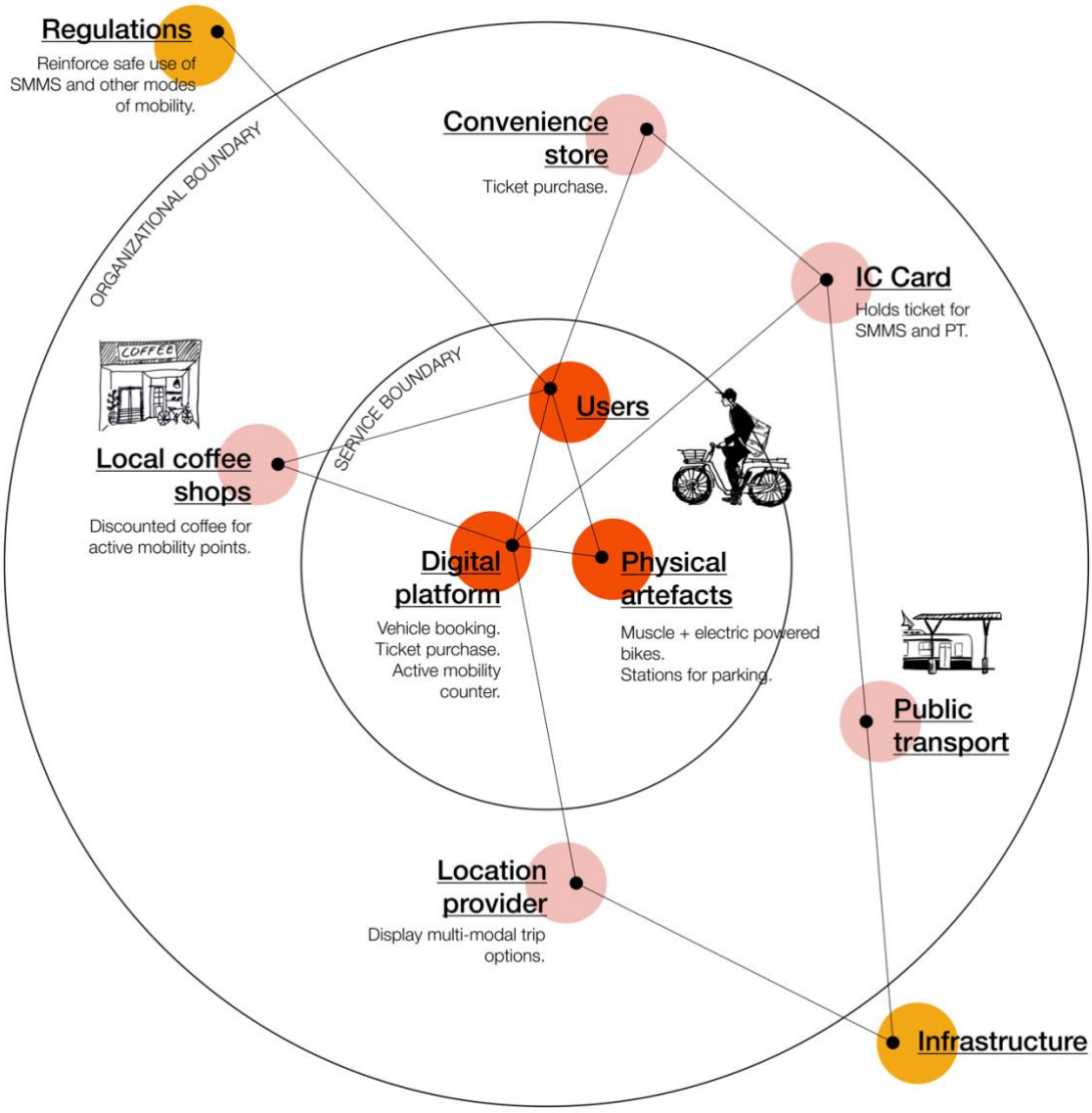


Figure 25. Service system map

6.7 Service Blueprint

This section describes the Integrated Active Mobility system through a service blueprint. The service blueprint showcases the customer journey in more detail, displaying the physical evidence that the user interacts with during the service use. Furthermore, it displays the service's frontstage actions, which are the parts of the service experience visible to the user. It also evidences the backstage actions, which are the service provider's actions that are not visible to the customer. Finally, it highlights the support processes of collaborating organisations and institutional elements. The service blueprint is shown in the following spreads.





Physical evidences

Location provider

SMMS Station

Convenience store SMMS App

Customer actions

- Unexpected need for SMMS
- Need for first/last mile
- Need SMMS commute

→ Plan trip → Locate SMMS station → Arrange payment method

Frontstage actions

- One time payment
- Integrated in PT monthly pass
- SMMS monthly pass

Backstage actions

Set up ticket on IC Card

Support process

Display multi-modal trip options

Display PT, SMMS, walking, etc.

Calculate time, cost, etc.







SMMS App

IC Card

Bike

Switch SMMS App

Lock

Pre- book

Tap card

Start riding

Muscle power

Electric power

(Pause ride)

Display available vehicles for booking

Bike unlocks

Display indicates readiness

Display timer

Display activity level

(Lock and unlock bike)

Process booking, reserve vehicle

Start timer for short trips

Collect trip data

Vehicle speed cap

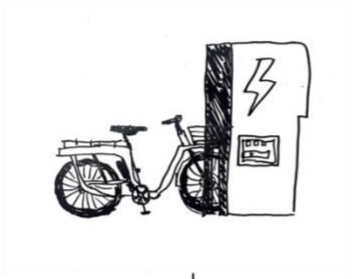
Calculate active mobility

Street speed limit

Compulsory insurance

Traffic signals

Infrastructure



Physical evidences

SMMS Station

IC Card

SMMS app

Customer actions

Find station

End ride

After ride

Frontstage actions

Lock bike

Display points

Display price

Backstage actions

GPS ensures correct parking location

Deduct money from IC card

Charge and change batteries

Big data AI-based relocation calculation

Support process





SMMS App | **Local coffee shop** | **SMMS App** | **Local coffee shop**

Want to claim rewards → Open app → Find affiliated coffee shop → Use points → Buy coffee

Display points

Present discount coupon

Collaboration with coffee shops to support local community

Deduct points from coupon

Collect trip data

Vehicle speed cap

Local small business

Deduct discount amount from price

6.8 Result Summary

This section summarises the Integrated Active Mobility system, connecting the key attributes to user needs and sustainability objectives. Table 7 lists the key attributes, the user needs and the sustainability objectives.

Table 7. List of key attributes, user needs, and sustainability objectives included in the system

Key attributes
Hybrid vehicles
Stations
Digital platform
Pause ride
Pre-book
Point system
IC card
Integrated planning
Safety regulations
Infrastructure
User needs
Need to be able to use SMMS in combination with public transport more seamlessly
Need the freedom to choose mode of transport throughout the day
Need trust in accessibility when needed
Need to feel trust in SMMS
Need SMMS to be more approachable
Need a vehicle to identify with
Need to feel like every day travel is used for something valuable
Need to feel in charge
Sustainability objectives
Increased utilisation rate
Encouragement of more responsible use
Substitution from cars to more sustainable modes of mobility

Figure 26 visualises the connections between these factors. The key attributes are presented on the vertical axis, and the user needs are presented on the horizontal axis. The sustainability objectives addressed through this are visualised as coloured dots, where each colour represents a sustainability objective.

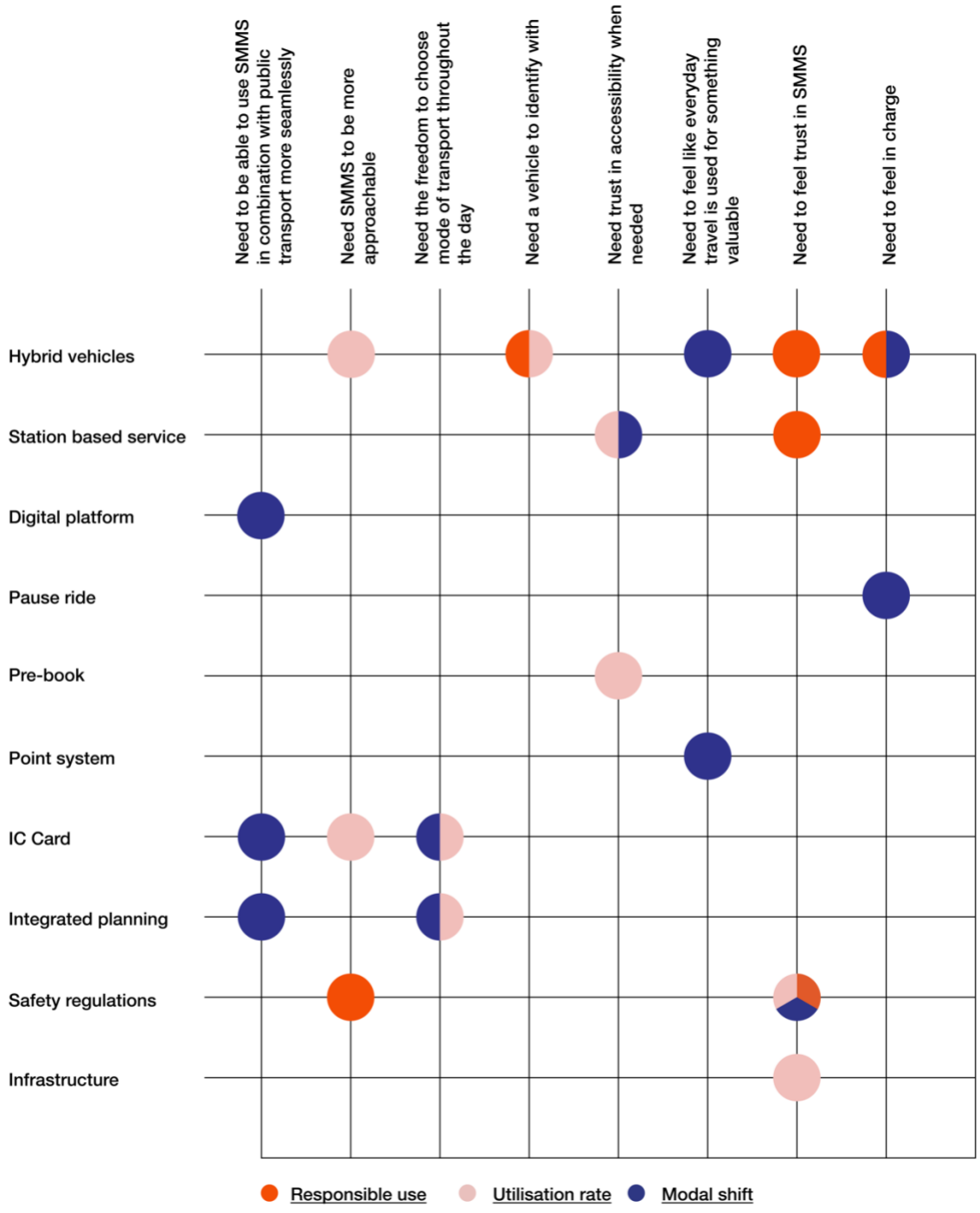


Figure 26. Service attributes and their correspondence to user needs and the sustainability objectives

7. Discussion

In this chapter, the results will be interpreted and discussed. The discussion covers the purpose and research questions, key findings and interpretations, sustainability objectives, methodology, the mobility landscape in Stockholm, and recommendations for future work.

7.1 Purpose and Research Questions

This section discusses the project's purpose and whether the research questions were successfully addressed.

The purpose of this project was to contribute to the development of more sustainable transportation options in Stockholm with the aim of understanding how shared micro-mobility services can be designed to support Stockholm residents in adopting more sustainable mobility practices. This was achieved through a literature study, qualitative interview insights and ideation supported by experts in the field. Through the literature study, we identified problems with current micro-mobility solutions, which informed the development of objectives for positive environmental impact. The objectives included increased utilisation rate, encouragement of more responsible use and promotion of substitution from cars to more sustainable modes of mobility.

However, we did not consider certain factors during the ideation process, namely efficient recharging and rebalancing, which were framed as two of the key factors determining the environmental impact of SMMS (Clemm et al. 2024). Since operational strategies, not user choices, heavily influence efficient rebalancing and charging, qualitative user research was deemed less effective in addressing these aspects.

The first research question was, ‘What role do micro-mobility services play in Stockholmers' mobility practices?’ This question was answered through insights from the interview data. The identified use cases provided an overview of how micro-mobility services are used in Stockholm today and revealed a situation where SMMS is mostly used spontaneously.

The second research question was: ‘What shared micro-mobility service attributes are required to enable more sustainable mobility practices in Stockholm?’. We used an iterative approach to address this question, combining a literature study, qualitative interviews, a co-creation workshop, and final ideation. The result is a service system concept proposing a set of service attributes and a conceptual service system design. This result should be treated as a proposal for a micro-mobility service system that has the potential to contribute to a more sustainable mobility landscape in Stockholm. It is, however, important to acknowledge that the result is on a conceptual level, meaning it has not yet been verified for usability and environmental benefit in a real-world setting. It is also worth noting that the result is based on interview data solely from Stockholm residents and can not necessarily be generalised in other contexts, such as other cities.

7.2 Key Findings and Implications

This chapter discusses the result use cases and the key attributes of the new SMMS system.

7.3.1 Use cases

Based on the insights from the interview data, four main use cases were outlined as a response to the first research question, 'What role do micro-mobility services play in Stockholmer's mobility practices?' These were running late, late nights, joyriding, and avoiding crowds.

The research identified the spontaneous use of the service as an area for improvement where SMMS could be better integrated into everyday mobility practices and better facilitate first- and last-mile adoption. However, this spontaneous use does not necessarily mean that SMMS is a competitor to public transport or active modes of mobility since it can act as a complementary service offering additional value that public transport cannot, such as late-night transportation or a fast option that facilitates arriving on time. Furthermore, as addressed in the expert interviews, SMMS can make up for the shortcomings in public transport and infuse a sense of safety.

However, as we saw that the design of the current SMMS mainly caters to trips that substitute walking (Hollingsworth 2019), we decided to design a service system that would better integrate into the everyday mobility practices revealed in the interview study. The opportunity arose to better integrate SMMS with public transport networks while keeping the option to use it spontaneously for unexpected needs. To address this area of improvement three visionary use cases were formed: *First/last-mile journeys, Commuting journeys* and *Unexpected journeys.* These aim to nudge the user to not only use the service spontaneously, but also to foster a use that also caters to routine integration while keeping the option to use the service for unexpected journeys.

7.3.1 Service attributes

This section examines the proposed service system, discussing whether user needs are met and how the service attributes address the sustainability objectives. The section is structured around the three spaces for innovation that the 'How might we' statements address: '*Intentional use*', '*Respect and Accessibility'*, and *'Valuable commuting'.*

Intentional use
Through the interview study, we identified the opportunity to design an SMMS that is used with more intent. To address the issue of SMMS mainly being used spontaneously and not being a viable option for day-to-day commutes, several service attributes were proposed, including IC card, integrated planning, station-based service, and pre-booking.

Firstly, the key features of IC cards and integrated planning jointly address the user need for seamless multi-modality, approachability, and freedom of mode choice. By offering monthly payment, the economic decision that otherwise must be made before every micro-mobility trip is eliminated. Integrating monthly payment with public transport payment and using the IC Card payment system for both modes could streamline the user experience for first and last-mile adoption. Additionally, it addresses the user need of freedom to choose between modes throughout the day. Making SMMS easier to use in combination with public transport can foster

modal shifts and increase utilisation rate, allowing new practises to form, where public transport and micro-mobility can be used interchangeably and consecutively. The use of IC cards offers a seamless multi-modal experience, promoting modal shift and offering easier transfers and registrations. These features empower users with greater control over their commutes.

The Integrated Active Mobility system offers stations as a key feature. A station-based service enables a consistent access point, facilitates trip planning and encourages strategic use. This contrasts with free-floating models, where users are required to search for available vehicles, possibly in different locations every time. High vehicle density favours a free-floating model, as docking stations wouldn't restrict users from taking the vehicle to their desired destinations. However, a station-based system can be more conducive to integrating SMMS into daily routines by enabling intentional trip planning and fostering habitual use. Therefore, this service attribute has the potential to increase vehicle utilisation rates and can lead to a more viable option for multimodal commutes, potentially making e-bikes a more attractive alternative to personal car use. Similarly, having a pre-booking option can facilitate planning, enforce trust in accessibility, and make it possible for the user to have their vehicle ready upon arrival at the station.

To conclude, more integrated use helps streamline first and last-mile adoption, potentially making the combination of SMMS and Public transport more attractive than personal car use. Integrating the service system into daily commutes will furthermore increase the utilisation rate.

Respect and accessibility
The second area for innovation targets reckless user behaviour towards e-scooters and addresses concerns about their accessibility to the broader public. This target area was addressed through the key features of a station-based service, hybrid vehicles, safety regulations, and infrastructure.

Informed by interviewees' concerns about e-scooter safety and user behaviour, the option of using e-bikes for the micro-mobility service presented itself as an opportunity that is accessible to a broader demographic and less susceptible to reckless behaviour. E-bikes inherently offer greater stability and a more familiar riding experience than e-scooters, potentially leading to broader user adoption and increased utilisation rate. Furthermore, according to interviewee feedback, bicycles are perceived as a more practical transportation mode than e-scooters, which can be seen as recreational "toys". As informed by the interviews, bikes are potentially less likely to be misused, such as careless parking, reckless riding, and mistreatment.

Implementing a station-based service system requiring trips to end at designated stations could encourage responsible use. The design of the proposed stations should incorporate clear visual cues to guide users on proper e-bike parking, fostering a neat and orderly arrangement. Stations can reduce cluttering in public places around the city and reduce vehicles being thrown around wrongfully. Furthermore, as informed by one of the stakeholder interviews, a heavier, bulkier vehicle is more challenging to toss aside in inappropriate places. Finally, informed by the interviews, bikes were claimed to look sportier and more aesthetically pleasing, providing more intangible value for the user and an image that more people could potentially identify with.

Safety regulations are an essential part of the service system, addressing the need for respect for shared micro-mobility services. This attribute further addresses all sustainability objectives, utilisation rate, responsible use, and modal shift. More safe use and respect for SMMS will mitigate SMMS's bad reputation, providing a service that more people will be willing to try.

Increasing accessibility will potentially attract users who do not identify as e-scooter riders, potentially attracting car-bound commuters.

Finally, infrastructure also addresses the need for respect for SMMS. Better bike lanes and traffic networks will encourage more people to use active mobility and micro-mobility, promoting increased utilisation rates.

Valuable mobility experience
The last space for innovation stemmed from the key insight that people feel like commuting is a chore, subtracting time for other, more self-fulfilling activities. Addressing the user's needs for both rewarding everyday mobility and feeling in charge, the proposed service offers hybrid vehicles and a point system.

Hybrid bikes give the user the option to combine daily exercise with transportation. As identified in the interviews, this has the potential to lower the threshold for users seeking to commute by bike by removing the need to deal with the hassles of personally owning a bike in Stockholm.

Micro-mobility services have the potential to give the user more control over their commute, as they are not bound by time schedules or tied to the public transport network. Offering a service that is viable to use as a daily commute can potentially attract car riders who avoid public transport for these reasons. Essentially, the micro-mobility service offers several advantages that mirror car usage: flexible departure times, uninterrupted journeys, and the freedom to stop and resume rides at will. By replicating these user-valued aspects of car travel, e-bikes, if implemented effectively, have the potential to encourage a reduction in personal car use.

Lastly, the point system provides extra gratification, adding by offering discounts at local coffee shops. This could emphasise the service as a positive contribution to the city as it supports smaller businesses and gives the user an incentive to use the service and explore the city away from the underground subway and the congested car traffic.

To conclude, these features could help facilitate a modal shift from cars to SMMS, which would connect to the sustainability objective of promoting substitution from cars to more sustainable modes of transport.

7.3 Contribution to Sustainability

The practice-oriented approach allowed us to examine the everyday mobility practices of Stockholm residents, providing a holistic perspective on how people move about in their day-to-day lives. We investigated how skill gaps influence usability and how the social nature of mobility influences mode choice and cultural conventions on how to move about in Stockholm. This approach lifted the project's focus to address not only shared mobility services and users of shared mobility services but also how people move about in their day-to-day lives. In this sense, theories of social practice were successfully used to shift the focus from the product to the underlying practice, as proposed by Scott, Bakker, & Quist (2012). Ultimately, we could explore what role SMMS have and could have in this existing web of practices.

However, the service system proposed will probably only lead to incremental changes in mobility practices and carbon dioxide emissions. The attributes proposed in the service system aim to promote more sustainable mobility practices for Stockholmers. If all three sustainability objectives are met, the system has the potential to achieve this goal. However, the proposed suggestion does not question the amount of mobility in Stockholm and does not intervene in the web of practices producing the need for mobility itself.

The attributes proposed can mainly be sorted into two of the practice interventions suggested by Spurling and McMeekin (2015), 'recrafting resource-intensive practices' and 'substituting practices'. Providing hybrid vehicles with the option of using muscle power and incentivising active mobility is an example of an attribute that falls under the first category. The integration of SMMS with public transport falls under substituting practices, aiming to facilitate first and last-mile adoption. The only attribute that can be linked to the third, and according to Spurling, the most effective framing of 'changing how practices interlock' is the point system providing discounts in local coffee shops. This attribute can strengthen local relationships, fostering a culture of moving about in the local area. However, this will likely only lead to incremental changes over a longer time.

On the other hand, one interesting finding was that the interviewees rarely seemed to enjoy their commutes. Utilising a social-practice-oriented perspective allowed us to create a service system that could change the meaning of moving about. Instead of just moving to get to a destination, this system proposes a way of moving where the mobility itself is a part of the 'why'. The suggested service system aims to provide a rewarding mobility experience, bridging material and skill gaps for active mobility commutes identified in the interview study.

Utilising the sustainability objectives was a way to provide an overarching goal and to include both user experience and sustainability objectives in the ideation process. However, this might have limited the solution space, fostering the ideation of attributes that only address the current need for mobility. Modal shift and utilisation rate were two important goals of the ideation and were identified as key factors influencing the environmental impact of SMMS (Clemm et al. 2024). While these objectives served their purpose for ideation, it is difficult to determine where these new users come from. For the objective of utilisation rate to be effective, modal shift must be carefully crafted.

While the sustainability improvements from this service system might be incremental, the resulting concept aims to enable new, more sustainable mobility practices to form in Stockholm.

7.4 Method

This section addresses potential limitations associated with the qualitative data collection in this project and how this could have influenced the result. It also discusses the demographics of the researchers who attended the co-creative workshop.

Firstly, the research was conducted in Stockholm, where e-scooters exist in greater concentration than other micro-mobility options. This may have influenced interviewee responses, potentially leading to a bias towards e-scooter experiences. While the interview guide aimed to broaden the discussion beyond e-scooters, the readily available e-scooters might have served as a more prominent reference point for participants.

The sampling methods for the pop-up interviews were contextual and random, meaning they were conducted in areas where many people travelled by different modes, and people were approached at random. This yielded a balanced demographic in terms of age and gender. However, there is a lack of e-scooter riders in the demographic. This is mostly because the interviews took place in February, a winter month in Stockholm. During this time, micro-mobility services generally experience a decline in ridership due to weather conditions. This seasonal factor led to a lower representation of e-scooter riders among the pop-up interview respondents. Research conducted in seasons with higher micro-mobility usage could potentially yield a more balanced perspective.

Furthermore, there were other difficulties in recruiting e-scooter drivers for pop-up interviews since the current design of the services caters to people in situations of time pressure. The e-scooter users we encountered were often stressed or paying by the minute as they rode, which obstructed a peaceful and rewarding conversation.

Snowball sampling was utilised for the in-depth interviews, and while this technique recruited many relevant participants, it may have introduced a bias towards demographics similar to the initial interviewees. With more time, the snowball effect could have yielded a broader demographic representation. For both the pop-up and the in-depth interviews, the interview target group was broad, including users and non-users of micro-mobility. This broad interview target group may have limited the captured range of perspectives.

Lastly, the recruitment of attendees for the co-creative workshop resulted in participants with diverse nationalities and a range of research backgrounds. None of the workshop participants had prior knowledge of the micro-mobility landscape specific to Stockholm. This could have benefitted the result from an unbiased perspective on the Stockholm case. However, it also presents potential challenges, such as knowledge gaps regarding the specific context and the risk of proposing solutions less applicable to the Stockholm case. These challenges were addressed by providing the participants with background information on Stockholm’s mobility and the insights gathered from Stockholm. Ultimately, the participants' diverse research backgrounds and nationalities benefitted the ideation process with a broader range of perspectives.

7.5 Stockholm Mobility

This project examines mobility practices in Stockholm and how SMMS can contribute to a more sustainable mobility landscape in this context. It is important to note that Stockholm is a city that is bound by public transport to a high degree. Public transport as a mode is, in most circumstances, better from a sustainability perspective than shared micro-mobility services. Because of this it is important that SMMS can co-exist with public transport and not compete with it. While it is not within the scope of this project to criticise the existence of SMMS, the question arises if micro-mobility is necessary in Stockholm or if it just adds an unnecessary element to the mobility landscape that produces additional CO_2 emissions. This might be the case in the way micro-mobility services are designed in Stockholm today. However, looking at Stockholm from a long-term and visionary perspective, where the infrastructure prioritises people over cars, we argue that shared micro-mobility services can play a key role.

In the frame of reference, it is identified that people who live in the outer city are more likely to be car-bound than people in the inner city. This is a challenge for implementing a SMMS

that successfully substitutes cars, since distances are longer, and people live less dense in these areas. In these areas, free-floating SMMS are often not economically feasible because of the required vehicle density and low usage compared to operational costs. On the other hand, a station-based service could be challenging as it is difficult to achieve a station distribution that offers a convenient walking distance for all residents. To effectively capture a potential user base in these areas and ensure that the new SMMS system caters to both inner-city and outer-city residents, further investigation into tailored implementation strategies for these distinct contexts is necessary.

7.6 Future Work and Recommendations

This section discusses recommended future work and the challenges of implementing the Integrated Active Mobility system concept. While the proposed service system concept offers a promising vision for a more environmentally friendly micro-mobility option in Stockholm, its realisation requires further exploration across several aspects.

The integrated active mobility system represents a proposal for a new micro-mobility service in Stockholm. However, the result remains on a conceptual level and has not yet been tested on users for usability. Usability testing can identify potential issues and refine the service concept for real-world application. Testing the service system concept together with users is a crucial step to ensure that user needs are met, and the service achieves its functionality. This could be done by prototyping the service, performing user tests, and iterating the concept based on user feedback.

In the Integrated Active Mobility system, changes beyond the immediate service boundaries are proposed. The feasibility of these external changes, such as modifications to public transport and station infrastructure, requires further evaluation. Additionally, collaboration with various actors, including the municipality, public transportation, and local coffee shops, is essential for successful implementation of the service system. Evaluating the potential challenges and opportunities for collaboration is necessary to ensure the feasibility of the concept.
When iterating and evaluating the feasibility of the Integrated Active Mobility system, a detailed design should be considered. This includes bike design, station design and digital platform interface.

Another crucial aspect in the detailed design that presents an opportunity for further research is station planning. This planning should ensure that stations are connected to public transport hubs and that the density and locations of the stations best cater to user needs. The Integrated Active Mobility system must also unlock routes that are difficult to reach by public transport, allowing it to function as an extension of the public transit network when needed.

The environmental implications should be further evaluated to ensure that the Integrated Active Mobility system actually contributes to decreased carbon emissions compared to today's case. While this project relies on existing research and LCA data, the methods used have limitations. Within the scope of this project, the real-life effects of the proposed service cannot be assessed. It is particularly hard to predict user substitution patterns.

For future recommended work, a consumer choice models with life cycle simulation and environmental evaluation should be done. Consumer choice models can be used to simulate

real-world user behaviour patterns, including substitution effects. This can provide a more accurate depiction of how the proposed service would impact existing travel practices and the corresponding environmental consequences. By integrating LCA within the simulation, the environmental impact of the entire service system can be evaluated. Combining consumer choice models with LCA allows for a more holistic evaluation of the environmental impact. These methods consider the broader environmental footprint associated with user choices and resource utilisation.

Lastly, the economic feasibility should be assessed, and the prices should be set after evaluating revenue streams and operational costs. This is necessary to ensure the service's long-term financial sustainability.

8. Conclusion

This project explores how to design micro-mobility services to support sustainable mobility practices in Stockholm, proposing a service system with ten key attributes. These attributes, working together, creates a shared micro-mobility system that supports more sustainable practices through three sustainability objectives: modal shift, increased utilisation rates, and more responsible use.

Extensive user research using a mixed-methods approach highlighted the current role of shared micro-mobility services as a response to research question one, namely ‘What role do micro-mobility services play in Stockholmer's mobility practices?’.

Four primary use cases were revealed: spontaneous usage in moments of time pressure, late-night rides, joyrides, and avoiding crowds.

The purposed service attributes offer practical suggestions that can be implemented to support sustainable mobility, answering to research question two, ‘What shared micro-mobility service attributes are required to enable more sustainable mobility practices in Stockholm?’.

The service system poses an opportunity for implementing a new type of micro-mobility service that make everyday mobility a rewarding experience and offers hybrid vehicles with a point system to incentivise active mobility. This approach encourages modal shifts for users dissatisfied with their current commutes, transforming a chore into a rewarding activity. The e-bikes and stations will jointly reduce reckless behaviour by encouraging clutter-free parking and provide vehicles with more intangible value. Integrating with public transport through a station-based layout and integrated planning and payment system streamlines first- and last-mile adoption, increasing utilisation rates and promoting modal shift. Additionally, the system supports respectful use through safety regulations and infrastructure.

The proposed service system is likely to contribute to a reduction in carbon dioxide emissions through shaping new mobility practices, but the extent of this impact might be incremental. Additionally, it is difficult to predict the modal shift patterns that would emerge from the usage of the service system design. Further work includes usability testing to ensure the system meets user needs and preferences, and life cycle assessments combined with consumer choice models to evaluate the environmental impact compared to existing shared micro-mobility services in Stockholm. Furthermore, the system does not question the existing amounts of mobility, it only seeks to reduce the resource intensity of existing practices and promoting modal shift towards lower-emission transportation modes.

In conclusion, this thesis project successfully addressed the initial research questions. It contributes to a deeper understanding of current shared micro-mobility service practices, complementing quantitative studies on use patterns. It also provides practical suggestions for designing shared micro-mobility services to improve environmental impact, aligning with current research on factors influencing the environmental impact of these services.

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Appendix A: Pop-up Interview Questionnaire

Hello,

We are master's students from KTH and we are writing our master's thesis on how to design shared micro-mobility services for more sustainable mobility practices.

We want to know more about how people travel within Stockholm and would like to ask a few short questions about your current trip.

Q1: Can you tell us about the journey you are currently on?

- Why did you choose this means of transport?
- What works well and what works poorly?

Q2: How have you planned your trip?

Q3: What do you think is needed to accomplish this journey? Knowledge? Artifacts?

Q4: How to behave? Is there anything you can't do?

Q5: How do you view shared e-scooter and e-bike services?

Appendix B: Demography of Interview Sample

Table 2 Pop-up Interview Sample

N	Gender	Age	Transport Mode	Place
P1	F	A2	Roslagsbanan	Tekniska
P2	F	A1	Car	Tekniska
P3	M	A2	Bus	Tekniska
P4	F	A4	Roslagsbanan	Tekniska
P5	F	A5	Roslagsbanan	Tekniska
P6	F	A1	Metro	Tekniska
P7	M	A4	Roslagsbanan	Tekniska
P8	M	A5	Roslagsbanan	Tekniska
P9	F	A4	Roslagsbanan	Tekniska
P10	M	A3	Bus	Tekniska
P11	F	A4	Bus	Bergshamra
P12	M	A5	Metro	Bergshamra
P13	M	A3	Metro	Bergshamra
P14	F	A2	Metro	Tekniska
P15	F	A5	Metro	Bergshamra
P16	F	A6	Metro	Bergshamra
P17	M	A3	Metro	Bergshamra
P18	F	A2	Metro	Tekniska
P19	M	A1	Metro	Ropsten
P20	F	A2	Bike	Tekniska
P21	M	A4	Car	Valhallavägen
P22	M	A6	Walk	Valhallavägen
P23	M	A2	Walk	Östermalm
P24	F	A3	Bike	Vasagatan
P25	F	A3	Tram	TC
P26	M	A3	Bus	Hötorget
P27	F	A5	Bike	Hötorget
P28	M	A4	Personal e-scooter	Hötorget
P29	M	A4	Bus	Sickla
P30	M	A5	Commuter train	Stockholm södra

P31	F	A5	Bike	Skanstull
P32	F	A5	Bike	Blecktornsparken
P33	M	A4	Bike	Medborgarplatsen
P34	F	A4	Commuter train	Stockholm södra
P35	M	A2	Shared e-scooter	Götgatan
P36	M	A3	Bike	Skanstull
P37	F	A5	Bike	Skanstull
P38	M	A2	Metro	Gamla stan
P39	F	A1	Bus	Jarlaplan
P40	F	A1	Walk	Jarlaplan
P41	M	A2	Metro	Gamla stan
P42	M	A1	Bus	Jarlaplan
P43	M	A2	Metro	Rådmansgatan
P44	F	A1	Walk	Jarlaplan
P45	F	A4	Bus	Norra djurgårdsstan
P46	M	A1	Walk	Jarlaplan
P47	M	A2	Metro	Rådmansgatan
P48*	M	A1	Metro	Rådmansgatan
P49	M	A2	Metro	Stadion
P50	M	A2	Metro	Midsommarkransen

Table 2. In-depth Interview Sample

N	Name	Gender	Age	Residence area
I1	Stina	F	A2	Värmdö
I2	Martin	M	A2	Malmö
I3	Lily	F	A2	Bergshamra
I4	Josefine	F	A2	Sundbyberg
I5	Josef	M	A2	Vasastan
I6	Helena	F	A4	Södermalm
I7	Elsa	F	A2	Gärdet
I8	David	M	A3	Fruängen
I9	Vera	F	A2	Norra Djurgårdsstaden
I10	Daniella	F	A2	Hammarby Sjöstad
I11	Alexander	M	A2	Värmdö
I12	Fredrik	M	A2	Knivsta
I13	Sebastian	M	A3	Liljeholmen
I14	Viktor	M	A2	Södermalm
I15	Olle	M	A1	Hammarby Sjöstad and Sickla
I16	Ivar	M	A2	Hammarbyhöjden

Appendix C: In-depth Interview Guide

Introduction:

We are master's students in product design at KTH. In our thesis, we are investigating shared micro-mobility services, i.e., services for renting lighter vehicles such as e-scooters and electric bicycles for shorter distances within urban areas. With this interview, we aim to understand how sustainable travel habits within Stockholm can be addressed in the future. We will not judge or value your answers; we are only interested in how you think and feel.

Your participation is voluntary and anonymous; you can cancel the interview anytime. In future compilations, your identity will not be identifiable. I will record the interview if that is okay with you, and it is mainly to help me remember what was said during the interview. The recording will be used for scientific purposes and treated confidentially

Demographics

- How old are you? *(Note/ask gender)*
- Describe your living situation.
    - Why do you live in Stockholm?
- Can you tell us about your everyday life?
    - Do you ever work from home?
    - What do you like to do in your spare time?

Mobility in general

Explain: Travel mode includes walking, biking, public transport, car, etc.

- Which modes do you use for traveling within Stockholm?

Note which travel modes; A, B, C…

- For what purposes do you use A, B, C...? Why?
    - Tell me about what influences your choice. Why?
- Which modes of transport do you like the most? Why?
- When do you find it difficult to travel within Stockholm?
    - Have you ever had to use a mode of transport you didn't want to use in the first place?
- Have your travel habits changed in recent years? If so, how and why?

Explain: We will now go deeper into car usage.

- Do you have a driver's license and/or a car? Why/Why not?

If car owner:

- When do you use your car? For what purpose?
- Is there something specific that makes you choose the car? Why?
- Is there anything that makes you choose not to use your car? Why?

If not car owner:

- Would you like to own a car?

    - Why don't you own a car?
- Have you ever owned a car?
    - Why not anymore?
- Do you ever need a car?
    - When do you need it?
- How do you access a car if you need it?

For all:
- How do you view car usage?
    - How do you think society views car usage?
- What emotions do you associate with driving?

Shared Micro-Mobility Services

Describe SMS and limit the scope to Stockholm (bike-scooter-car etc.)
- How does your experience of renting differ from owning?
    - What do you prefer?
- Do you think there is any difference in behavior? How and why?

Limit to platforms for e-scooters & electric bicycles
- Have you used a shared e-scooter or e-bike service?

If no or very rarely (otherwise skip this part)
- Why not?
    - What do you think would need to change for you to use it?
- What do you think about e-scooters & e-bikes?

If yes to e-scooter (otherwise skip this part)
- Tell me more: For what purpose? How often?
    - What makes you choose the e-scooter?
    - Can you think of other situations when it could be appropriate to use an e-scooter? Why is that not applicable today?
- Do you tend to plan ahead of your trips? How and why?
    - What other mode would you have used if the e-scooter was not available?
- How does it feel to use an e-scooter?
- What works well? Why?
- What does not work as well? Why?
- Is there anything that makes the use difficult or annoying? What and why? *E.g., Price, security, use, weather, app, other traffic, cityscape*
    - What would you have changed?
- How do you view people who use e-scooters?
    - Who do you think these services are aimed at?
    - Are there any unwritten rules?

If yes to electric bike (otherwise skip this part)
- What services have you used?
- Tell me more: For what purpose? How often?
- How did you experience the use?
- What works well? Why?

- What does not work as well? Why?
- Is there anything difficult or annoying?
    - What would you change?
- How do you view people who use e-bike services?
    - Who do you think these services are aimed at?

Sustainability

Explain: This section is about sustainability in general, not necessarily mobility.

- What does sustainability mean to you?
- How engaged in sustainability are you on a scale of 1-10? Why?
- Have you made any changes in your life to live more sustainably? What?
- Are you doing something that is not sustainable?
    - How does that feel?
- Do you think shared micro-mobility services can help people live more sustainably?
- What kind of services do you see as being able to replace cars in the future?
    - What could the vehicles look like?
    - Who is the sender? *E.g. Private company, Municipality*

Dreams and expectations

Explain: Now we want to know more about your dreams and hopes for the future.

- Can you tell us about a pleasant or great trip you have made within Stockholm?
- Can you paint a dream picture of the future of sustainable travel in Stockholm?

Outro

Thank you very much for your time and participation. Do you have anything else you would like to add? If you want us to share our final report with you, can we take your email address and send over the results when the report is published?

Appendix D: Workshop Outline

Part 1: Introduction, 40 minutes.

Start-up time, 13:00 (10 min)

Introduction, 13:10 (15 min)
- Presentation and purpose
- What potential do YOU see for shared micro-mobility to make a positive environmental impact?

Introductory lecture 13:25 (10 min)
- Goal, problem, interventions
- Stockholm mobility and how SMMS are used in Stockholm today

Opportunities 13:35 (5 min.)
- HMW statements, problems, and opportunities

Part 2: Ideation, 70 minutes.

Start up 13:40 (12 min)
- Divide groups, assign HMW statements

Warm-up 13:52 (3+2 min)
- Do the opposite!

Brainwriting 13:57 (5+2 min)

Build on each other 14:04 (15+2 min)

Prioritize 14:21 (10+2 min)
- Usability VS Feasibility

Compile intervention cards 14:33 (15+2 min)

Part 3: Results, 40 minutes.

Presentation 14.50 (30 min), 10 min/group
- 5 minutes presentation
- 5 minutes discussion

Wrap-up 15.20 (10 min)

Appendix E: Workshop Templates



Figure 1, Reverse brainstorming template



Figure 2, Brainwriting template



Figure 3, Clustering template

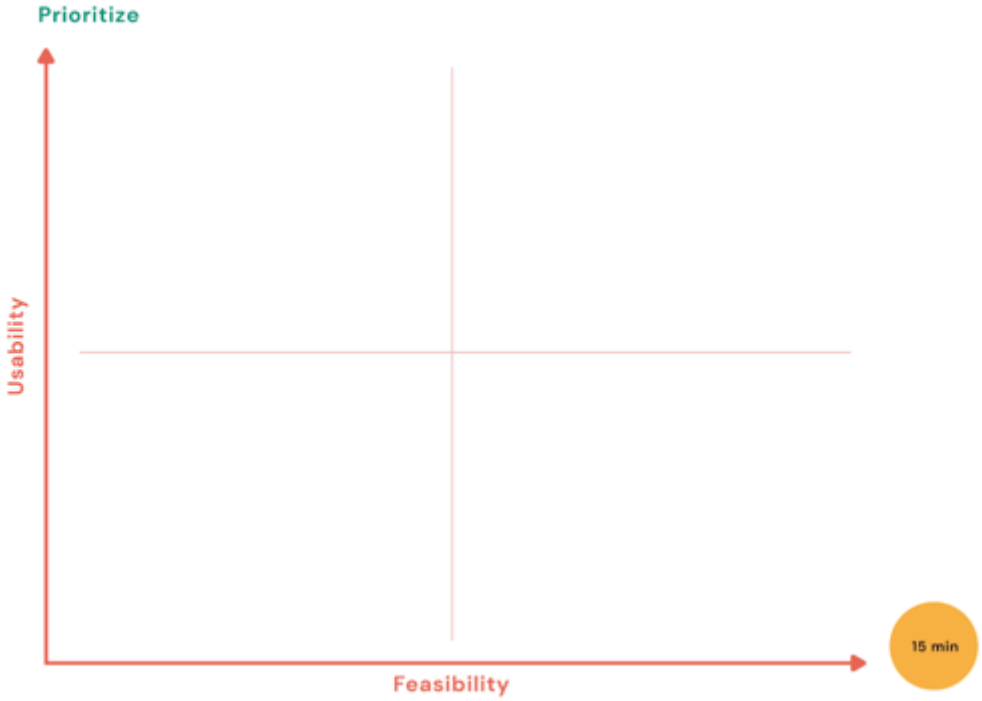


Figure 4, Prioritizing template

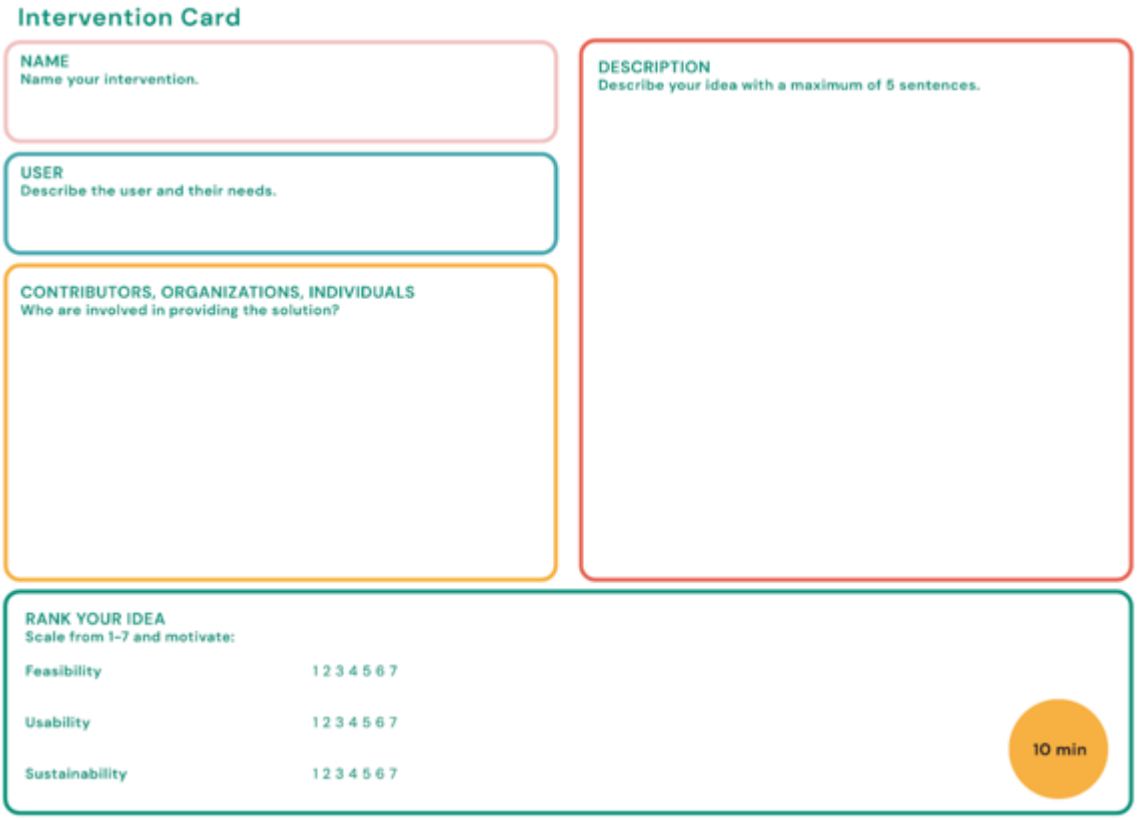


Figure 5, Intervention card template

Appendix F: HMW Cards

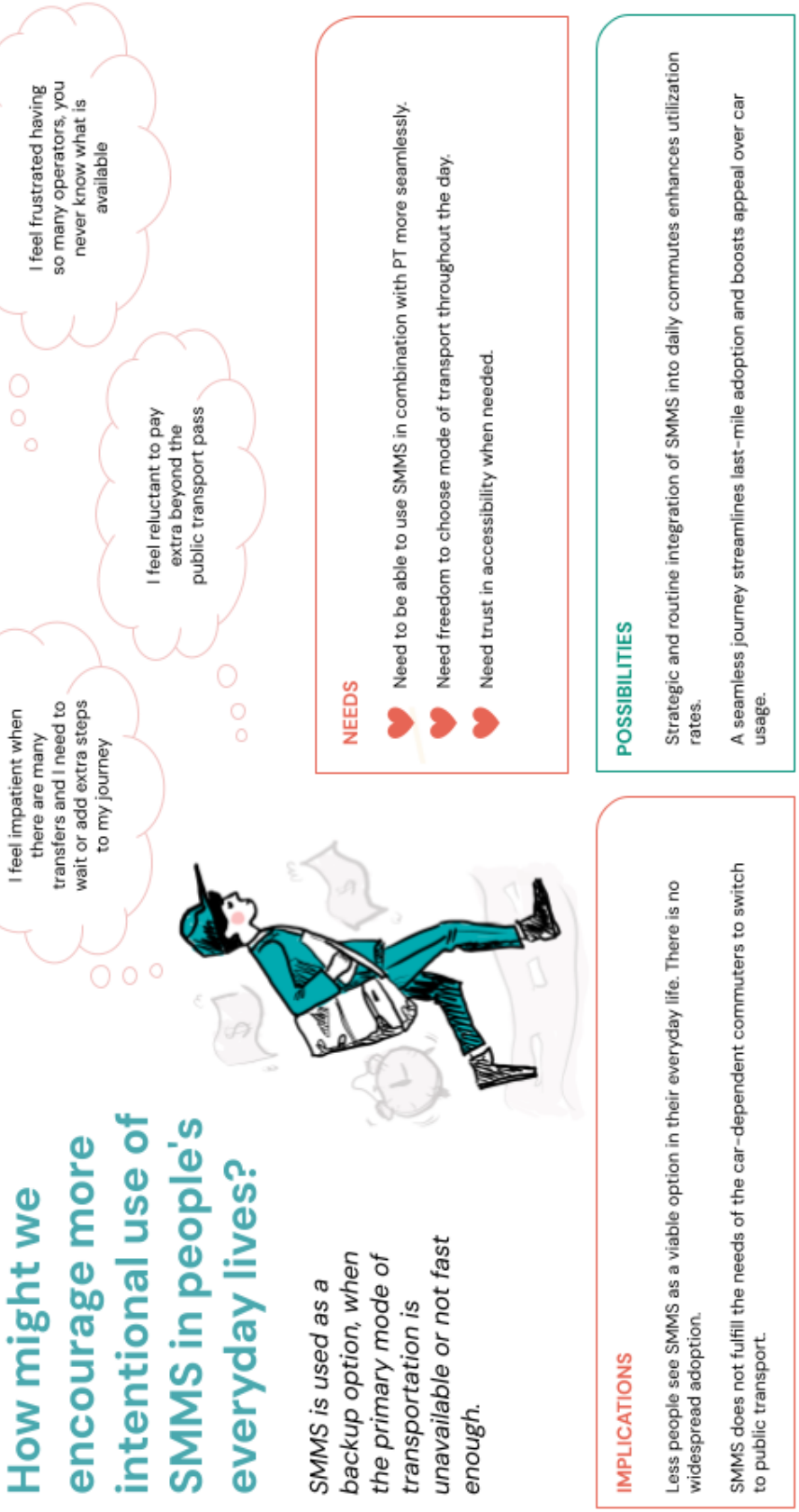


Figure 1, HMW1: How might we encourage more intentional use of SMMS in people's everyday lives?

How might we make people use micro mobility services with more respect?

SMMS have a bad reputation because history of reckless use of e-scooters.



I feel scared when I see people drive around recklessly, and I dislike seeing ragged e-scooters lying around in the streets

I don't like the look of riding an e-scooter. I don't want to be associated with that image

It feels dangerous to jump on a vehicle I cant master... you can really hurt yourself

NEEDS

- Need to feel trust in shared micro-mobility services.
- Need SMMS to be more approachable.
- Need a vehicle to identify with.

IMPLICATIONS

People have e negative attitude towards e-scooters and SMMS due to safety concerns, cityscape and image.

No widespread adoption.

POSSIBILITIES

More people are willing to try SMMS because they have a positive view of the service.

Vehicles are treated with greater respect, extending their lifespan.

Figure 2, HMW2: How might we make people use micro-mobility services with more respect?

Figure 3, HMW3: How might we use SMMS to make everyday mobility feel like a valuable experience?

Appendix G: Workshop Results, Integrated Smart SMMS



Figure 1, Intervention card, integrated smart SMMS

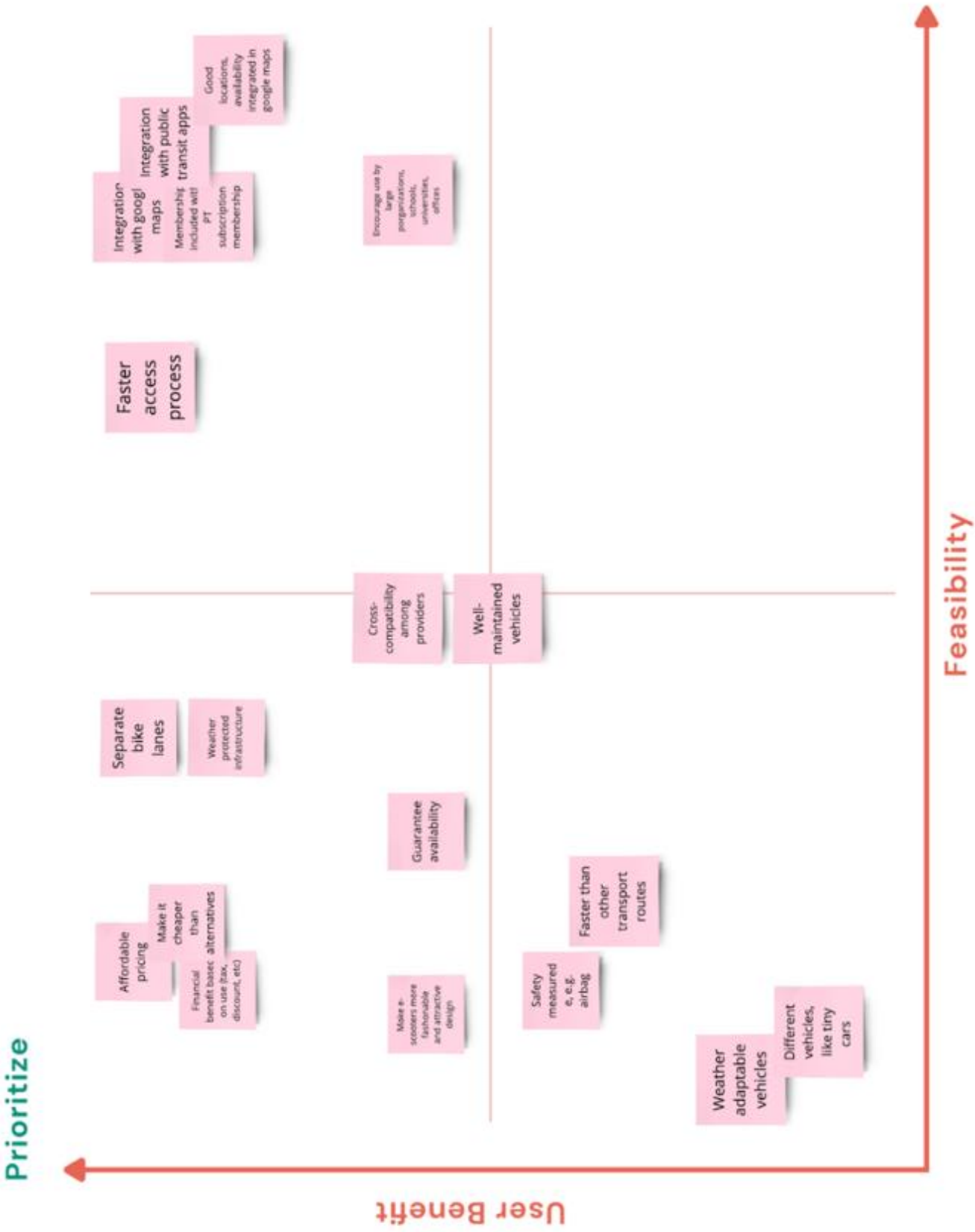


Figure 2, Solution set, integrated smart SMMS

Appendix H: Workshop Results, Safety Regulations

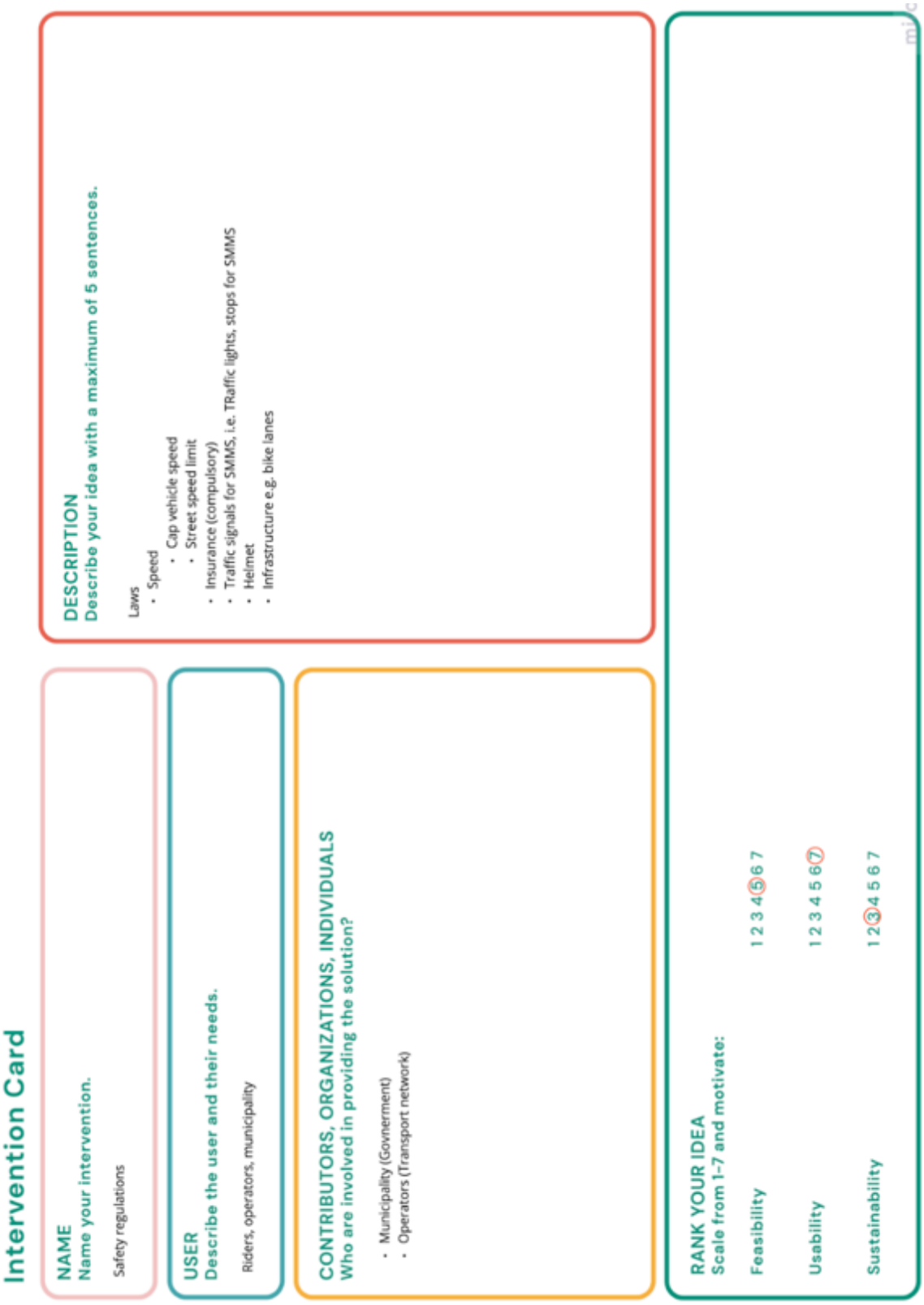


Figure 1, Intervention card, safety regulations

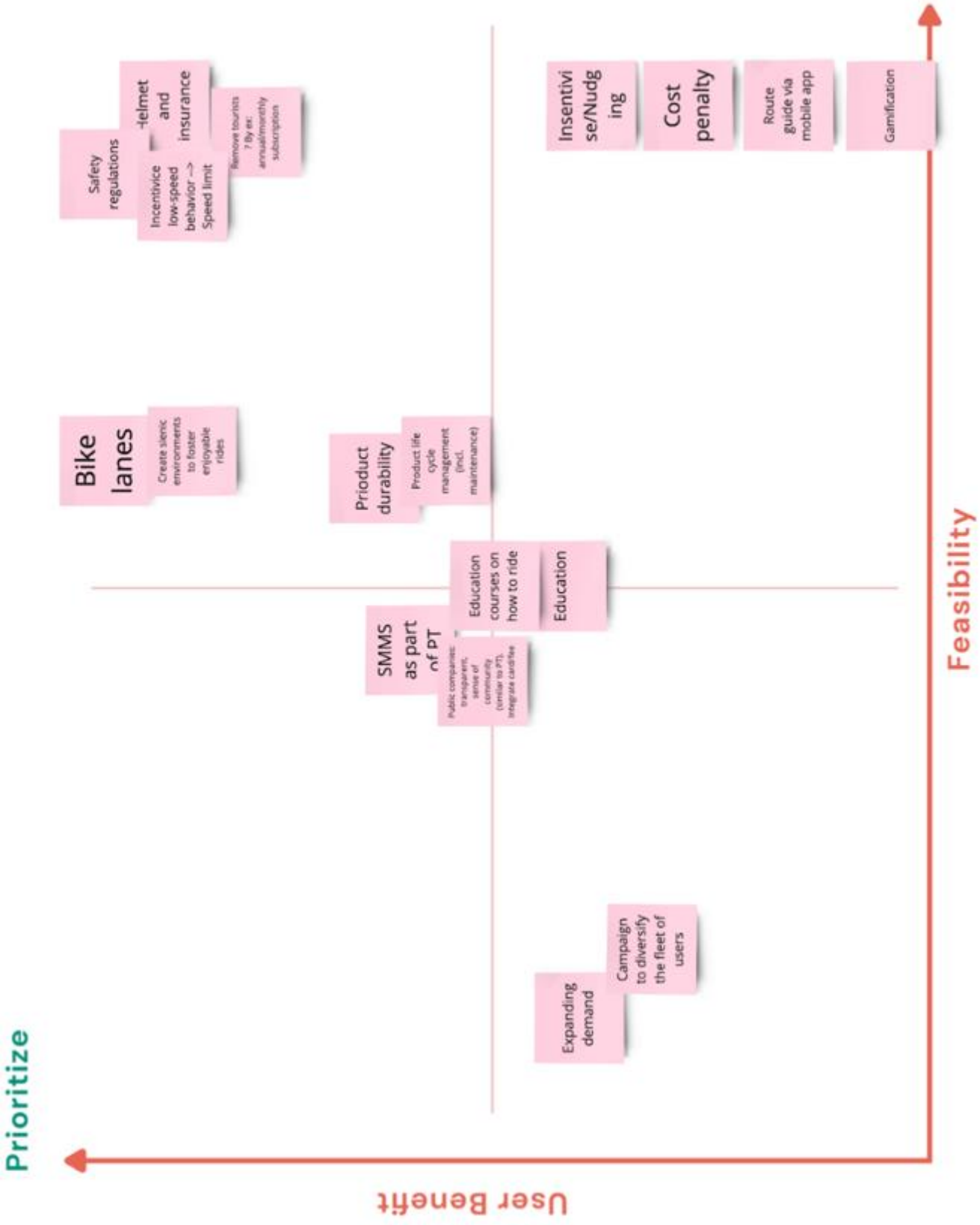


Figure 2, Solution set, Safety regulations

Appendix I: Workshop Results, Incentivising Active Mobility

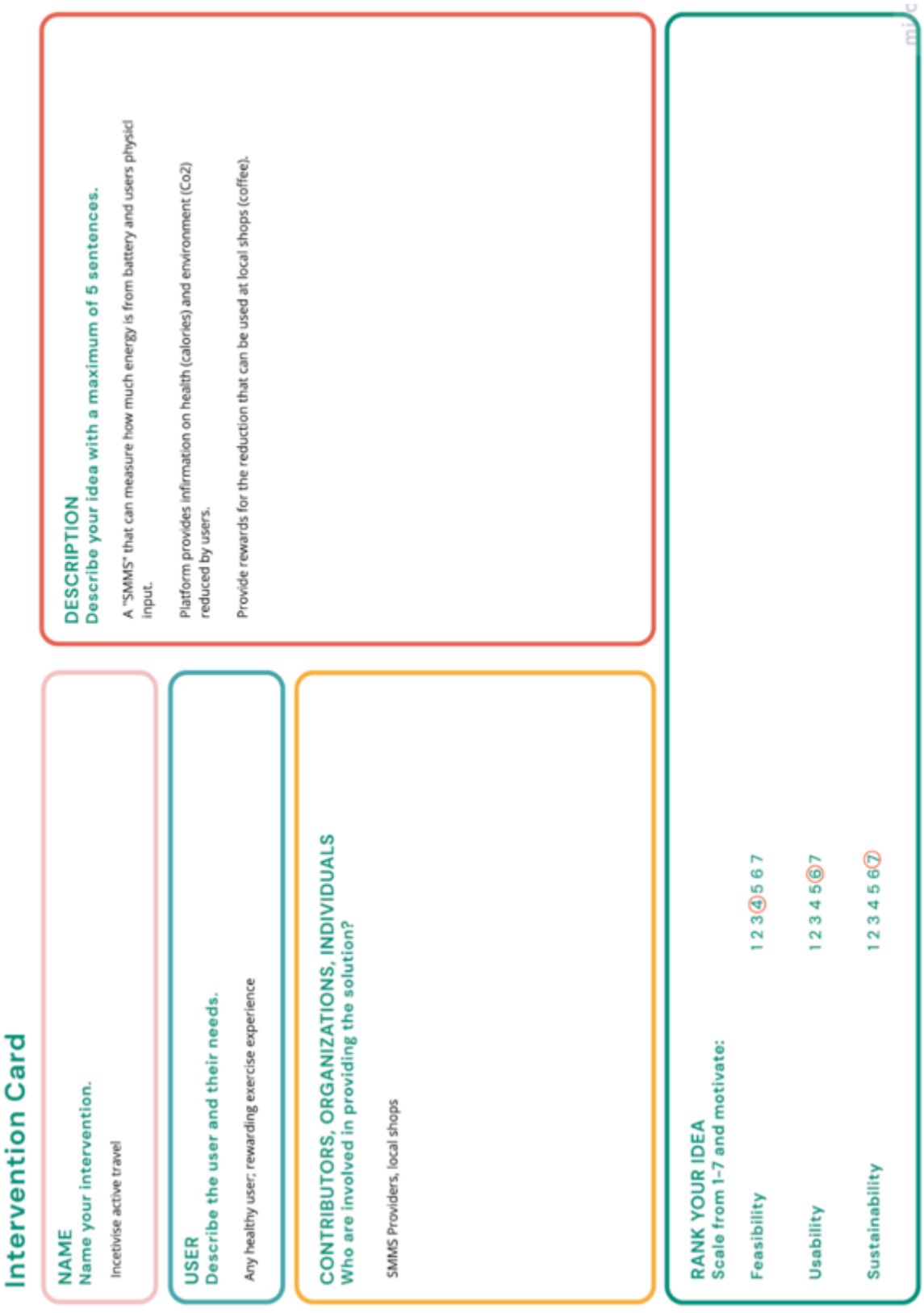


Figure 1, Intervention card, active mobility

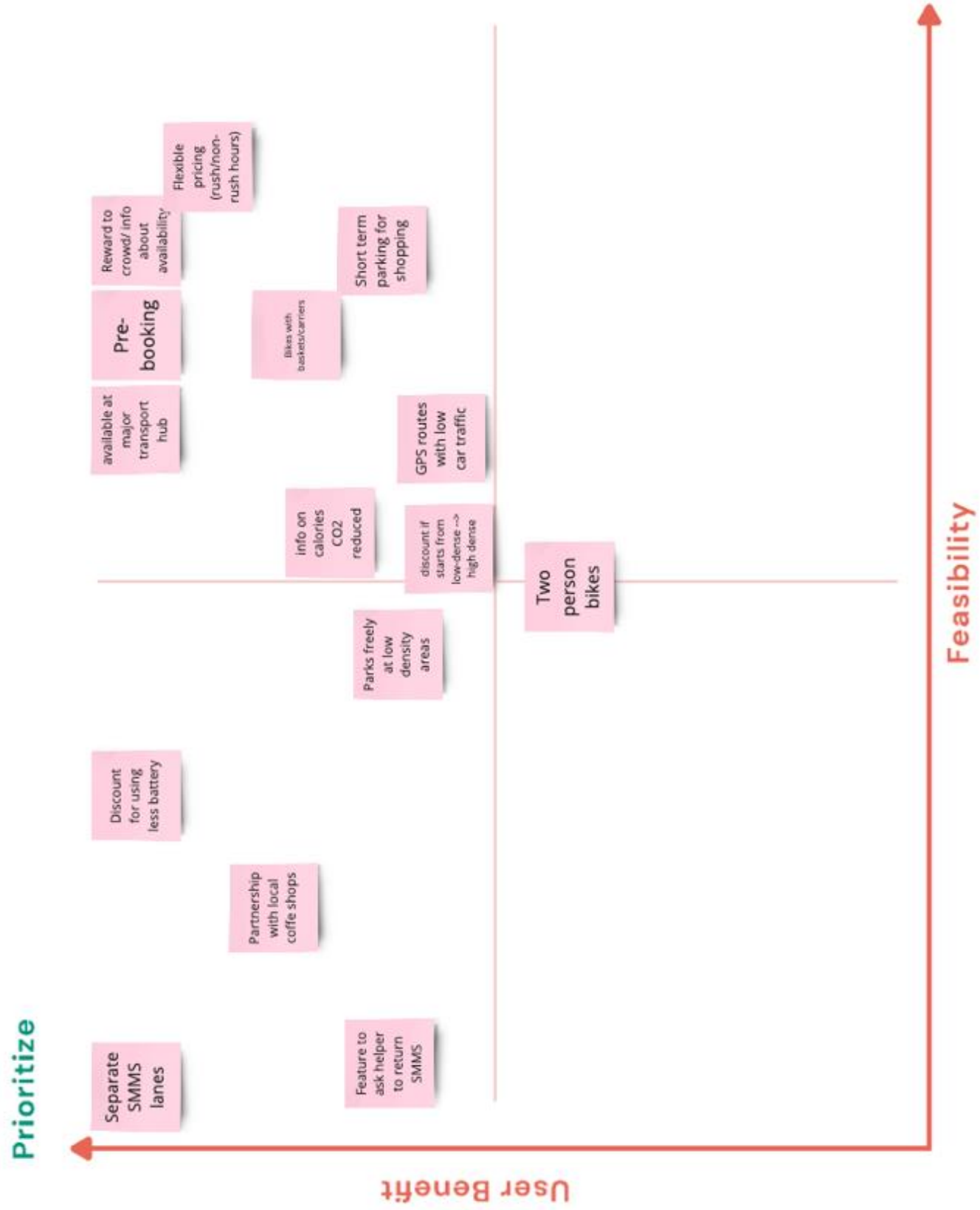


Figure 2, Solution set, active mobility

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